The TFPT Prime-Front RH Program: Certified Finite Reductions and the Remaining Arithmetic Edge

TFPT verification programme — `rh/` workspace

2026-09-02 (research documentation, no RH claim)

Abstract

We document the current state of the TFPT prime-front program on the Riemann Hypothesis: a chain of *certified finite reductions* that compresses the positivity of the finite Galerkin compression of the localized Weil operator ("the wall") into exact integrable-systems dictionaries — IIKS tau functions, a half-filling moment geometry, a bordered Riemann–Hilbert dictionary, and a coupled two-term tau recursion whose budget telescope reduces the entire fiber positivity to one terminal inequality $q_N < 1$. The certified part is machine-checked at the exact-identity level (**[E]** in the repository ledger); what remains open are two precisely named edges (**[O]**): the terminal cross-ratio inequality cofinally, and a globally oriented prefix resummation — since r273 both are corollaries of one master target, the augmented prefix positivity of Section 17.1. We also catalogue the measured no-gos (the honest negatives), position the program relative to the external unconditional result arXiv:2608.13637, and report the Lean 4 formalization status of the provable core. *This document is research documentation. Nothing in it is evidence for or against the Riemann Hypothesis in either direction. NO RH CLAIM.*

Status (binding). Research documentation, **no RH claim in either direction**. Status markers follow the repository ledger `verification/status_ledger.csv` (the ledger wins on any disagreement): **[E]** = certified finite theorems (machine-checked exact identities with mutation controls), **[O]** = honestly open. The reduction mincut of the program is **base 4 / refined 5** and is unchanged by everything reported here (dictionaries and adjudications move no edge). All numerical statements below are *finite, window-level* statements about the concrete verification objects; none extends to ζ itself.

Contents

1 The window object and the claim chain

1.1 The window object

The program studies one concrete finite object per *window* w: the Galerkin compression of the localized Weil operator onto a window of the von-Mangoldt double comb (prime side μ, smooth/archimedean side ν), together with three control worlds (SCRAMBLE: randomized weights; EPSTEIN: an Epstein zeta comb, off-line zeros; SMOOTH: a smooth density). “The wall” of window w is the statement that this compression stays positive (quasi-definite) through the whole window — a finite, machine-decidable statement per window. The measured ladder spans 42 rungs with terminal degrees $N = 142$ to 878.

1.2 The equivalence chain (all [E], typed as equivalences)

The certified dictionaries identify the wall, exactly and finitely, in five languages. Each identification is an *equivalence*: it moves no edge of the mincut, it changes the coordinates.

$$\begin{aligned}
\text{wall of } w \quad &\Longleftrightarrow \quad \text{quasi-definiteness of the signed defect measure } \tilde{\mu} = \mu - \nu \quad \text{(v955)} \\
&\Longleftrightarrow \quad \text{all Christoffel ratios } r_n = h_n(\tilde{\mu})/h_n(\mu) > 0 \quad \text{(v955, Toda/Hirota)} \\
&\Longleftrightarrow \quad \text{survival of the full free moment window at half-filling} \quad \text{(v956)} \\
&\Longleftrightarrow \quad \text{bordered PSD: } \begin{bmatrix} G_N & u \\ u^{\mathsf{T}} & B \end{bmatrix} \succeq 0 \text{ i.e. wall + budget} \quad \text{(v958)} \\
&\Longleftrightarrow \quad \text{budget inequality } \textstyle\sum_n \rho_n < B \text{ with } \rho_n = F_n^2/h_n \quad \text{(v959)} \\
&\Longleftrightarrow \quad \text{ONE terminal inequality } q_N = \rho_{N-1}/(5/7) < 1 \quad \text{(v959, r258).}
\end{aligned}$$

1.3 Mincut

The reduction mincut stays **base 4 / refined 5**: the two named open edges of Section 17 (fiber and base/orientation), plus the frozen upstream contracts, none of which is moved by a dictionary. The wall itself is contracted as `PRIME.CASE.PAIRCORR.CONTRACT.01` [O]. A parallel *conditional* Gabor-wavepacket route (Section 17.4) does not move this mincut: its live Lean criterion is positivity of `gaborZeroSide` (hpos*); the historical wrapper `gabor_inputs_to_mathlib_rh` is VACUOUS (r600); the r541 numeric precheck is not a completed separation.

2 Certified finite theorems

All theorems in this section are [E]: machine-checked exact identities (sympy-symbolic and/or exact rationals, with f64+mpmath wards and firing must-fail mutations), promoted into the load-bearing suite. We state the mathematical content; the certification boxes carry the provenance.

2.1 The IIKS tau dictionary (v955)

Theorem 2.1 (Tau identification, finite IIKS set). *On the measured heavy rungs the wall determinant is the exact IIKS Fredholm tau function of the discrete* 2×2 *problem: (i) Sylvester as a full characteristic identity* $\det(I - sQ_{\text{state}}) = \det(I - sE_{\text{node}})$ *on the whole* s*-grid; (ii) the* s*-family factorizes exactly through the* s*-dressed port,* $\det(I - sE) = \det(I - sR)\,\det(I - s\,D_P(s))$ *with* $D_P(s) = P + sX(I - sR)^{-1}X^{\mathsf{T}}$*; (iii) the diagonal is the confluent limit of the same generator formula (constructed kernel, no completion divisor); (iv) the dressed resolvent is IIKS-integrable* *again, with source-explicit generators via the Schur transport $\hat{F} = F_p + X(I-R)^{-1}F_b$; (v) the Fredholm/JMU variational identity $d \log \det(I - sK(t)) = -s \operatorname{Tr}[(I-sK)^{-1}\, dK]$ holds in two independent times; (vi) the relative tau transfer $\log \tau_{h+1} - \log \tau_h = \log \det[I - (I - K_\uparrow)^{-1}\Delta K]$ is exact across all dimension changes. The h-step is exactly rank one and carries a predictive degree-1 Lax dynamics; the closing direction is h, not s.*

Theorem 2.2 (Toda/Hirota dictionary and sign adjudication)**.** *With $\tau_{w,n} = D_n(\mu - \nu)/D_n(\mu)$ (Hankel-determinant quotient) and $r_n = h_n(\mu-\nu)/h_n(\mu)$, the Hirota coefficient $H_n = r_n/r_{n-1}$ comes from a scaled signed Stieltjes recursion consuming only node positions and weights; the flag Hirota identity holds in Schur innovation form, and*

$$\textit{all } r_n > 0 \iff \textit{all } h_n(\mu-\nu) > 0 \iff \textit{quasi-definiteness of } \tilde{\mu} \iff \textit{the wall},$$

with $h_n(\tilde{\mu}) = \|\pi_n\|_\mu^2 - \|\pi_n\|_\nu^2$ an explicit difference of two positive norms. On MAIN *all* 184 *ratios r_n are measured positive (min* $+0.3666$*); the controls flip at $n = 25/21/27$.*

Certification. Ledger: `PRIME.PORT.TAU.FINITE.IIKS.01 [E]; PRIME.PORT.TAU.NOPOLE.COFINAL.01 [O].`
Module: `verification/v955_tau_iiks_toda_dictionary.py`.
Probes (embedded byte-exact, re-run at promotion): `tau_symbolic_probe.py` (18/18, SPEC `fb04cd7ce9fb306b`), `lax_conditioned_probe.py` (15/15, SPEC `93fd9759db7e7b3c`), `hirota_sign_probe.py` (17/17, SPEC `d78e236bf633de7b`); rounds r224–r226.

2.2 The half-filling geometry (v956)

Theorem 2.3 (Half-filling law and free moment window)**.** *Let $S = \#\mathrm{supp}(\tilde{\mu})$ be the union support count of the signed defect measure. Then on all five measured windows the window cap obeys the exact law*

$$N_w = \lceil S/2 \rceil = \frac{S+1}{2}\Big|_{\mathbb{N}},$$

i.e. the wall lives exactly at half-filling *of the union support: H_n consumes the moments $m_0, \ldots, m_{2n-2}$, an S-atom signed measure has exactly S free moment parameters, and the largest Hankel window inside them is $n = (S+1)/2$.* MAIN *survives the* entire *free moment window plus $O(1)$ forced-tail degrees (0/2/2/3/1 on the five windows), while the controls die at* 0.11–$0.15\, N_w$*: free-window survival is not generic for signed measures — it is the arithmetic. Beyond the cap the wall dies immediately (first r-flip at $N_w + O(1)$, no growth with N):* the wall is maximal.

Theorem 2.4 (Four-path dictionary, dual reversal, L-gauge)**.** *Moment prefix = Jacobi chain = Euclidean chain = value chain: one object in four exact source-pure coordinate systems ($m_k = h_0(J^k)_{00}$ exactly in rationals; parameter budget $2N-1$). The dual signed measure carries the mirrored Jacobi chain $\alpha_m^\# = \alpha_{S-1-m}$, $\beta_m^\# = \beta_{S-m}$ (exact), and the original and dual FIK problems are connected by an explicit h-free L-gauge whose one blind spot is exactly the wall's orientation. The control-world collapse is a genuine false pivot of the Euclidean remainder algorithm, localized without Hankel or tau data (first wrong pivot on* EPSTEIN *exactly at the known flip $n = 25$).*

Certification. Ledger: `PRIME.FREEMOMENT.JFRACTION.01 [E]; PRIME.JACOBI.DUAL.REVERSAL.01 [E]; PRIME.FREEMOMENT.POSITIVEPREFIX.01 [O].` Module: `verification/v956_signedmoment_halffilling_duality.py`.
Probes: `holedual_probe.py` (14/14, SPEC `a823d4be3f0b1c06`), `pontryagin_maxpos_probe.py` (17/17, SPEC `b062906cb458da2a`), `jfraction_probe.py` (17/17, SPEC `124cda6f00caeeca`), `rhp_midpoint_probe.py` (13/13, SPEC `90d2c5a8926d820a`); rounds r228–r231.

2.3 The bordered-RHP readout dictionary (v958)

Theorem 2.5 (Bordered dictionary)**.** *The bordered Hankel end-object (bordered PSD = wall + budget) has an exact Riemann–Hilbert dictionary: (i) the Uvarov border column* $C_n^\sigma(z) = \int \hat{\pi}_n \, d\sigma/(z-x)$ *carries the border readout* F_n *as its* z^{-1} *coefficient; (ii) the budget is an RHP functional of the terminal data alone,* $S_{n-1} = \iint K_n \, d\sigma \, d\sigma$ *with the CD kernel* $K_n(x,y) = \left(Y_n^{-1}(y) Y_n(x)\right)_{21}/(y-x)$*; (iii) the border is a Schlesinger rank-1 insertion: the* 3×3 *RHP is unipotent/triangular over a canonical* 2×2 *core with a one-dimensional budget cocycle (no irreducible* 3×3*, no growing rank;* $\det Y_3 = -h_{n-2}F_{n-1}$*); (iv) the centering congruence with* $c = F_0/h_0$ *eliminates the geometric rank-1 head algebraically (bordered PSD iff centered PSD with* $B^\circ = B - \rho_0$*), and the extensive tail compresses to one terminal CD readout; (v) three exact error-evaluation formulas hold, including* $\delta h_n = (R_1)_{12}$ *with* $R = YM^{-1}$ *and* R_1 *computable as the contour integral* $\frac{1}{2\pi i}\oint (R(z)-I)\,dz$*; (vi) the PSD base theorem (module-own, exact rationals): for any positive measure the Gram/Hankel matrix is PSD and every border readout* $T = t^\mathsf{T} G_N t \geq 0$*; a signed measure breaks the minor chain — any bordered-positivity route using only positive-basis Gram structure proves the wrong statement.*

Certification. Ledger: `PRIME.PORT.RHP.BORDERED.READOUT.01 [E]`; `PRIME.PORT.RHP.FULLSOURCE.BASEFIBER.01 [O]`. Module: `verification/v958_bordered_tau_readout_dictionary.py`.
Probes: `bordered_hankel_probe.py` (23/23, SPEC `c21e15b5852126b9`), `bordered_finite_rank_probe.py` (29/29, SPEC `886c65618145b838`), `border_centering_probe.py` (24/24, SPEC `767d16cff674f312`), `targetreadout_error_probe.py` (20/20, SPEC `b76487877383f645`), `base_gauge_constant_probe.py` (22/22, SPEC `26276cb2758152e9`), `schlesinger_pairing_probe.py` (26/26, SPEC `e5f0498c98112da4`); rounds r244–r253.

2.4 The coupled-tau terminal dictionary (v959)

Theorem 2.6 (Coupled recursion and Riccati drain; v959 S0.1)**.** *With* $a_n = c_n^2 h_n$ *and* $b_n = -(c_n F_n)^2$ *the pair*

$$\tau_{n+1} = a_n \tau_n, \qquad \tau_{n+1}^{\mathrm{aug}} = a_n \tau_n^{\mathrm{aug}} + b_n \tau_n$$

implies, wherever the divisions are defined, the Riccati drain

$$D_{n+1} = D_n - \frac{F_n^2}{h_n}, \qquad D = \frac{\tau^{\mathrm{aug}}}{\tau}.$$

The coefficient b_n *is a manifest negative square (source-sign-pure in every world);* a_n *reads the h-chain* — the base alone carries the sign, the border only nonnegative magnitude.

Theorem 2.7 (Bilinear tau form; v959 S0.2)**.** *Pure ring algebra gives the Wronskian-type bilinear identity*

$$\tau_n^{\mathrm{aug}} \, \tau_{n+1} - \tau_{n+1}^{\mathrm{aug}} \, \tau_n = (c_n F_n)^2 \, \tau_n^2,$$

whose right side is a manifest square: the fiber increment is a positive tau cross-ratio, and the chain telescopes exactly to $D_N = B - \sum_{n<N} \rho_n$*.*

Theorem 2.8 (Terminal reduction and consequence gate; v959 S0.3–S0.4)**.** *With the r243 budget definition* $B = S_{N-2} + \frac{5}{7}$ *(by construction),*

$$D_N = \tfrac{5}{7} - \rho_{N-1}, \qquad \textit{and} \qquad D_N > 0 \iff q_N = \frac{\rho_{N-1}}{5/7} < 1 :$$

the entire *fiber positivity is one terminal inequality. The corner ray* $B(t) = S_{N-2} + t \cdot \frac{5}{7}$ *makes the terminal pivot affine in* t *with machine root* $t^* = q_N$*. The provenance of* $5/7$ *is sober: it is the inverted r241 floor* $7/5$ *of the* measured $\min h_{N-1}/F_{N-1}^2 = 1.4278$ *(margin* 0.028*), typed* FLOOR_IMPORTED *— its source-pure derivation is the named terminal task.*

Theorem 2.9 (Positive-prefix firewall; v959 S0.5)**.** *For a signed rational measure the prefix moment kernel H_p is positive definite iff the whole h-prefix $h_0, \dots, h_{p-1}$ is positive (exact Sylvester chain, $\det H_p = \prod_{i<p} h_i$). Firewall table:* MAIN *$p_{\max} = 184 = N$ with zero negative modes (the only positive-prefix world);* SCRAMBLE/EPSTEIN/SMOOTH *continue indefinitely at $p_{\max} = 21/25/27$.*

Remark 2.10 (The q-census and the covariant ratio; r258)*.* Across the full ladder (42 rungs, $N = 142..878$): $q_N < 1$ everywhere, $\min/\mathrm{med}/\max = 0.0015/0.4188/0.9805$, minimal margin 0.0139; trend FLAT (gap slope +0.41, Spearman +0.36) — *the wall margin does not shrink with* N. The sharpest single finding: Spearman$(2\log F_{N-1}, \log h_{N-1}) = +1.000$ — border-readout square and terminal norm are rank-perfectly covariant; the driver is neither F nor h but the quotient itself. Any terminal proof must bind the *ratio*.

Certification. Ledger: `PRIME.PORT.RHP.COUPLEDTAU.TERMINAL.01 [E]; PRIME.PORT.COUPLEDTAU.TERMINAL_CROSSRATIO.01 [O]; PRIME.PORT.FULLSOURCE.PREFIX_RESUMMATION.01 [O]`. Module: `verification/v959_coupledtau_terminal_dictionary.py`.
Probes: `baseborder_factorial_probe.py` (21/21, SPEC `c7aa9f46889414c4`), `coupledtau_probe.py` (26/26, SPEC `73d8247f6de36a2b`), `budget_anatomy_probe.py` (27/27, SPEC `28026ae6b0750098`), `parametrix_pass_probe.py` (19/19, SPEC `e79da4f43d63213a`); rounds r256–r259.

2.5 Kernel-Loewner positivity at $L = 0.3$ (v1017)

Theorem 2.11 (Kernel-Loewner floor at $L = 0.3$)**.** *Let $h \in L^2(\mathbb{R})$ be supported in $[-0.3, 0.3]$ (so $2L = 0.6 < \log 2$ and the classical prime term is empty). Write $g = h * h(-\cdot)$ and*

$$Q_W(h) = \int_0^{2L} k(x)\big(g(0) - g(x)\big)\,dx - c_L\, g(0) + \Pi(h),$$

with $k(x) = e^{x/2}/\sinh x$ and $\Pi(h) = 2\big(\langle h, \cosh(x/2)\rangle^2 - \langle h, \sinh(x/2)\rangle^2\big)$. Then

$$Q_W(h) \;\geq\; 2.1{\times}10^{-3}\,\|h\|_2^2.$$

The enclosed floor is $2.122{\times}10^{-3}$ as a float64 *machine check (not interval arithmetic, not an* **[E]** *exact/interval certificate): a quintic C^2 cutoff books the near-diagonal as κ_w, a 401-dimensional Legendre section of a degree-48 Chebyshev surrogate is enclosed by Bernstein HS bounds, and the exact tail is charged three times. Independently, the zero-extension identity $2(g(0) - g(x)) = \|h - \tau_x h\|^2_{L^2(\mathbb{R})}$ holds exactly over $\mathbb{Q}$ on two step functions (boundary-strip mass strictly positive), and doubling c_L produces a negative certificate budget.*

This is a *single-length* operator inequality. It is not $\lambda_*(L) \geq 0$ for general L. Round r496 is a method no-go for the same compact-tail completion at $L = 0.8$ (Section 15). No RH claim.

Certification. Ledger: `PRIME.RDAGGER.KERNEL_LOEWNER.01 [Numerical/certified]`. Module: `verification/v1017_kernel_loewner_positivity.py`.
FLOAT64 certificate (Gauss–Legendre + Chebyshev/Bernstein + HS tail), not interval arithmetic, not **[E]**.
Probes (stay experiments-side; module re-derives): `kernel_loewner_probe.py` (r494, 17/17, VERDICT `PROVED(c=2.1e-3)`, SPEC `2e6d71138dff4540`), `kernel_redteam_probe.py` (r495, 11/11 smoke / `CONFIRMED(c=2.1e-3)`, SPEC `431251a9bcfd40a1`). Boundary: `kernel_loewner08_probe.py` (r496, `NO_GO` compact-tail@$L = 0.8$).

3 The two-branch theorem and the surface closure (v960)

Rounds r260–r263 organize the one remaining fiber inequality; rounds r268–r271 close its 42-rung surface. Since wave 4 the whole set is promoted as `v960` (certification box at the end of this section).

3.1 The exact decomposition (r263)

Write the terminal object in scaled coordinates: $Z_w = r_{N-1}$ (the driven terminal readout; identity $Z_w \equiv r_{N-1}$ gated at 1.8×10^{-14}), $M_w = \sqrt{5/7}$, and let U_w be the sealed target-blind triangle certificate (the C1b terminal triangle $|r_{N-1}| \le |t_{N-2}| + |a'||r_{N-2}| + |b'||r_{N-3}|$, consuming prefix data only; $t_k = T_k/\sqrt{h_{k+1}}$ is the border-comb drive). Then $q_N < 1$ splits into two branches on the scalar $g_w = M_w - U_w$:

- **Generic (cheap) branch $g_w \ge 0$:** the triangle inequality alone closes the target, *without any cancellation*: $|Z_w| \le U_w \le M_w \Rightarrow q_N \le 1$ (strict with strict slack). Measured: exactly 35/42 rungs, including *both* mains (w9 $g = +0.442$, w13 $g = +0.212$).
- **Exception branch $g_w < 0$:** exactly the seven rungs $\{kz15, 20, 22, 36, 38, 39, 52\}$, all with $Z_w < 0$, all *drive-dominated* ($|t_{N-2}|$ the largest triangle term), minimal modulus slack $M - |Z| = 0.027$ (kz15). The missing decade is genuine cancellation between the border-comb drive and the chain terms, at one degree per window.

3.2 The binding retype (r263)

ONE_DEGREE_PER_WINDOW_LOCALIZATION + TRIANGLE_GENERIC_BRANCH
+ PAIRING_CANCELLATION_EXCEPTION_BRANCH

A finite sum term per window is *not* a finitely solvable problem — the unbounded window quantifier makes the family of growing finite comb sums the same arithmetic wall. The eventual-triangle hope is refuted: exceptions occur in all terciles and the deepest rung kz52 ($N = 878$) is itself an exception. The measured cancellation degree separates sharply: $\kappa = |r_{N-1}|/\text{triangle} = 0.427$ on misses vs 0.998 on covers.

3.3 The S6 requirement and the RHP dictionary (r264)

The campaign charter (frozen, SHA `7b9e751d`; 7 exceptions, 5 kills, 2 blocked paths) states the remaining goal verbatim:

$$\textbf{S6:} \qquad Z_w = Z_w^{\mathrm{RHP}} + E_w, \qquad |Z_w^{\mathrm{RHP}}| + |E_w| < \sqrt{5/7} \quad \text{on the branch } g_w < 0,$$

or directly the cross-ratio positivity. Round r264 delivered the exact dictionary $Z^{\mathrm{RHP}} =$ (border-column readout)/(Y1 root): F_{N-1} from the rank-1 Uvarov pairing, h_{N-1} from the Y_1 entry, the base constant cancelling exactly (worst 9.2×10^{-9} on all 42+2 worlds, sign 44/44) — the target package is entirely in RHP language. The data signature of the exception branch sits in the drive column: S1_TERMDRIVE $|t_{N-2}|$ separates at 0.959, while the bare comb/source geometry is blind (0.14/0.18).

3.4 The surface closure (r268–r271): 42/42 certified

The exception branch is now closed *as a surface*:

- **Edge truncation (r268).** The outer-hull triangle bound `b3F0.20` delivered the *first source-pure certification of an exception rung ever* (kz22, margin +0.04), with the remaining six gaps measured *sub-paircorr* (0.18–0.44 dec, all demands < 1.0 dec, no wall flag). Structurally closed en route: the L-gauge path collapses at the nodes exactly ($\hat{\pi}^{\#}_{S-1-k}(x_j) = w_j L'(x_j)\hat{\pi}_k(x_j)/h_k$) — the dual degree of the drive is N, not small.
- **Block alternation (r269).** Adjacent sign-run pairs as exact pair sums, then the triangle on the pairs: gains +0.26 to +0.41 dec on *all* 42 rungs and certifies six of the seven exceptions source-purely (kz20/22/36/38/39/52; full ladder 37/42); *no* detector fires (paircorr demand max +0.07 at bar 1.0 — adjacent-run pairing is local phase structure).

- **The kz15 interval certificate (r270).** The whole bordered chain in outward-rounded interval arithmetic (mpmath, dps 640, 202 steps, ∼548 digits of dependency consumption, end width 1.5×10^{-92}, all 810 source atoms exactly converted): $\sup |Z_{kz15}| < \inf \sqrt{5/7}$ *strictly*, margin +0.02680; the dps-halving ward destroys the certificate as demanded. Honestly typed an exact *finite* certificate (`EnclOK` class) — no mechanism, no target-blind class evidence.

- **The universal pair theorem (r271).** The fixed form `c2PAIR` as a theorem: hypotheses H1–H4 are structural/source-side (exact pair decomposition, alternation, fixed pairing, boundary term) and are *proved in Lean* (`rh/lean/RH/PairBound.lean`, no `sorry`); H5 — the margin $|Z_{\mathrm{local}}| + \varepsilon < \sqrt{5/7}$ — is the *single* window-dependent hypothesis, the formal entry gate of the cofinal step.

The milestone. Together: the 42-rung surface census $q_N < 1$ is *certified on all* 42 *rungs* as a finite fact — 41 mechanism-certified target-blind (35 cheap via the two-branch theorem, 6 exceptions via the sealed phase bounds) + kz15 exact-finite; independent cross-validation (r270-b1 vs r271-b2LEVEL2 identical on separate implementations: kz39 0.002 / kz15 0.06 dec). The surface is a census + certificate set, *not* a cofinal theorem: the window quantifier stays (Section 17).

3.5 The cofinal state (r272/r273, typed measurements)

The r271 scaling alarm $\mathrm{sp}(N, \varepsilon) = +0.67$ is dissected exactly (r272): pair count $\alpha = +1.01$, amplitude $\beta = -0.81$, bound-side cancellation $\gamma = -0.01$ (exactly flat) — while the *truth* side runs opposite ($\gamma_{\mathrm{true}} = +0.453$; the truth margin *rises* with N, $+0.264 \to +0.414$): sealed adjudication BOUND_COARSENESS_CONFIRMED — the bound loses cancellation, not the world; H5 is contradicted by no measured truth quantity. The quantified flip condition: a *non-adjacent/global* mechanism (address c3) must supply $N^{-\delta'}$ with $\delta' > 0.21$ of the measured available 0.45 (TRUTH_ALLOWS). The r273 Euler surgery ladder shows the required cancellation is *generic* (no surgery collapses γ_{true}) while *every* surgery kills the free-prefix positivity at degrees 33–101: *the wall, not the cancellation rate, is what is arithmetically special about* MAIN.

Certification. Ledger: `PRIME.PORT.COUPLEDTAU.SURFACE_CLOSURE.01 [E]; PRIME.PORT.COUPLEDTAU.TERMINAL_CROSSRATIO.01 [O] (cofinal front)`. Module: `verification/v960_terminal_surface_closure.py`.
Probes (embedded byte-exact, sealed smoke stage; full 16-hex SPEC SHAs pinned in the module and in `rh/INVENTORY.json`): `terminal_crossratio` (31/31, `36eee7b4`), `terminal_triangle` (26/26, `db32d31f`), `cancellation_adjudication` (26/26, `151a9176`), `drive_local_asymptotics` (28/28, `0c4df408`), `phase_bulk_bound` (29/29, `f5578bb9`), `kz15_boss` (29/29, `8834a1c2`), `universal_pair_theorem` (28/28, `66f61c8a`), `l2_scaling_anatomy` (28/28, `3fb76b65`), `euler_mechanism` (24/24, `8cb1804e`), `kyp_memory` (24/24, `88ebdb00`); rounds r260–r275.

4 The orientation dictionary and the Maslov census (v961)

Theorem 4.1 (Wronskian dictionary; r274, v961 S0.1–S0.2)**.** *On every measured window the monic orthogonal chain $\hat{\pi}_n$ and its second-kind chain q_n satisfy the base Casoratian identity*

$$\hat{\pi}_{n+1}(z)\, q_n(z) - \hat{\pi}_n(z)\, q_{n+1}(z) = h_n \qquad (c' = 1,\ \textit{identically in } z),$$

the right solution is triple-constructed with exact coincidence (residue route; downward recursion with the exact Dirichlet boundary $q_S = 0$ — the node polynomial is the right boundary condition*; dual chain + L-gauge), the h-free midpoint form $\hat{\pi}_{n+1}\hat{\pi}^{\#}_{S-1-n} - \beta_{n+1}\hat{\pi}_n\hat{\pi}^{\#}_{S-2-n} = L(z)$ is the node polynomial (constant in n: the r231 sign-blindness as an identity — it provably carries no* *orientation), and the augmented Casoratian is a polynomial (border poles cancel exactly) with the telescope*

$$D_{n+1} = B - \frac{W_n^{\mathrm{aug}}}{W_n^{\mathrm{base}}}, \qquad W_n^{\mathrm{aug}} = h_n S_n,$$

exact against the independent r263 route. The honest c-typology: $c_n = 1/W_n^{\mathrm{base}}$ is source-pure and positive on the free prefix, but carries the entire *orientation — the dictionary relocates the orientation question, it does not solve it.*

Theorem 4.2 (Maslov census; r277)**.** *Rule R2 = the interlacing/reality structure of the Jacobi zeros passes the sealed blind examiner protocol: blind 42/42 rungs + all mains* SAFE *at full depth; the controls fire* exactly *at flip+1 (*26/22/28*); R2 is provably* SAFE *on every positive prefix ($\beta > 0 \Rightarrow$ symmetrizable $\Rightarrow$ real + Cauchy-strict), is* not *h-equivalent (*79 *pattern mismatches vs* 78 *h re-entry pivots) and never heals: a one-way break detector — the cofinally monotone object the proof plan needs.*

Remark 4.3 (The honest refutation; r277)*.* The *raw atom-Sturm census is not the right winding quantity*: on MAIN $c_n \equiv n$ breaks at $n = 56/48$ at throughout-positive h (zeros first escape the atom hull, then pair inside single atom gaps) — the classical separation theorem for positive measures fails genuinely for the signed comb. STURM_CHAIN_VERIFIED is honestly *not* awarded; the named consequence is the *contraposition* form: an independent index obstruction must forbid the R2 break before half-filling on MAIN — exactly there the `MainWindow` arithmetic must enter (the oriented midpoint theorem, Section 17).

Remark 4.4 (The metric firewall; r276/r278, typed measurements)*.* The r273 firewall map is refined to a graded *continuum* law ($D \sim \theta^b$, b 0.04–1.09; no jump; single operations typically survive to depth 0.88–1.00). The property ranking puts *support exactness* first (jitter at 2% of the local atom gap already costs 3/4 of the depth) and the Euler family structure last: the arithmetic firewall is more precisely a *metric* firewall. The exact position gradients (Hellmann–Feynman class, identity fine type: $d \log h_n / du_j = \langle \dot{w}_j, \hat{\pi}_n^2 \rangle / h_n$ — the polynomial variation drops by orthogonality) localize the lethal positions (u-profile PREDICTIVE, sp $= -0.82$; the top-3 atoms are the small primes $2, 3, 5$), but the local stability law is PERTURBATIVE_ONLY: θ^* sits 25–170× below the smallest measured dose (the 0.02 collapse is a nonlinear cascade across tent kinks; L_D window-specific, no uniform lemma candidate). The MAIN positivity itself stays the open center.

Certification. Ledger: `PRIME.PORT.RHP.MIDPOINT.ORIENTATION_DICTIONARY.01 [E]; PRIME.PORT.RHP.MIDPOINT.ORIENTED_THEOREM.01 [O]; PRIME.PORT.WALL.METRIC_FIREWALL.01 [O]`.
Module: `verification/v961_midpoint_orientation_dictionary.py`.
Probes (embedded byte-exact, sealed smoke stage; full 16-hex SPEC SHAs pinned in the module and in `rh/INVENTORY.json`): `wronskian_dictionary` (32/32, `56e8a03e`), `minimal_firewall` (25/25, `ed17d79f`), `maslov_census` (29/29, `3858fd16`), `metric_stability` (31/31 full record / 30 smoke gates, `7031200f`); rounds r274–r278.

5 The half-filling pinning theory (v962): four theorems, four refutations, and the north star in reinstform

Wave 5 (rounds r279–r281) freezes the pinning anatomy as a small mathematical theory — the reviewer demand "not just as experiments". All four theorems are machine-certified at the exact-rational level (module-own re-derivation in pure Fractions plus the embedded sealed probes).

Theorem 5.1 (Moment counting; r281, v962 T1)**.** *The pivot h_n consumes the moments $m_0..m_{2n}$. An S-atom signed measure has exactly S free moments $m_0..m_{S-1}$ (Vandermonde bijection);*

every m_k with $k \geq S$ is forced by the monic node-polynomial recurrence $\sum_{i=0}^{S} c_i m_{k-S+i} = 0$. Hence the free *pivots are exactly $h_0..h_{N_w-1}$ with $N_w = \lceil S/2 \rceil$, and h_{N_w} is the first forced pivot:* ***half-filling is the end of the free moment space****. (Gated as arithmetic for all $S = 2..2000$; freedom demonstration exact in rationals: $dm = e_{S-1}$ moves h_{N_w-1} alone and shifts the first forced moment by exactly $-c_{S-1}$; the wrong-L must-fail is loud. Lean:* `moment_counting_free_pivots`*, proved.) The reviewer question "why half-filling" is answered by counting.*

Theorem 5.2 (Crossing budget; r279, v962 T2)**.** *For a nondegenerate signed measure ($h_n \neq 0$ for $n < S$), Jacobi's minor-sign rule and the Sylvester congruence $G_S = VWV^{\mathsf{T}} \sim W = \mathrm{diag}(w_j)$ give*

$$\#\{n < S : \ h_n < 0\} \ = \ S_- \ = \ \#\{j : \ w_j < 0\}$$

exactly: the total Maslov crossing budget over the full algebraic continuation is the negative-atom count, world-blindly *(w9* 104*, w13* 98*, controls* 141/94/6*, kz15* 121*, kz52* 551*). Lean:* `crossing_budget` *(stated, v962 reference).*

Theorem 5.3 (Two-sided parity; r279, v962 T3)**.** *With $e_j = \mathrm{sign}(w_j)$ and $L'(x_j) = (-1)^{S-1-j}|L'(x_j)|$ on sorted nodes, the union polynomial $Q_n = \hat{\pi}_n \hat{\pi}^{\#}_{S-1-n}$ satisfies $\mathrm{sign}\, Q_n(x_j) = e_j (-1)^{S-1-j} \mathrm{sign}(h_n)$ at every node and degree; hence the union zero occupancy of every open atom gap is* odd *in every weight-agreement gap (never empty) and* even *in every disagreement gap — at every degree, h-blind. The census bilanz $c_n + c^{\#}_{S-1-n} = (S-1) - |D| + 2\,\mathrm{scD}(n)$ holds identically (exact via Fraction Sturm chains at every degree; proved by exhaustion over all* 87376 *sign-vector pairs $k \leq 8$).*

Theorem 5.4 (Main window reduction; v962 T4)**.** *Given Theorem 5.1 (the free window ends at N_w) and Theorem 5.2 (the budget is fixed at S_-), the entire open statement of the wall is the localization* $\min C \geq N_w$*, and this is exactly equivalent to*

$$\forall\, n < N_w : \ h_n > 0$$

(free-window quasi-definiteness; Lean: `main_window_reduction`*, proved). No size, cancellation or orientation question remains: one placement question.*

Remark 5.5 (The north star, reinstform)*.* The campaign's open center is, by Theorem 5.4, *why is the signed prime moment form quasi-definite up to the maximally free order?* In Lean this is the fog-free central `sorry free_window_positivity` (`rh/lean/RH/Window.lean`); via Theorem 5.4 it is the base half of the master theorem `augmented_prefix_positive` (`master_implies_free_window` is proved). Since wave 6 the open center has its *canonical form*, lemma L* (Section 6): carried on `PRIME.LSTAR.SUBORDINATION.01` **[O]** (the predecessor contract `PRIME.PORT.RHP.FULLSOURCE.QUASIDEFINITENESS.01` is consumed by `v963`).

Remark 5.6 (The four refutations; named negative gates)*.* (N1) NO_UNIVERSAL_O1_PINNING (r281, exact): the one-negative toy measure has $\min C = S-1$ and offset $N_w - 2$, *unbounded* in S — any $O(1)$ upper pinning must consume the comb structure; the only provable ceiling today is the pigeonhole $\min C \leq S - S_-$ from Theorem 5.2, and the measured pinning constant $C \leq 5$ is a 42-rung *measurement* (the Lean guard `upper_pinning_not_universal` carries an exact 4-atom instance with machine-checked Hankel minors). (N2) NO_EXTREMALITY (r280, measured + exact witness): three structured mp-confirmed gradient directions push the w9 first crossing $184 \to 185$ *past* half-filling; a 4-atom one-negative measure has $\min C = N_w + 1$ exactly — N_w was the free-window bound, never an absolute maximum. (N3) NO_GENERIC_MASLOV_OBSTRUCTION (r279, exact): three rational counterexamples refute the candidate obstruction "two-sided compatibility forbids a crossing before half-filling" — the two-sided machinery is world-blind classical structure and provably carries no arithmetic. (N4) NO_SIMPLE_OFFSET_LAW (r281, measured): six sealed source-pure offset candidates reach $\max |\mathrm{sp}| = 0.273 \ll 0.75$ over the 42-rung census, all world-blind.

Certification. Ledger: `PRIME.WALL.HALFFILLING_PINNING_THEORY.01 [E]`; `PRIME.PORT.REPRESENTATION.CONTEST.01 [O]`; `PRIME.PORT.RHP.FULLSOURCE.QUASIDEFINITENESS.01 [O]`.
Module: `verification/v962_halffilling_pinning_theory.py`.
Probes (embedded byte-exact, sealed smoke stage; full 16-hex SPEC SHAs pinned in the module and in `rh/INVENTORY.json`): `oriented_theorem` (32/32, `9107709b`), `budget_localization` (29/29, `7abf7a20`), `halffilling_pinning` (28/28, `a0081572`); rounds r279–r281.

6 The L* reduction (v963): the canonical form of the open center

Wave 6 (rounds r282–r285 plus the standalone problem document) freezes the L* arc. The r283 attack reduces the entire free-window question — and with Theorem 5.4 the entire open statement of the wall — to *one* two-measure subordination scalar; the r282 contest closes the classical representation exits with exact reasons; the r285 round makes the decomposition bookkeeping exact. All exact statements are machine-certified at the exact-rational level (module-own re-derivation plus the embedded sealed probes).

Theorem 6.1 (The mu-frame reduction; r283, v963 A1–A3)**.** *Split the signed defect measure $\tilde{\mu} = \mu - \nu$ into its positive channel μ (S_+ atoms) and negative channel ν (S_- atoms, weights $v_k > 0$). With U the unit lower-triangular monic μ-OP change of basis, $UH_k(\tilde{\mu})U^\mathsf{T} = D_\mu - G$ where $D_\mu = \operatorname{diag}(h_i^\mu)$ and $G_{ij} = \int \pi_i \pi_j \, d\nu$; leading principal minors are preserved, so*

$$\operatorname{minor}_k(D_\mu - G) \;=\; D_k(\tilde{\mu}) \qquad \textit{exactly, at every } k \le S_+.$$

Consequently, with E_n the ν-dressed μ-Christoffel–Darboux kernel ($E_n = B_n B_n^\mathsf{T}$, $B[k,i] = \sqrt{v_k}\,\varphi_i(y_k)$),

$$h_0, \ldots, h_{m-1} > 0 \iff \lambda_{\max}(E_m) < 1,$$

and by monotone rank-one loading $\lambda_{\max}$ crosses 1 exactly *at $\min C + 1$. The pigeonhole ceiling $\min C \le S_+$ is re-proved in the frame (null-polynomial witness $\int p^2 \, d\tilde{\mu} = -\int p^2 \, d\nu < 0$), and capacity-as-counting is refuted by the exact rank-one pair (same rank, same support, opposite window fate; the deciding values are metric, 1.25×10^{-13} vs 1.25×10^{2}).*

Theorem 6.2 (The canonical form: lemma L*)**.** *By Theorem 6.1 the free-window quasi-definiteness of Theorem 5.4 is exactly the statement*

$$\boxed{\int p^2 \, d\nu \;<\; \int p^2 \, d\mu \quad \textit{for every real polynomial } p \neq 0,\ \deg p < N_w}$$

— equivalently $\lambda_{\max}(E_{N_w}) < 1$: the ν-channel is strictly μ-subordinate on the free polynomial window. This is a two-measure moment problem instead of a determinant cascade; it is not proved *— it is the open center in canonical form, registered as `PRIME.LSTAR.SUBORDINATION.01` [O] and stated fully standalone (no project vocabulary) in `rh/problem/lstar_problem.tex` for external mathematicians. Lean: `lstar_subordination` (`rh/lean/RH/Window.lean`, documented `sorry`); the direction L* ⇒ free-window positivity is* proved *(`lstar_implies_free_window`).*

Theorem 6.3 (Decomposition bookkeeping; r285, v963 B)**.** *With $\operatorname{maxdiag}_n = \max_k v_k K_n(y_k)$ and $\operatorname{assist}_n = \lambda_{\max}(E_n)/\operatorname{maxdiag}_n - 1$: $\lambda_{\max} = \operatorname{maxdiag} \times (1 + \operatorname{assist})$ exactly, the Christoffel sandwich $\operatorname{maxdiag}_n \le \lambda_{\max}(E_n) \le \operatorname{trace}(E_n)$ holds with its exact-rational consequences at every degree, and the budget equivalence $(1/\operatorname{maxdiag} - 1) - \operatorname{assist} > 0 \iff \lambda_{\max} < 1$ is sign-exact. L* thus splits into (D) the per-atom diagonal condition $v_k K_{N_w}(y_k) < 1$ and (C) the coherence condition — exact bookkeeping, no give-away bound.*

Remark 6.4 (The four-language elimination; r282, exact negative gates)**.** The contest "what *is h_n*?" is closed in all four classical languages, each with the exact reason (verdict CONTEST_ALL_DEAD):

Language	Verdict	Exact reason
K1 RHP / monodromy norm	RESTATEMENT	the SOS/$\lvert m\rvert^2$ structure exists iff the negative register is empty ($N_n \equiv 0$); the surviving clause $N < P$ is $h > 0$ verbatim (fake SOS misses h_2 by exactly $N_2 = 48360721965/70120631072$)
K2 Hodge / Kasteleyn orientation	NOT CONSTRUCTIBLE	full 2^S exhaustion: the coherent class is exactly $\{\pm \operatorname{sign} w\}$; value preservation fails by exactly 2×negative mass — an orientation exists iff $S_- = 0$
K3 de Branges / canonical system	RESTATEMENT	Hamiltonian PSD $\iff$ $\beta > 0 \iff h > 0$: exact bookkeeping, no second gestalt (the only independent shadow lags +1 everywhere)
K4 operator / dual pair	WORLD-BLIND BY THEOREM	$h_n h^{\#}_{S-1-n} = 1$ exactly at every degree: the dual pair is sign-synchronized identically, no second condition

The common deep reason (the program finding): *every classical positivity language forces positivity exactly for the positive measure class* — MAIN is signed, and its free-window positivity is a genuine arithmetic statement without a known structural carrier.

Remark 6.5 (Measured anatomy and the honest margin decay; typed, not laws). (i) *One atom vs collective* (r284): on w9 the pure single-atom Christoffel bound crosses at $n_{\text{DIAG}} = 187$, only two degrees above the true crossing $185 = \min C + 1$, while all four controls die collectively far below their single-atom degree; the near-breaking direction is 99.7% two shallow-u ARCH background atoms below the first prime (PR 1.89 vs control 4.97..9.72; SMALLP__DEPLETED, ONE__BAND); the shielding geometry is world-blind and the Nyquist/resolution formula fails on all four controls by 1.5–3 octaves. (ii) *Where the arithmetic sits* (r285): (D) separates at window scale (controls' $n_{\text{DIAG}} = 50..91 \ll 184$); the binding edge atom's kernel growth is *sub-classical* ($p = 0.38$ vs bulk law 1, classical coverage 0/42 — any (D) proof must be a discrete bound); MAIN's wall assist is the extreme low outlier of both conservation-gated replicate ensembles (pct 0.00); and the round delivers the first two positively MAIN-separating detectors of the program (assist-at-crossing 0.0195 vs dead 1.69..2.99; random-sign $z = -3.15$ destructive vs dead $+4.95.. + 12.44$ constructive). (iii) *The margin decay* (stated neutrally): all 42 ladder margins $1 - \lambda_{\max}(E_{N_w})$ are strictly positive, but the family minimum is 1.4175×10^{-7} (anchor $z = 233$, $S = 1717$) while the flagship $z = 16$ carries the *largest* margin 1.68×10^{-4} — a decay of ∼3 orders over the tested range, driven by the maxdiag → 1 trend of the per-atom budget. At this cut the unbounded-family version of L* was genuinely uncertain in both directions; the r286 margin-scaling round has since landed and is consumed in wave 7 (Section 7): resolved *harmless* — no counterexample on the extended ladder, the $O(1)$ census offset survives.

Certification. Ledger: `PRIME.LSTAR.REDUCTION_DICTIONARY.01 [E]; PRIME.LSTAR.SUBORDINATION.01 [O]`.
Module: `verification/v963_lstar_reduction_dictionary.py`.
Probes (embedded byte-exact, sealed smoke stage; full 16-hex SPEC SHAs pinned in the module and in `rh/INVENTORY.json`): `representation_contest` (30/30, `0f9954b8`), `fullsource_quasidefiniteness` (26/26, `4cf5ea53`), `lstar_two_measure` (30/30, `687112a8`), `christoffel_decomposition` (33/33, `9781e6d6`); rounds r282–r285; standalone document `rh/problem/lstar_problem.tex` + `verify_lstar_instance.py` (12/12).

7 The L* coherence census (v964): the margin law, the van der Corput theorem, and the metric-only adjudication

Wave 7 (rounds r286–r289) freezes the coherence arc. The r286 round resolves the margin-decay warning of Remark 6.5(iii); the r287 round gives the L2 lemma its first *named classical theorem candidate*; the r288 round maps the carrier of the destructive coherence; and the r289 round adjudicates the diophantine question — *against* number theory.

Theorem 7.1 (The van der Corput inequality; r287, v964 S0, module-own exact)**.** *For any real sequence $P_1, \dots, P_m$ and any window $H \geq 1$, with $A(h) = \sum_j P_j P_{j+h}$,*

$$\Big|\sum_{j=1}^{m} P_j\Big|^2 \;\leq\; \frac{m+H-1}{H} \sum_{|h|<H} \Big(1 - \frac{|h|}{H}\Big) A(h),$$

an exact, constant-free, arithmetic-free finite theorem (proof route gated in v964: the Fejér window-sum identity $\sum_{|h|<H}(1-|h|/H)A(h) = \frac{1}{H}\sum_t W_t^2$ with W_t the sliding H-window sums, the H-fold covering identity $\sum_t W_t = H \sum_j P_j$ and Cauchy–Schwarz over the $m+H-1$ window positions; the wrong prefactor m/H breaks exactly on $P = (1,1,1,1)$, $H = 2$: $14 < 16$). Measured on the 42-rung ladder at the frozen window $H = \lceil \sqrt{m} \rceil$: $\delta' = +0.309 > 0.21$ world-blind *(admissible: L2 is the generic half), certification* 6/7 *exceptions* +38/42 *rungs; only the kz15 razor blocks* 7/7 *(miss* 0.10 *dec) and kz15 is exact-finite certified per r270. The F1 discrepancy/Erdős–Turán frame dies loudly ($\delta' = -0.21$, 0/7: both factors grow). Registered as* `PRIME.PORT.L2.VDC_LEMMA.01` ***[O]****: the remaining input is the* chain origin *of the P-variance scaling — per the r287 autocorrelation map the cancellation is already absorbed into the P_j magnitudes (on* 39/44 *worlds the block sequence* reinforces *at root scale), so the frames win through the $\sqrt{m}$ count economy, not lag structure.*

Remark 7.2 (The margin law resolved harmless; r286, typed census)*.* The ladder is extended by 15 new anchors ($N_w = 942..1218$, 40% beyond the old family cap): *all* margins are mp-sign-safe positive (minimum $+1.806 \times 10^{-8}$ at $z = 529$; every anchor mp dps-30/45 staggered; builder cross-check $\leq 2.9 \times 10^{-15}$) — no counterexample; the $O(1)$ census offset survives beyond the cap (off distribution $\{0{:}1, 1{:}10, 2{:}2, 3{:}1, 4{:}1\}$, max +4). The decay law is a *flattening* power law ($\alpha \approx 3.05$ vs S; halves 3.67/2.07, curvature −1.60) driven by the cancellation sharpening $c_w \to 1$, and the HARMLESS reading is quantified: the margin falls *slower* than the local eigenvalue loading speed (slope $-3.33 \geq -4.44 - 1.0$) — the decay is the kinematic signature of the fixed $O(1)$ census offset at growing eigenvalue density, not an approach to a crossing. The terminal ratio q_N is reconciled (r258 FLAT reproduced: two different objects), and the frozen 42-rung fit *predicts* all 15 extension margins inside the sealed 1-decade band (EXTRAP-CALIBRATED 15/15).

Remark 7.3 (The carrier map; r288, typed measurements)*.* The destructive coherence of the L* coordinate has a named carrier class: the neg fraction 0.503 of the E-kernel sign field is globally coin-like but sharply band-structured — fold distance 1–2 is 87% positive (sampled in phase), distance 3–4 is 80% negative (antiphase); the wall destructivity is carried by the *antiphase next-nearest ARCH–ARCH pairs* ($X_v = -0.0517 = +0.222 - 0.273$, $z_v = -3.149$ exactly reproduced, $C_{\text{off}} = -0.1046$), with *total* control reversal (all four controls constructive at their own crossing, $C_{\text{off}} = +0.30..+1.00$). The coherence lives in *finest alignments* far below the median phase resolution: the chain zeros co-move with the atoms (turn rate 0.24 per unit dose = the static prediction 0.23) yet z_v flips already at dose 0.005. Honest negatives: SAMPLING_BLIND (the sinc/CD predictor is only partial, $0.728/0.878 < 0.90$), SOURCE_SEPARATOR_NOT_FOUND (plain phase dispersion world-blind and ensemble-generic), DIFFERENT_OBJECTS (the r287 magnitude absorption and the kernel destructivity are two coordinates).

Remark 7.4 (The metric-only adjudication; r289, the central refutation)*.* The diophantine question — does the wall consume the number-theoretic exactness of the $\log n$ positions? — is adjudicated by the sealed three-way rule (rational keeps + shuffle loses + ladder does not keep at θ_{eff}), with the twin table as the executioner:

Twin	**depth**	**crossing**	z_v	**band 3–4**	**verdict**
MAIN	1.000	185	−3.149	−0.105	—
rational (dens ≤ 56801, $\lvert du\rvert \leq 2.1 \times 10^{-9}$)	1.000	185	−3.149	−0.105	**keeps, identic**
shuffle r0/r1/r2 (θ_{eff} 0.18/0.12/0.12)	0.201/0.212/0.239	38/40/45	all > 0	> 0	loses, metric-expla

The rational twin replaces every tent center $\log n$ by a rational $p/q \times \Delta$ (CF convergents; weights and cells bitwise preserved), destroying *every* exact log-linear-form relation — and keeps the full r288 signature *identical* (mp dps-60 ward $\rho_{184} = 0.99983248 < 1 < 1.00003660 = \rho_{185}$ confirmed at the 1.7×10^{-4} margin): the effect of the diophantine structure is *null.* The jitter ladder places the metric coherence threshold between 10^{-3} and 3×10^{-3} of the local gap; the shuffle's $\theta_{\mathrm{eff}} = 0.12$ is ∼40–120x beyond it, so its loss is fully metric-explained. The only sub-gap entry point is exactly identified: the tent-split fractions $\{\log n/\Delta\}$ (completeness identity 4.2×10^{-14}; with $\Delta = \log 2/46$ exactly, the fractions are $\{46 \log_2 n\}$ and the eight on-node 2-power atoms give exactly $\binom{8}{2} = 28$ exact resonances — module-own exact in v964). The Baker need is documented, never applied: the flip dose needs fraction structure at exponent ~ 8.9 while Baker/Matveev-class two-log bounds deliver $\sim 7 \times 10^4$ — ∼7900x too weak *and* unnecessary per the verdict. *Verdict* METRIC_ONLY: the firewall reads the fraction *profile* metrically, not its arithmetic — every profile of 10^{-3}-gap precision carries (the rational twin proves it); the why-MAIN search continues in profile space, and the precise open front is: *which functional of the fraction profile forces the destructive coherence?* (The r290 profile-functional round was in flight at the wave-7 cut; rounds r290–r295 are consumed in wave 8, Section 8.)

Certification. Ledger: `PRIME.LSTAR.COHERENCE_CENSUS.01 [E]; PRIME.PORT.L2.VDC_LEMMA.01 [O]`.
Module: `verification/v964_lstar_coherence_census.py`.
Probes (embedded byte-exact, sealed smoke stage; full 16-hex SPEC SHAs pinned in the module and in `rh/INVENTORY.json`): `lstar_margin_scaling` (29/29, `0a44ac4e`), `l2_deterministic_cancellation` (25/25, `761d88fa`), `destructive_coherence` (31/31, `da46f7ee`), `arch_kernel_diophantine` (42/42 record, `91cdc2b1`); rounds r286–r289; the standalone document `rh/problem/lstar_problem.tex` carries the updated "what is known" state.

8 The curvature arc (v965): the geometry of the working set and the functional kill catalog

Wave 8 (rounds r290–r295) freezes the curvature arc — the answer to the wave-7 question "*which functional of the fraction profile forces the destructive coherence?*" in its honest form: the *geometry* of the working set is now quantitatively measured, and the *closed-functional* search is sealed negative over five class families. Nothing here is a theorem about L*; the arc is a measured census plus a catalog of sealed rejections, and the rejections are the load-bearing content.

Remark 8.1 (The basin geometry; r290, typed measurements). In the exact θ_{eq} LAG coordinate (analytic reference REF $= \frac{1}{2}\sum_j m_j g_j/\Delta = 125.75$; inversion identity gated at 10^{-12}), the working set around MAIN is a *soft-shouldered, strongly anisotropic tube*: the kill fraction over 16 conservation-gated random directions is 0.38/0.62/1.00 at distances $5 \times 10^{-4}/10^{-3}/2 \times 10^{-3}$ (NEAR radius $\sim 5 \times 10^{-4}..2 \times 10^{-3}$ gap-equivalent, consistent with the r285 ensemble percentile 0.00); every onset is a soft shoulder, no cliff. The *world-directed* axes kill 5–50x earlier than random (onsets $4 \times 10^{-5}..2.3 \times 10^{-4}$ against the r289 jitter threshold $10^{-3}..3 \times 10^{-3}$): the rim is highly direction-structured. The SMOOTH (arithmetic) axis is a privileged killer (onset 1.36×10^{-4}) yet *orthogonal* to the first-order wall gradient ($\cos = -3 \times 10^{-5}$; the midpoint is already control-like, $\min C = 29$): its lethality is collective, not gradient-linear. The r280 ridge is *real*: $\min C = 185$ — one degree above MAIN — at extension factors 1..8 (θ_{eq} $1.5 \times 10^{-4}..1.2 \times 10^{-3}$, inside a tube where half the random directions already kill), first death at factor 16.

Remark 8.2 (The budget mechanism of the lift; r291/r293, typed). The ridge lift is a *first-order budget phenomenon*, not a distinguished carrier: a conservation-gated sub-direction lifts iff its side-selected linearized wall budget $-\sum_{j \in A} c_j \times$dose crosses one sharp threshold in $(1.280, 1.291]$ — perfect separation over all 18 matched-dose cases (PRIME/HEAD/XIPOS lift, POWER/MID/TAIL/XINEG do not; $k_{\min} = 9$, top atoms $2, 3, 5, 13, 11, 4, 29, 7, 89$: small-prime head-heavy, not one-atom), with exactly one overdrive retraction (TOP6 at factor 8). The budget itself is an exactly additive linear functional (the partition telescope is gated module-own in exact Fractions, with the dropped-atom mutant loud). There is *no fixed point*: the step-controlled iteration saturates on the one-degree plateau $\min C = 185$ (never 186) while the axis decoheres; and the lift is MAIN-specific (EPSTEIN is first-order flat, $\theta_{\text{up}} = 5.03 \times 10^{10}$, and never lifts). The r293 flip anatomy closes the mechanism at the singularity: all 8 selected flips are *simple zeros* of h_{184} ($\alpha = 1.000 \pm 0.003$ both sides; the doubling law $|h(t_0+2d)|/|h(t_0+d)| = 2^{\alpha}$ is gated module-own exactly, with the double-zero mutant giving exactly 4), and the TOP6 retraction *is* a second simple h_{184} zero at $f_{\text{ret}} = 7.107$ (lift window 1.226..7.107) — on the side the monotone budget prognosis is structurally blind to. But MSTAR_NO_LAW: the per-direction thresholds $m^*_{\text{dir}} = b \times f^*$ scatter 1.1295..1.3931, straddling the r291 bracket (a fixed-dose artifact); no fixed safety distance to the singularity exists in budget, θ_{eq}, L2 or curvature units (spreads 3.0/3.0/3.8/10.2 vs bar 1.25).

Remark 8.3 (The curvature spectroscopy; r292, typed). The wall Hesse form at MAIN over the sealed 29-direction set (diagonals at three dyadic steps plus the full 406-pair polarization matrix $H(u, v) = [d^2(u+v) - d^2(u-v)]/4$, whose exact skeleton is gated module-own in Fractions with the wrong-prefactor mutant loud):

Finding	**Value**	**Reading**
all 29 diagonals	negative	a genuine quadratic valley
top eigendirection	$92.5\,\%$ of $\sum |\lambda|$	HESSIAN_LOW_RANK(1)
λ_{top} (per unit L2)	-0.418	the valley scale
top axis	DENS05 +1.00 / DENS04 −0.71 / DENS03 +0.69	a pure on-support DENS combination
SMOOTH alignment	$|\cos| = 0.07 < 0.7$	*not* the arithmetic axis
ridge position	L2-diagonal rank 28/29	RIDGE_CURV_FLAT (the r291 expectation
EPSTEIN contrast	curvature ratio 5.4×10^{-15}	second-order structureless too
m^* through jets	q_R median -0.911 (assist); $B_2 = 27.6$, $B_3 = 150.9$	THRESHOLD_NOT_EXPLAINED + RETRAC

The curvature valley of the wall is real, rank-1 dominated, and its axis is a DENS combination — neither the SMOOTH axis nor the ridge; the m^* threshold and the retraction are near-flip nonlinearities invisible to jets up to third order.

Remark 8.4 (The functional kill catalog; r290–r295, the sealed negatives). Five class families of closed profile functionals were sealed before evaluation and all fail the honest bar:

Family	**Round**	**Best** $|\text{sp}|$	**Bar / verdict**
1: local profile scalars	r290	0.263	0.5; ALL_FUNCTIONALS_BLIND
2: non-local / low-rank	r291	0.471	size baseline 0.881; NONLOCAL_BLIND
3: curvature two-form (mixed metric)	r292	0.884	baseline 0.907; CURVATURE_BLIND (gap 0.023)
4: metric-reconciled two-form F10	r293/r294	home win +0.024	partial 2/5 < 4/5; F10_FRAGILE
5: hardened partial-free $|\text{sp}|$ form	r295	wins 14/20	HARDENED bar fails 2 clauses; F10_SP_MAJ

The r294/r295 decision bars are themselves gated module-own as exact decision logic (every clause has tipping mutants in both directions). After five families, *no closed predictive profile functional of the working set exists over the honest bar*; the question is re-registered as `PRIME.LSTAR.CLOSED_FUNCTIONAL.01` [O] with the named open fronts (loss-corpus forensics, the rank-2 DENS core, why L2).

Remark 8.5 (The honest F10 story — a case study in sealing discipline)*.* The F10 arc is the program's cleanest demonstration of why bars are sealed *before* evaluation. (i) r292: F10 loses the mixed-metric contest by 0.023 — the near miss flags the metric mix. (ii) r293: with three metrics sealed *first*, F10 beats the baseline of its own metric ($+0.024 = 0.884 - 0.860$, and the mixed margin $-0.023 = 0.884 - 0.907$ is re-derived from the same pins: MIX_IS_CAUSE) and carries partial information (+0.423) — a promotion candidate is flagged, nothing promoted. (iii) r294: on five fresh corpora the $|\mathrm{sp}|$ win replicates 5/5 but the partial channel does not (median $0.299 < 0.3$, 2/5): F10_FRAGILE, promotion *rejected.* (iv) r295: on twenty fresh corpora even the partial-free form wins only 14/20 (median +0.028, min −0.079): F10_SP_MAJORITY — the r294 5/5 was itself top-of-distribution, exactly as the r293 partial was (R293_LUCK: the composition-matched subsample explains part, +0.067 gain, and misses the level clause by 0.004). A documented regularity (19/25 combined census), not a theorem — and the wave freezes exactly this: the regularity *with* its rejections. F10 is not promoted.

Certification. Ledger: `PRIME.LSTAR.CURVATURE_ARC.01 [E]`; `PRIME.LSTAR.CLOSED_FUNCTIONAL.01 [O]`.
Module: `verification/v965_lstar_curvature_arc.py`.
Probes (embedded byte-exact, sealed smoke stage; full 16-hex SPEC SHAs pinned in the module and in `rh/INVENTORY.json`): `profile_functional` (32/32, `f953dd71`), `ridge_anatomy` (30/30, `bb512c17`), `curvature_form` (36/36, `050821ff`), `metric_reconciliation` (43/43, `33c44cc6`), `f10_stability` (42/42, `88c6fd1e`), `f10_sp_hardening` (44/44, `4d7d8095`); rounds r290–r295; F10 stays UNPROMOTED per the sealed r294/r295 bars; no round in flight at this cut.

9 The L2 reduction chain (v966): the DENS fork closed, the generic-half target reduced to one inequality

Wave 9 (rounds r296–r300) freezes two things. First, the reviewer's hard identity fork on the r292 curvature axis is *closed honestly*: r296 adjudicated DENS_WORLD_BLIND under sealed rules — the coupling number $\cos(e_{\mathrm{top}}, \nabla\lambda_{\max}) = +0.394$ sits below the sealed 0.40 bar (noise max 0.331; the miss is 0.006, reported as such), the λ_3 control couples *harder* (−0.574: what coupling exists is a near-crossing eigenvalue-*band* property, not top-specific), every arithmetic candidate of the hash-sealed library stays ≤ 0.38, and the only above-chance structure (the 4-dim moment-gradient subspace, 0.825) holds on both dead controls (0.571/0.603) — construction-adjacent low-moment smoothness, so the sealed world-downgrade fires. The DENS-identity lane is closed and, per the pre-adjudicated fork, the resources moved to the L2 front. Second — the load-bearing content of the wave — the four-round chain r297→r300 *reduces* the $\delta' > 0.21$ target of the generic half (the vdC L2 lemma, Section 7) to **one remaining inequality**:

$$\underbrace{\text{NEFF\_TARGET}}_{\substack{\text{open at the wave-9 cut}\\ \text{(wave 10: retyped, v967)}}} \Longrightarrow \underbrace{\mathrm{sl}_D \leq \sigma^*}_{\text{exact algebra (r300)}} \Longrightarrow \underbrace{\sigma \leq \sigma^*}_{\text{ratio structural (r300)}} \Longrightarrow \underbrace{\text{r297 target}}_{\text{margin 0.198}} \Longrightarrow \underbrace{\text{vdC (v964-S0)}}_{\text{proved}} \Longrightarrow \delta$$

Every link is exact or structurally settled except the first; exceptions stay finitely certified (6 via the r287 F2 certificates, kz15 exact-finite via r270). **No RH claim**; nothing here is a cofinal theorem — every slope is measured on the 42-rung ladder.

Theorem 9.1 (The target equivalence; r297, exact composition)**.** *Let* $\mathrm{sl}_{c2} = \mathrm{slope}(\varepsilon_{c2}/M)$ *and* $\mathrm{sl}_{\mathrm{pref}} = \mathrm{slope}((m{+}H{-}1)/H)$ *be the ladder slopes of the r287 vdC bound and* $\sigma = \mathrm{slope}(S_F/M_W^2)$

the input-scaling slope. By the exact additivity $\mathrm{slope}(\varepsilon_{F2}/M) = \frac{1}{2}[\mathrm{sl}_{\mathrm{pref}} + \sigma]$*, the statement* $\delta'(F2) > 0.21$ *is* equivalent *to the target inequality*

$$\sigma \;\le\; \sigma^* \;:=\; 2\,(\mathrm{sl}_{c2} - 0.21) - \mathrm{sl}_{\mathrm{pref}} \;=\; -0.516,$$

and σ^* *composes back to* $\delta' = 0.21$ exactly *(gated module-own in Fractions for arbitrary rational slope pairs; the pad-dropped mutant composes to exactly* 0*). Measured:* $\sigma = -0.714$ *— the truth clears the target with* 0.198 *sigma margin (*0.099 *in* δ'*):* what is missing is not room, it is provenance.

Theorem 9.2 (The sign-preserving transfer; r298, exact identity)**.** *Let* β *(border) and* ω *(window) be the two sealed measures,* $\Delta := \beta \ominus \omega$ *their exact signed difference, and* $B(u,v) = \frac{1}{H}\sum_t W_t(u)W_t(v)$ *the frozen positional Fejér block form. Then* $S_F = B(\beta,\beta)$ *and*

$$S_F \;=\; B(\omega,\omega) \;+\; B(\Delta,\,\omega+\beta)$$

identically *(bilinear difference-of-squares; gated module-own in exact Fractions on the r298 hand toy* $3 = 1+2$ *and a second rational witness at* $H = 2,3$*; on the real chain the identity holds to* 8.8×10^{-16} *of scale on 47 worlds). Nothing passes through magnitudes: the* $|\Delta|$ *mutant breaks the identity loudly (exactly* 4 *on the toy,* ∼*1.7 of scale on the real chain). Measured adjudication* TRANSFER_DOMINANT*: the window main term is* empty *(med* -3.94 *dec, slope* -1.386*), the cross term negligible (med* -1.4×10^{-4}*), so* $S_F \approx B(P\Delta, P\Delta)$ *—* the vdC input is the Fejér energy of the difference measure itself*, and* σ *is its decay exponent. r299 adds the structural find: the two unions share their full support on 42/42 rungs (*$\Delta_{\mathrm{fresh}} \equiv 0$*) —* Δ *is a pure c-value difference measure on one shared node set, with no pointwise c-convergence (cconv* 0.86*, slope* $+0.045$*): the decay is aggregate cancellation.*

Theorem 9.3 (The participation identity; r300, exact algebra)**.** *For the one positive vector* $|P\Delta|$ *define* $D = \sum_j P\Delta_j^2$*,* $L_1 = \sum_j |P\Delta_j|$*,* $m_x = \max_j |P\Delta_j|$*,* $n_{\mathrm{eff}} = L_1^2/D$ *and* $\mathrm{fill} = D/(m_x L_1)$*. Then* $D\cdot n_{\mathrm{eff}} = L_1^2$ *and* $D = m_x L_1 \cdot \mathrm{fill}$ *identically, hence at the ladder-ratio level*

$$\mathrm{sl}_D \;=\; 2\,\mathrm{sl}_{L1} - \mathrm{sl}_{\mathrm{neff}} \;=\; \mathrm{sl}_{mx} + \mathrm{sl}_{L1} + \mathrm{sl}_{\mathrm{fill}}$$

exactly (gated module-own in Fractions with ratio multiplicativity; the signed-sum mutant breaks by exactly 8/3 *on* $(1,-1,1)$*). On the record:* $\mathrm{sl}_D = 2(+0.196) - (+0.963) = -0.571 = -0.542 + 0.196 - 0.225$ *(devs* $6.7/4.4\times10^{-16}$*) —* the entire diagonal decay is participation growth*, and the r299* DIAG_TARGET *is an exact reparametrization of* NEFF_TARGET*:* $\mathrm{slope}(n_{\mathrm{eff}}) \ge 2\,\mathrm{sl}_{L1} - \sigma^* = +0.908$*, measured* $+0.963$*, margin* 0.055 *exact in* both *coordinates (machine-checked).*

Theorem 9.4 (The kernel envelope; r300, exact inequality)**.** *The Fejér kernel satisfies* $F_H(\theta) \le \min(H,\, 1/(H\sin^2(\theta/2)))$ *for all* θ *(pure algebra:* $\sin^2(H\theta/2) \le 1$*), with exact equality witnesses on both branches (as a rational function of* $s = \sin^2(\theta/2)$*:* $H = 2$ *at* $s = 0$ *and* $s = \frac{1}{2}$*;* $H = 3$ *at* $s = 0$ *and* $s = \frac{1}{4}$*; the halved-envelope mutant breaks at the witness). Consequence (measured):* $B/D \le R_{\mathrm{env}}$ *with* R_{env} *med* 1.61 *and* falling *(*$-0.122 \le +0.05$*) on 47/47 — adjudicated* RATIO_BOUNDED_STRUCTURAL*: the ratio half of the r299 rest pair is settled at the structural level (honesty clause:* R_{env} *still consumes the measured spectrum shape; the lobe-width heuristic is refuted,* $\mathrm{sp}(B/D, q_{50}) = -0.47$*).*

Remark 9.5 (The magnitude no-go catalog; r297/r299/r300, sealed negatives)*.* Every magnitude route to the input scaling was closed with a named break locus — the signed structure carrying $\sigma = -0.71$ is what $|\cdot|$-chains cannot see:

Route	Round	Composed vs $\sigma^* = -0.516$	Break locus
B1 pair-identity (gapmax $\times\, \varepsilon_{c2}$)	r297	δ' -0.112	max pair gap falls -0.535, need ≤ -1
B2 node-density ($2\,\mathrm{maxM} \times \mathrm{sumM}$)	r297	δ' -0.097	mass imbalance grows $+0.244$
B3 Fejér/Parseval (per-run CS)	r297	δ' -0.097	measure transfer + signed-to-abs gap
ET/Abel ($D_{\mathrm{pos}} \times V$, frequency level)	r299	$+1.948 = 1.504 + 0.444$ exact	position/kernel factor *grows*
chain-norm CS on $\lvert dw\rvert$	r300	-0.384 (miss 0.133)	the $\lvert dw\rvert$ identity census *breaks* (cross
max $\times L_1$ factorization	r300	-0.346 (miss 0.170 exact)	the fill decay -0.225 is invisible

The r300 anatomy localizes *why*: the max is fully atom-controlled (m_x/maxatom med 1.07), so the irreducible loss of every max $\times$mass bound is the profile's internal evenness loss (fill, -0.225); and the proven window sum rule does *not* attach to the difference measure (the $\lvert dw\rvert$ census breaks) — the chain-norm shortcut is closed honestly.

Remark 9.6 (Two world-separating classes; r299/r300, disclosed)**.** The lane now carries two disclosed WORLD_SENSITIVE classes, the first world-separating signals of the L2 side: the *O-sign class* (r299: MAIN's $P\Delta$ pair field *reinforces*, $O < 0$ on 13/42 only — O_POS — while both broken-arithmetic controls cancel, O_NEG) and the *fill class* (r300: MAIN is FILL_LOW, both dead controls FILL_HIGH). Both are typed disclosed findings on control rungs, not gates, and neither is used as a proof premise.

Open problem 9.7 (The NEFF target; the one remaining inequality)**.** Prove $\mathrm{slope}(n_{\mathrm{eff}}) \geq 2\,\mathrm{sl}_{L1} - \sigma^* = +0.908$ for the participation number $n_{\mathrm{eff}} = L_1^2/D$ of the aggregated difference profile $P\Delta$ (measured $+0.963$, margin 0.055, thin) — source-pure and window-independent. By Theorem 9.3 this is exactly the diagonal decay $\mathrm{sl}_D \leq \sigma^*$, and through the chain above it closes $\delta' > 0.21$ on the generic half. The named bridge to provable terrain is *correlational, not causal*: the carrier count grows where the boundary-position discrepancy falls ($D_{\mathrm{rank}}(\Delta)$ slope -0.117, $\mathrm{sp}(D_{\mathrm{rank}}, n_{\mathrm{eff}}) = -0.81$) — a proof must turn this correlation into a mechanism. Ledger `PRIME.L2.NEFF_TARGET.01` [O]; the executing round r301 was in flight at the wave-9 cut and *not* consumed there. In Lean no new statement is added: NEFF_TARGET is a measured ladder-slope aggregate (a halves log-slope estimator over 42 rungs with a threshold derived from measured slopes), not an exact finite identity or a matrix theorem, and a universally quantified ladder form would be refutable exactly like the pre-r273 universal edge forms (the machine-checked guard pattern of `RH/Counterexamples.lean`); the hole stays `lstar_subordination`. *Wave-10 resolution (Section 10): rounds r301–r304 executed and consumed in* `v967` *— the r303 regress audit retyped this inequality: NEFF/UNIF/ATOM are one algebraic slack $S = \sigma^* - \mathrm{sl}_D = +0.0547$ (Theorem 10.2), and the contract stands retyped to the documented stop state (Remark 10.6).*

Certification. Ledger: `PRIME.L2.REDUCTION_CHAIN.01 [E]; PRIME.L2.NEFF_TARGET.01 [O]`. Module: `verification/v966_l2_reduction_chain.py`.
Probes (embedded byte-exact, sealed smoke stage; full 16-hex SPEC SHAs pinned in the module and in `rh/INVENTORY.json`): `dens_identity` (39/39, `ffb413c8`), `vdc_chain_provenance` (27/27, `e42a76eb`), `window_border_transfer` (28/28, `05e831be`), `fejer_decay` (32/32, `f432e944`), `diag_target` (31/31, `55218b5d`); rounds r296–r300; the DENS lane closed honestly (DENS_WORLD_BLIND); the chain reduced to NEFF_TARGET; r301 in flight at that cut (consumed in wave 10, `v967`, Section 10).

10 The cascade closure (v967): the regress audit, the retyping and the documented lane stop

Wave 10 (rounds r301–r304) is the honest end of the reduction cascade — *the rejections and retypings are the content.* Rounds r301/r302 executed the wave-9 rest targets to their end: r301 proved the count/uniformity identity $n_{\mathrm{eff}} = n_{\mathrm{act}}/(1 + CV^2)$ exactly with the *perfect* count link $n_{\text{act}} = m$ on 42/42 (the effective-carrier count *is* the constructive level-2 block count; the whole Rényi order family grows at one exponent — echt anti-concentration; the margin is jackknife-stable) and relocated the rest onto UNIF_TARGET; r302 found the normalized block profile PROFILE_STATIONARY (KS $0.043 \ll 0.125$ over a $3\times$ depth range) with an exact $1/N$ transient onto $m_2^\infty = 1.973$ (held-out dev 0.002; the r299 pointwise-cconv negative was the wrong convergence notion), proved the coherence identity below, and fired UNIF_DERIVED(B1+A2) — the first DERIVED verdict of the lane, every hypothesis typed MEASURED — relocating the rest onto ATOM_TARGET. Then the reviewer set the regress audit, and r303/r304 closed the lane. **No RH claim**; nothing here is a cofinal theorem.

Theorem 10.1 (The coherence identity; r302, exact algebra)**.** *On the shared bulk support with atom c-differences dc, block sums P_j, atom energy $Q = \sum dc^2$, atom mass $L_1^a = \sum |dc|$, block energy $D = \sum P_j^2$ and block mass $L_1 = \sum |P_j|$, let $\chi = D/Q$ (the in-block coherence factor),* $\text{surv} = L_1/L_1^a$ *and $n_{\text{eff}}^{\text{atom}} = (L_1^a)^2/Q$. Then*

$$1 + CV^2 \;=\; \frac{n_{\text{act}}\,\chi}{\text{surv}^2\, n_{\text{eff}}^{\text{atom}}} \qquad \textit{and} \qquad \chi \;=\; 1 + \frac{2}{Q}\sum_{k\geq 1} T_k$$

identically, where T_k is the within-block lag-k cross sum (gated module-own in exact Fractions on two rational toys, $\chi = 5/7$ and $5/19$; on the real chain the wards hold at 6.0×10^{-16} and 4.7×10^{-16} on 47 worlds; the lag-0 double-count, swapped-D/Q and factor-1 mutants break by exactly $5/7$, $48/35$ and $1/7$). Measured: $\chi = 0.63$ destructive *and falling — the lag-1 anti-correlation $\rho_1 = -0.22$ of the dc profile is its atom-level mechanism.*

Theorem 10.2 (The margin invariance; r303, exact algebra — the regress audit)**.** *Let* NEED $= 2\,\text{sl}_{L1} - \sigma^*$, UNIF_NEED $= \text{sl}_{\text{nact}} -$ NEED *and* ATOM_NEED $= \text{sl}_{\text{nact}} + \text{sl}_\chi - 2\,\text{sl}_{\text{surv}} -$ UNIF_NEED*. Under the three gated product identities (r300: $\text{sl}_D = 2\,\text{sl}_{L1} - \text{sl}_{\text{neff}}$; r301: $\text{sl}_{\text{neff}} = \text{sl}_{\text{nact}} - \text{sl}_{CV^2}$; r302: Theorem 10.1 at the slope level) the four target margins*

$$m_D = \sigma^* - \text{sl}_D, \quad m_{\text{NEFF}} = \text{sl}_{\text{neff}} - \text{NEED}, \quad m_{\text{UNIF}} = \text{UNIF\_NEED} - \text{sl}_{CV^2}, \quad m_{\text{ATOM}} = \text{sl}_{\text{natom}} - \text{ATOM\_NEED}$$

are pairwise equal *— the cascade slack is **one algebraic number** (gated module-own in exact Fractions: all four margins = 3/50 on the rational slope set; the halved-need mutant breaks the $D \leftrightarrow$ NEFF invariance by exactly $\text{sl}_{L1} = 1/5$). Measured: $m_D = m_{\text{NEFF}} = m_{\text{UNIF}} = m_{\text{ATOM}} =$* $+0.0547$ *(invariance devs $\leq 9 \times 10^{-16}$), the core being $S = \sigma^* - \text{sl}_D$, i.e. the r299* DIAG *margin. The σ level is* not *the core ($\sigma^* - \sigma = +0.1976 = S +$ ratio surplus $+0.1429$ exact), and the 1/2-conversion conjecture is* refuted *(*$0.0988 \neq 0.0547$*: the factor 2 lives only in* NEED *and cancels in every margin difference).*

Remark 10.3 (The retyping — a methodological lesson). Theorem 10.2 *retypes* the cascade r297→r302: six rounds of exact reduction were *coordinate finding* — an exact dictionary locating one measured slack in progressively sharper coordinates — not six proof steps. The sealing discipline is what caught the reparametrization trap: each round had disclosed its own clause as "an exact reparametrization" (r300-B3, r301-B1, r302-B1), and the sealed regress audit turned the disclosures into an adjudicated verdict (REGRESS_CONFIRMED) instead of letting the lane count the same slack four times. The structure gains of the levels (count link, stationarity, coherence identity, χ destructive, lag-1 trace) *stand*; but the hard reviewer rule binds from r303 on: *a round counts as progress only if it adds new information, not if it re-expresses the same slack in a new coordinate.* The [E] status of the wave-9 exact identities is unchanged — the reading is refined.

Remark 10.4 (The first causal coordinate and the two-scale mechanism; r303/r304, typed measurements). The r303 mechanism test (1008 sealed marginal-preserving builds: bitwise multiset permutations with ρ_1 steered to matched / zero / flipped targets) delivers the lane's first *causal* coordinate: the ladder is monotone in ρ_1 (χ 0.630 $\to$ 0.764 $\to$ 1.029 $\to$ 1.342; end margin $+0.055 \to +0.057 \to +0.032 \to -0.044$) — *flipping the mixing sign kills the target inequality*. But ρ_1 alone is insufficient (MIXING_INSUFFICIENT: the matched family misses the destructive χ level by 0.134), and r304 names the split exactly: matching $\rho_{1..8}$ closes the χ *level* (0.652 vs 0.630, $|d| = 0.022 \leq 0.05$) at the price of the *slopes* ($0.028/0.027 > 0.02$) — the exact complement of the r303 family. **The mechanism is two-scale**: the level lives in the short lags ($k \leq 3$, lag-2-dominated, exactly attributed via Theorem 10.1), the slopes in the rung-wise ρ_1 trend; no lag-matching family reproduces both. The graduated reviewer ladder is monotone in χ but *not* in the margin (honest negative: absolute-level targets do not kill the inequality — the r303 kill came from per-rung flipped targets). Bycatch at theorem grade: $n_{\mathrm{eff}}^{\mathrm{atom}} = (L_1^a)^2/Q$ is a pure *marginal* functional (invariance 10^{-15} on all $2100 + 1008$ builds) — stationarity and mixing are complementary, not redundant.

Open problem 10.5 (The period-4 comb — the named open object)**.** The centered lag profile $\rho_{1..16}$ of the *dc* profile on the shared bulk support is a *stable, world-specific period-4 comb* (med $-0.222/-0.140/+0.089/+0.130$, then strong lags at every $k = 4m$ $(+0.13..+0.16)$ and $k = 4m+2$ $(-0.14..-0.15)$ up to 16; halves-stable 3/3; EPST/SCR differ), *not* a decaying tail: the sealed short-range rule finds no $k_0 \leq 8$. The reviewer condition splits exactly: the net covariance $NC(16) = 1 + 2\sum_{k \leq 16} \rho_k = 0.712 < 1$ *holds* (net-negative) while summability-with-small-tail *fails* ($\mathrm{SUM}(16) = 1.563$); the full lag sum is the zero-sum tautology ($1 + 2\sum_{\mathrm{all}} \rho_k \equiv 0$ for mean-removed finite profiles, gated live), so only the truncated NC carries content. The sign pattern of $S_{1..4}$ is $-/-/+/+$ with Fractions-exact certificates on both smallest rungs (kz18, kz23). *What is the comb?* Its derivation from the three-term recursion, its world-specificity and its invisibility to χ (the within-block shares die by $k \sim 4$) are measured facts without a mechanism — the comb is registered as the named open object of the stopped lane, not as a target.

Remark 10.6 (The documented lane stop; r304 consequence (iii))*.* The sealed reviewer stop case fired (LAW_LONGRANGE): the **global-profile mixing route of the L2 lane is documented closed**. The clean equivalence state at the stop: *L2 on the generic half* $\Leftrightarrow$ *anti-concentration of an explicit block field with long-range period-4 structure* (core slack $S = \sigma^* - \mathrm{sl}_D = +0.0547$ measured); return later with new tools — per the hard rule, no further coordinate change counts as progress. What stands honestly for a future attempt: the exact identity dictionary (count, coherence, χ-lag), the two-scale mechanism split, $NC < 1$, the $-/-/+/+$ sign pattern, and the marginal-functional invariance of $n_{\mathrm{eff}}^{\mathrm{atom}}$. This is a documented stop, not a defeat theorem: the period-4 comb is a measured finite-window object (42 rungs); no cofinal law is claimed for or against. In Lean no new statement is added: the stop carries no new exact target form (the comb and the two-scale split are measured ladder aggregates, not exact finite identities, and a universally quantified ladder form would be refutable — the machine-checked guard pattern of `RH/Counterexamples.lean`); the hole stays `lstar_subordination`.

Certification. Ledger: `PRIME.L2.CASCADE_CLOSURE.01 [E]; PRIME.L2.NEFF_TARGET.01 [O, retyped to the documented stop state]; PRIME.L2.REDUCTION_CHAIN.01 [E, reading refined]`. Module: `verification/v967_l2_cascade_closure.py`.
Probes (embedded byte-exact, sealed smoke stage; full 16-hex SPEC SHAs pinned in the module and in `rh/INVENTORY.json`): `neff_target` (32/32, `6f8cc404`), `unif_target` (30/30, `36df9424`), `atom_target` (26/26, `375e9f2b`), `shortrange_law` (37/37, `2cc5d23f`); rounds r301–r304; the cascade retyped as an exact dictionary (REGRESS_CONFIRMED, core $S = +0.0547$); the global-profile mixing route documented closed (LAW_LONGRANGE); no round in flight at this cut.

11 The architecture adjudication (v968)

Wave 11 (rounds r305–r316 incl. r310b) is the *architecture day*: after the wave-10 lane stop the reviewer set a three-lane contest (A: block-Green / constructive PSD-G rule; B: paired-cone / rank-one stream; C: fixed head) in parallel with the sharpest new fiber attack (the pointwise Rényi-3 law) and a Lean reconstruction of the formal layer. The day is frozen *in the reviewer's binding four-level structure* — each level is a different epistemic type, and the module `v968_architecture_adjudication.py` gates the composition so that the assignment is machine-checked, not editorial. **No RH claim**; nothing here is a cofinal theorem.

11.1 Level 1: real formal theorems (Lean, proved)

The formal layer was rebuilt in three Lean rounds (r305, r310, r310b; `rh/lean/RH/Closure.lean` and `RH/Source.lean`); the artifacts are already merged and green — *no Lean change ships with this promotion.* The sorry census fell 9 → 5 with clean bookkeeping: the two *true* formal window-local holes now stand alone (`lstar_subordination`, `terminal_positive_main`), next to the `crossing_budget` mathlib gap (not mathematical; v962-certified), the definitional source bridge (opacity-forced) and the pair-coordinate refinement `pair_margin_main`.

Theorem (`RH/Closure.lean`, **r305**)	**Content**	**Status**
`lstar_terminal_implies_master`	L* + terminal $q_N < 1 \Rightarrow$ full master positivity (Schur bordering + principal submatrices; the Riccati backward drain in matrix coordinates)	proved
`augmented_prefix_positive`	the former master `sorry` — now a *corollary*, no third arithmetic axiom	proved (corollary)
`free_window_positivity`	the base half in fog-free coordinates	corollary
`terminal_positive_main`	$0 < B \wedge q < 1$ (the border/fiber half; r258 TERMINAL__Q__LAW form)	the TRUE hole (so
Inertia layer (r305)	`psd_base`, `positive_prefix_firewall` (incl. a from-scratch Sylvester criterion `posDef_of_leading_det_pos` — mathlib v4.29.1 has none), `sylvester_pullback`, `half_filling_boundary`	proved (4 of 5; cros

Theorem (`RH/Source.lean`, **r310/r310b**)	**Content**	**Status**
`primePow_index_injective`, `nodes_injective`	unique factorization ⇒ strict node monotonicity on the unfolded prime-power chain	proved
`buildPrimeWindow_source_exact`	source exactness by `rfl` *through derivation* (nodes = $\log p^k$, weights = $\Lambda(p^k)$ via mathlib `vonMangoldt`)	proved
`foldedWindow_mass` (+ comb/arch channels)	mass conservation of *any* folding map via `Finset.sum_fiberwise` — aggregation is a quotient, not a projection	proved
`buildPrimeWindow_support_canonical`	the full four-stage support chain for every folding map	proved
`predefined_family`, `cofinal_prime_windows`	constructive atom census; cofinality via Euclid	proved
`finite_forms_converge_to_weil`	the *strong* stabilization form: the finite comb form *equals* the Weil tsum prime side from anchor $\geq \lceil e^b \rceil$ (exact stabilization, not mere pointwise convergence)	proved
`mainWindow_iff_builtFromPrimeSource`	the definitional source bridge (opacity-forced)	document

Also on this level: the r309 *rank-one update identities* (Section below, frozen per reviewer case 3 as `PRIME.SOURCE.RANKONE.UPDATE.IDENTITY.01` [**E**]) and the r306 Hill monotonicity + Rényi-3 ⇒ N_2 bridge (two Lagrange witnesses, Fractions dev 0; re-proved module-own in the v968 exact section S0, Lean-ready statement printed by the probe).

11.2 Level 2: certified finite statements

Statement	**Certificate grade**	**Round**
Rényi-3: $\sum_j q_j^3 \le C(\log m)^2/m^2$ with $C = 1.069$, $A = 2$ on all 57 rungs, growing reserve (trend -0.322)	record-pinned, 57/57; the Hill/Lagrange chain exact (Fractions dev 0)	r306
Fixed-head census: the *full* flat arch edge restores macroscopic reserve (min 2.19×10^{-3}, median amplification 3.1×10^4) on 77/77 windows	record-pinned census	r307
Block-Green identity $Q = \sum \Delta^\mathsf{T} G \Delta + \frac{5}{7}t^2$ on the finite objects	symbolically exact (toy t^2 pin 4/7); f64 4.0×10^{-14} on w9, 57/57 rungs	r308
Strictness of the source cone: $C_w \subsetneq \mathrm{span} \cap S_+$	*fully exact* rational in-span Farkas certificates (MINI#0, SM1#0); 0/16 in-span PD samples decompose	r311
Paired-cone soundness: the sealed mass-ratio cone, 0 violations on 20714 steps (certifying 56 %)	record-pinned; sound, not complete	r309
Fold multiplicity $\equiv 2$ (exact class bound) + 93 % far-triple cancellation ($TC_{\mathrm{far}} = 0.069$, falling)	exact / record-pinned on 57/57	r313
The signed cubic identity $\sum_j \lvert x_j\rvert^3 = \Delta F + C_{\mathrm{collision}} + C_{\mathrm{boundary}}$ with *vanishing* boundary ($F_1 \equiv 0$, cubic algebra)	decided in *exact Fractions on the real w9 data* (every deviation the rational 0); f64 4.5×10^{-17} on all 57 rungs	r314
The 21 small-m certificates ($C_{\mathrm{small}} = 1.0694$ = the r306 shallow maximum)	individually certified rungs	r316

11.3 Level 3: negative architecture decisions

The negative decisions are the day's load-bearing content — verbatim, with the break loci named:

Decision	Content (break locus)	Round
FIXED__HEAD__DEAD	no fixed $r \leq 8$ exists (best reserve 6.5×10^{-6}, four decades under the sealed 10^{-2} bar; reserve decays with S in the r286 law class) — the one-atom anatomy is proof-relevant as a *coordinate*, refuted as a *fixed dimension*	r307
PAIRED__CONE__NO__INDUCTION + PAIRED__CONE__WORLD__BLIND	the B1–B4 adjudication: B2 only partial (56 %), B3 (real induction $K_j \to K_{j+1}$) not reached, B4 (world semantics) missed — **B does not overtake A**; the only separating scalar is the friendly mirror (the r243 wall reading itself)	r309
r308 discriminator demarcated	the r308 world separation is *entirely* the budget sign: the Δ-sparsity graph is chordal on every size (incl. w9 $S = 367$), the trivial common dual $Y_t = e_t e_t^\mathsf{T}$ closes the r308 stalls two-sidedly	r311
COEFFICIENT__SIGN__WALL	**Lane A closes as a cone language** with the mechanism *named*: block-psd membership without rank-one-SDD membership, obstructed at the budget/border sign (Farkas mass on the border/budget row: t-share 0.583, 0.725 at deg28; the entire obstruction in 2.4 % SDD-forbidden negative cross-mixings)	r312
FLOQUET__EXPANDING	the four-step monodromy on the split cubic mode space *expands* (med sr 1.0600, 0/57 contracting, exponent +1.917) — after r304 + r313 no transfer-operator contraction form remains open at chain level	r313
TRIPLE__TYPE__MAJORANTS__WRONG	T2/T3 hold 0/57 but T1 falls at kz55/kz53 and T4 at kz55 (trends falling) — composition refuted in form; type constants not world-portable	r313
PHI3__ALL__BLIND	all three presealed source-pure candidates fail; the *entire* break sits on kz55+kz67 (the same near-critical family as r306-$A \leq 1$ and r313-T1/T4); local cause named: FCIX 0.955/0.915 vs med 0.629 — exactly where the flux cancellation dies	r315
TWO__REGIME__DEAD	the anatomy is the find: the obstruction family cuts *across* the FCIX stratum (kz53 kills the L side at bulk-normal FCIX 0.654); the r306 first-5 protocol was *load-bearing*, not a shallow artifact — any tight pointwise majorant dies at kz53 under a mid-ladder freeze; kz55/kz67 are near-one-block worlds (TOP1 cube share 0.558/0.785 vs 0.18): concentration, not counting tightness, separates	r316

11.4 Level 4: open mechanisms

Open mechanism	Type / where it lives
The constructive PSD-G rule	closed as a *lane* (r312), open as a *question*; `PRIME.LSTAR.CLOSED_FUNCTIONAL.01` [O] carries the wave-11 lane closure note
The signed cubic flux theorem	the identity is exact (r314), no bound follows yet; the majorant chains transfer any future certification by algebra onto $\sum q^3$
The quantitative source functional	the class RECURSIVEDIFFERENCEPROFILE needs a *quantitative* membership functional (r313's sharpest honest negative: no boolean property separates like the r306 total bound)
The tensor mechanism	`PRIME.SOURCE.TENSOR.MECHANISM.01` [O]: does a source-pure cofinal control arise from the banked rank-one update identities? Registered *after* r313/r316 with the documented negatives
The Rényi-3 provenance	`PRIME.L2.RENYI3.PROVENANCE.01` [O]: the GO stands, the provenance is open (trilogy: identity exact, functional blind, two-regime dead); the R317 fork sits with the reviewer — a near-critical family coordinate (TOP1/QMAX source-pure in advance; kz53 needs a second coordinate) *or* a certified exception family (the r287-F2 pattern)
`crossing_budget`	a mathlib gap (Jacobi inertia / minor dictionary), not a mathematical one — v962-certified
The definitional source bridge	`mainWindow_iff_ builtFromPrimeSource`, opacity-forced
L* and the terminal	the two TRUE formal window-local holes (`lstar_subordination`, `terminal_positive_main`)

Remark 11.1 (The claim split — binding). Per the reviewer's explicit instruction the r309 outcome is *not* frozen under one "transfer" title. `PRIME.SOURCE.RANKONE.UPDATE.IDENTITY.01` [E] carries *only* the exact update identities (the determinant update dictionary, the Sherman–Morrison chain along the ordered stream, the exact $\pm(\eta \pm e_t)$ border split, the local Schur reserve $r = 1 - L_{bb} + L_{ab}^2/(1 + L_{aa})$, the signed source updates); `PRIME.SOURCE.TENSOR.MECHANISM.01` [O] carries the open question whether a source-pure cofinal control arises from them — registered *after* r313/r316 so the documented negatives (no chain-level contraction, no two-regime majorant) are part of the contract, not a surprise. The v968 four-level composition gate checks that the split is *not* hard-wired: a passing B2–B4 would hand the lane to architecture B, a firing r312 GO would keep Lane A open, a surviving two-regime bound would resolve the provenance — each tipping mutant changes the composed adjudication.

Remark 11.2 (Universality classes — base and fiber). The day sharpens a structural reading that frames every lane: the *base* is fine-metric — the r289 METRIC_ONLY adjudication and the twin invariance recur on every wave-11 surface (the rational twin keeps the full signature; the r307 twin metric, the r308 diophantine-trivialized twin and the r315/r316 twin bands are the strongest of the lane). The *fiber* is recursive — on every fiber law of the day EPSTEIN holds and SCRAMBLE breaks (the r306 bound, the r314 shares, the r315 candidates, the r316 majorants): the laws live in the recursion structure, not in block combinatorics. Base and fiber share the *algebra* (the identity dictionary applies to both) but not necessarily the *mechanism* — the reviewer's maturity assessment raised the program $4 \to 5$ (formal architecture 9.5) on exactly this reading; it is typed here as a **reviewer assessment**, not a machine-checked fact. The wave-12 adjudication supersedes this single scalar with a per-layer table (Remark 12.6).

Certification. Ledger: `PRIME.LSTAR.ARCHITECTURE_ADJUDICATION.01 [E]; PRIME.SOURCE.RANKONE.UPDATE.IDENTITY.01 [E]; PRIME.SOURCE.TENSOR.MECHANISM.01 [O]; PRIME.L2.RENYI3.PROVENANCE.01 [O]`. Module: `verification/v968_architecture_adjudication.py`. Probes (embedded byte-exact, sealed smoke stage; full 16-hex SPEC SHAs pinned in the module and in `rh/INVENTORY.json`): `renyi3` (27/27, `3bb365e1`), `fixed_head` (28/28, `ec2bb008`), `block_green` (30/30, `d5147850`), `paired_cone` (33/33, `f8d99877`), `blockgreen_nontriviality` (34/34, `fac7a8df`), `blockgreen_membership` (32/32, `6c32f749`), `renyi3_proof_fork` (32/32, `6505dd10`), `signed_cubic_flux` (29/29, `841b3196`), `phi3_functional` (29/29, `92d35a3a`), `two_regime_bound` (35/35, `5c28b12b`); rounds r305–r316 incl. r310b; the Lean rounds r305/r310/r310b are consumed as reports (artifacts in `rh/lean/`, re-verified by `lake build` and the RH suite); no round in flight at this cut — R317 waits on the reviewer fork.

12 Red team and the two forks (v969)

Wave 12 (rounds r317–r322) is the *red-team morning*: after the reviewer's asymmetry decision (the terminal has a proof path, L* needs a new idea; resource split 40/40/20) an independent extraction-chain audit ran against the formal layer while the two reviewer forks executed in parallel lanes. The module `v969_forks_and_redteam.py` freezes the six rounds; the audit findings U1–U3 are re-established *module-own in exact arithmetic* (rational witnesses, integer certificates — no Lean call), so the suite carries the inconsistencies independently of the Lean toolchain. **No RH claim**; nothing here is a cofinal theorem.

12.1 The extraction-chain red team (r319) and the repair (r320)

Round r319 reconstructed the chain from the two lemmata (L* + terminal) to Weil/RH independently and found *three kernel-checked type inconsistencies*, three levels above the boxes — not in the proofs, but in the statement *types* of the interface layer (the r273 disease, one level up):

- **U1 (bridge × terminal).** The r310b source bridge `mainWindow_iff_builtFromPrimeSource` as typed bound only nodes/comb/arch — never u, never B. A window with $B = -1$ therefore *represents the empty spec*, while `terminal_positive_main` demands $0 < B$: the two types refute each other for *every* interpretation of the opaque predicate.
- **U2 (bridge × L*).** At mesh level 0 the tolerance is log(anchor), so the *total node collision* (all nodes = 1) represents the $\{2, 3, 4\}$ spec — and $p = X - 1$ vanishes on the collided window, forcing the strict subordination to demand $0 < 0$. In v969 S0 the collision admissibility reduces *exactly* to three rational facts about e ($2 \le e$, $3 \le 2e$, $e \le 4$) on the certified series bracket $685/252 \le e \le 685/252 + 1/35280$.
- **U3 (pair margin × existence).** The pre-r320 `pair_margin_main` quantified (Z_{loc}, *runs*) freely, bound only by the split; the adversary $Z_{\mathrm{loc}} = |F| + 1$, *runs* $= []$ satisfies the split and demands $(|F| + 1)^2 < 5/7$ — false for every F since $(|F| + 1)^2 \ge 1 > 5/7$ exactly. The old type forces its source predicate *empty.*

In addition the audit named the **mesh-vs-anchor cofinality seam**: `cofinal_prime_windows` proves the *anchor* direction at fixed mesh, which is exactly the H_{cof} direction proved inadmissible in the carrier lane (mesh refinement needed, false floors measured); the v749 tower identification is the named open Lean target.

Remark 12.1 (The honest chain reading — binding). The audit's verdict on the composed claim is the most important sentence of the wave and is adopted verbatim as the binding formulation for every downstream surface: **"two lemmata ⇒ RH" does not survive the literal reading.** Literally, the two lemmata (L* + terminal) give *only window-local master positivity for the opaque, currently witness-less* `MainWindow`. The road to RH additionally needs: *(i)* the corrected source bridge with u/B/fold/arch fidelity (repaired in r320, one definitional `sorry` remains), *(ii)* the transport onto the real folded von-Mangoldt source, *(iii)* the mesh-cofinal v749 tower identification (to be formalized — the seam above), and *(iv)* the cited classics (form convergence *proved*, `finite_forms_converge_to_weil`; density; Dini continuity; the Weil criterion). Every earlier "two lemmata" phrasing is superseded by this reading.

Remark 12.2 (The three levels of the proof graph — binding reviewer adjudication, wave 12). The proof graph has three cleanly separated levels, and from wave 12 on the two arithmetic holes are named *window-local* on every surface:

- **Level A (local window mathematics, PROVED):** $L^*(w)+T(w) \Rightarrow \text{MasterPositive}(w)$ — cleanly typed after r305/r320. In the reviewer's words: *"For every canonical finite window, L^* and terminal positivity imply augmented prefix positivity. This is a window-local theorem."*

- **Level B (the universal window family — THE two arithmetic window-local holes):** $\forall w \in W_{\text{canonical}} : L^*(w) \wedge T(w)$ — the contracts `lstar_subordination` and `terminal_positive_main`.

- **Level C (extraction to Weil/RH, OPEN):** the source-exact realization of the *actual* source + an admissible transport theorem + a convergence/stabilization theorem in the correct directed order + density/continuity for the Weil criterion. *"Any implication to Weil positivity or RH additionally requires a source-exact realization, an admissible transport theorem, and a convergence or extraction theorem in the correct directed order."*

Two honest qualifications belong to this picture. First, **the false cofinality direction is documented, not solved**: the U3-adjacent seam of Section 12.1 (`cofinal_prime_windows` proves the anchor direction, not the H_{cof} refinement direction) is a named Level-C gap, and no wave-12 artifact closes it. Second, **the sorry census does not currently measure the global distance in full**: the missing Level-C extraction architecture appears in *no* `sorry`, because the correct theorem statements do not yet exist — the census of five counts the window-local layer only. *Wave-13 update (Section 13):* the second caveat is **partially discharged** — the r326 module `RH/Elementwise.lean` raises the census $5 \to 8$ with three *named typed Level-C statements* (arch/pole transcription classical, source completion definitional), so the proof graph now shows Level C as named statements; the distance is *named*, not closed.

Remark 12.3 (SourceExact: firewall now, construction theorem later). The reviewer's classification of the r320 `SourceExact` predicate is adopted: it is a *good firewall* (it seals the arch/border disease against the U-class adversaries), but in a final proof it must *not* stand as a free assumption. The named open Lean target is the target architecture `CanonicalPrimeWindow` plus *exactly one* construction theorem `sourceExact_buildPrimeWindow` — load-bearing for the global (Level-C) connection, not formalization polish.

Round r320 repaired the interface completely (all artifacts merged and green in `rh/lean/`; no Lean change ships with the v969 promotion): `RepresentsWindow/RepresentsSpec` retyped with u/B fidelity, the *separation discipline* ($2\delta <$ min node gap and $\delta <$ min comb weight; honest: mesh level 0 represents *nothing*) and `budget_pos`; the additional finding beyond the audit — the free spec fields arch/border carry the same disease — sealed by the new opaque `SourceExact` (r273 convention spec-side); the bridge lifted to $\exists s$, `SourceExact` $s \wedge$ `RepresentsWindow`... (the one definitional `sorry`; the earlier `Iff.rfl` promise corrected); `pair_margin_main` retyped onto the *canonical extraction* (`signRuns/terminalDrive/bulkRuns/edgeLocal` as definitions, `signRuns_sum` and `canonical_split` *proved* — no free $(Z_{\text{loc}}, runs)$ anymore); **three permanent sorry-free guards** (`old_bridge_terminal_inconsistent`, `old_bridge_lstar_inconsistent`, `old_pair_margin_forces_empty` in `RH/Counterexamples.lean`, $\forall P$ form, axiom census kernel-checked without `sorryAx`); and the **sorry-free witness** `witness_represents` (anchor 2, atoms $\{2,3,4\}$, mesh level 4 — the first level with satisfiable separation discipline, via the integer facts $2^7 < 3^5 < 2^8$; v969 S0 re-gates the reduction to $1/8 < 2/5$ exactly, and the level-0 failure). $\exists$`MainWindow` stays *deliberately* unprovable — two opaque predicates block exactly the adversaries. Sorry census $5 \to 5$ with two typed *retypes*; the two true *window-local* holes (`lstar_subordination`, `terminal_positive_main` — Level B of Remark 12.2) and `crossing_budget` are byte-identical.

12.2 The fiber fork: from the exception-family abort to the sliding bound (r317/r321)

Reviewer fork (b) executed first (r317, `exception_families_probe.py`, 38/38): the statement form "generic theorem + at most two source-pure exception families" under a hard gate sealed in advance. The gate fired **whac_a_mole(kz53, kz83)** by contract — the sealed blind gap rule *does* recover the spike family $B = \{kz55, kz67\}$ (the FCIX pair, gap 1.78, finite certificates $C_B = 1.0536$), but class A stays empty at bar distance 0.017 and kz53/kz83 stay uncovered on the complement. The census behind the abort is the round's find: the F_A top-3 are *exactly* kz53 (2.47), kz83 (2.39), kz67 (2.38) — one source-pure rank-local QMAX coordinate ranks the complete mid/deep near-critical family on top, but as a *continuum*: the exception-family form is the wrong statement shape.

Fork (a) in the r317-sharpened form (r321, `continuous_coordinate_probe.py`, 39/39) then fired **sliding_bound_go(G_{SQ})**:

Theorem 12.4 (Sliding cubic bound — *theorem candidate*, certified on the measured rungs only)**.** *Let $F_A(w)$ be the rank-local QMAX spike ratio (window $W = 5$, the r317 import) and q_j the normalized atomic $|P\Delta|$ profile of the window w with m blocks. On* every measured rung *of the 65-rung cofinal ladder with $m \geq 73$:*

$$\sum_j q_j^3 \;\leq\; 1.3056\; F_A(w)^2\,\frac{(\log m)^2}{m^2},$$

with the 21 small-m rungs certified individually ($C_{\text{small}} = 1.0694$). The constant is algebra-derived and source-pure*: $b = (\max_{\text{cal}} B)^2 = 1.1426^2$ with $B = \text{medloc} \times m/\log m$ — no target value enters the calibration — backed by the exact two-sided concentration bracket $q_{\max}\,\Phi_{H1} \leq \rho_2 \leq \Phi_{H1} = (F_A B)^2$ (warded live on 69 worlds; re-proved from scratch in v969 S0).* Status: a certified finite statement on the measured rungs (0/39 test violations; all four named r316/r317 violators kz53/kz83/kz67/kz55 inside at reserves 7.0..9.6), registered as the open contract *PRIME.L2.RENYI3.SLIDING_BOUND.01 [O]*. It proves no universal bound and no cofinal law.

Honesty, disclosed with the GO: the implied uniform constant $C_{\text{impl}} = g(2.47) = 7.97$ is $7.5\times$ looser than the r306 first-5 constant 1.069 — the round buys *form* (one gliding bound, no regimes, no exception families, all old violators inside), not sharpness. The pure-algebra transfer route does *not* close (B is not bounded by its calibration max: test max $1.4088 = 1.23\times$, trend $+0.122$ rising); and the sliding bound is *not* world-separating by itself (SCRAMBLE is a concentration spike too, covered at 5.21 — the rejection is carried by the r317 class side condition). The provenance question of the fiber is now *source-pure, local and split in two*: (a) is F_A bounded (measured max 2.47; the near-critical family is its top), and (b) what bounds the $q_{\max}$-share $\rho_2/(F_A^2 B^2)$ (the median-of-max shape route; the r302 M_2 stationarity 1.973 pins the second moment of the same shape).

12.3 The base fork: index language, sign regularity, and the basin lesson (r318/r322)

Round r318 (`indefinite_fork_probe.py`, 25/25) ran the new L* idea class — Pontryagin/Krein *index* vs *sign regularity* — as one representation test. Verdict **p2_main_specific**: the 2.4 % negative cross mixtures of the converged r308 Dykstra family are *not* structureless — on MAIN-class worlds the block residual lives lawfully on the *antiphase pair* $(D3, D4)$ with fixed sign -1 (12/12 rungs, twin exact), while the dead controls break by *pattern* on the arch-mean×border pair. The P1 index route closes *as language*: the bookkeeping is exact (window index defect 0/0/0 on the mains/twin vs 55/37/4/31 on the controls — the control flips *are* negative directions inside the window), but "$n_+(\text{window}) = N_w$" is a total restatement of L* (10/10 instances, both truth values realized), the global inequality is vacuous and the negative-subspace invariants are world-blind: a language gain, no lever. Classical sign regularity is dead on MAIN (PREMISE_FAILS_ON_MAIN: orientation-sensitive minor censuses are coin-flip ~ 0.50).

The dig round r322 (`antiphase_sign_law_probe.py`, 25/25) then adjudicated the fingerprint itself: **algorithm_artifact**. The sign law *and* the r312 97.6/2.4 cone anatomy are properties of the *least-norm-proximal Dykstra basin*, not of the psd solution set: the sealed random-start variants (staged to 20 000 steps) carry modal pair $(0, 2)$ at shares 0.310/0.264, while the canonical variants (least-norm start, long, zero start, swapped projection order) coincide within rel 2×10^{-7} and carry the law exactly. The r318 pattern dichotomy does *not* survive fair convergence (the budget-ablated *converged* EPST control carries the MAIN law *stronger* than MAIN, 0.742 vs 0.692); and the law is *not* identity-forced — the exact forced-component analysis (Fractions, machine-exact on the TOY4 calibrator 4/7) shows the $(D3, D4)$ sector is roughly half free with a *positive* forced value: the negative sign law lives entirely in the free/selection directions.

Remark 12.5 (The basin lesson). The r322 adjudication is a methodological lesson of the same class as the r303 reparametrization trap: a lawful, reproducible, position-structured fingerprint with total sign consistency (1.000; 92.3 % of all blocks) can still be an *algorithm artifact* — a property of the solver's selection geometry, not of the mathematical object. The sealed canonical-vs-random variant battery is the honest test, and it is now module-gated in v969 (the basin adjudication logic with tipping mutants). What survives as the dig yield: the named residual question — *why does the least-norm-proximal basin organize its indefiniteness on the antiphase pair?* (best reporting-grade hook: the block-local antiphase mass pairing, Spearman +0.81, below the sealed 0.9 bar) — and the r312 wall fact that no canonical source-pure selection rule is known.

Remark 12.6 (Typed reviewer assessment — wave 12; supersedes the wave-11 4 → 5 scalar). Where the wave-11 assessment placed one scalar (Remark 11.2), the wave-12 adjudication types the program per layer. It is a **reviewer assessment**, not a machine-checked fact:

Layer	**Rating**
Finite dictionaries	9.5/10
Formal window-local architecture	9/10
Terminal mechanism	6/10
L^* mechanism	2/10
Source/extraction architecture	2–3/10
Complete RH proof path	$\sim 3/10$
Audit quality	9/10

with the binding sentence: *"Stronger as a window-local research program; as a complete RH proof chain currently less far than previously assumed — the removal of a shortcut that did not exist."*

Remark 12.7 (Lane status after wave 12). L^* is internally **paused** (allocation 0 %, reviewer decision): the selection geometry is not pursued further (ALGORITHM_ARTIFACT, r322); re-opening only under the six documented conditions of the r322 round notes; as an *external* object, the two-weights sampling formulation $\|J_N\| < 1$ remains a memo candidate. Terminal is sharpened to the QMAX$\times M_2$ shape route (round R324 in flight — deliberately *not* consumed by v969). The extraction-repair fork R325 (towards `CanonicalPrimeWindow` + `sourceExact_buildPrimeWindow`, Remark 12.3) is in flight — deliberately *not* consumed by v969. *Wave-13 update:* both in-flight rounds are executed and consumed by v970 (Section 13); the lane status after wave 13 is Remark 13.3.

Certification. Ledger: `PRIME.REDTEAM.EXTRACTION_AUDIT.01 [E]; PRIME.L2.RENYI3.SLIDING_BOUND.01 [O]; PRIME.L2.RENYI3.PROVENANCE.01 [O update]; PRIME.LSTAR.SUBORDINATION.01 [O update].`
Module: `verification/v969_forks_and_redteam.py`.
Probes (embedded byte-exact, sealed smoke stage; full 16-hex SPEC SHAs pinned in the module and in `rh/INVENTORY.json`): `exception_families` (38/38, `04fbe5c0`), `indefinite_fork` (25/25, `f2d98683`), `continuous_coordinate` (39/39, `e68883ad`), `antiphase_sign_law` (25/25, `761b51d4`); rounds r317–r322; the Lean rounds r319 (audit) and r320 (repair) are consumed as reports (artifacts in `rh/lean/`, re-verified by `lake build` and the RH suite; the U1–U3 witnesses re-established in exact arithmetic in the module's S0); the false cofinality direction is documented, not solved; rounds R324 (terminal QMAX$\times M_2$) and R325 (extraction repair) are in flight and deliberately not consumed at this cut.

13 Elementwise extraction and the measured composition (v970)

Wave 13 (rounds r323–r327) executes the two lanes the wave-12 reallocation opened (55 terminal / 35 extraction / 10 formalization / 0 internal L*): the *extraction repair* answers the R319 cofinality seam with an elementwise quantifier architecture and implements it in Lean (r325/r326), and the *terminal lane* composes its chain to `terminal_positive_main` as an honestly typed MEASURED statement (r324 after a binding mid-round reviewer course correction, with the pre-work banked as r324-pre; r327 the follow-up anatomy). Round r323 — the planned L* fork — was *cleanly aborted* by the same reviewer adjudication at the end of its design phase, before any write access: no probe, no measurement, no `rh/` file, nothing counts as a round (post-abort suite 73/73 with 103/103 pins). The module `v970_extraction_and_composition.py` freezes the wave; the onset formula, the interpolation, the F_A identity, the group ledger and the m_0^* arithmetic are re-established *module-own in exact arithmetic* (pure Fractions; no Lean call), including the chain-honest bar of the independent chain audit r328B (the S3′ item below). **No RH claim**; nothing here is a cofinal theorem.

13.1 The extraction repair: elementwise stabilization (r325) and the Level-C architecture (r326)

Round r325 (`extraction_order_probe.py`, 18/18, SPEC `31277f91` final / freeze `57e50d36`) measured the machine-checkable halves of the three reviewer variants for repairing the RH-connection seam of Section 12.1 (the `rh/` cofinality theorems run in the *anchor* direction; H_{cof} demands the *mesh-refinement* direction, measured inadmissible with false floors). Verdict: **elementwise_stabilization_go** primary, with co-letters CANONICAL_MESH_REBUILD_GO + SIGNED_DEFECT_CORRECTION_GO(SIGN) + ANCHOR_ONLY_INSUFFICIENT. The repair statement is a *quantifier statement*, not a mesh theory: on the native dense class (dyadic step-function autocorrelations — the v749 “Weil form of step functions” class, elements sealed by seed) *all three channels* (comb/arch/pole) of the canonical tower windows stabilize *exactly* at the predicted finite anchor onset $\alpha^* = (n_g{+}1)D_0/2$ ($0.0/3.3\times10^{-18}$ above onset, $2.7\times10^{-2}..5.3\times10^{-2}$ below — a real onset) *and* the value is constant under dyadic mesh refinement (comb 3.3×10^{-18}, arch 1.5×10^{-15}, pole 2.0×10^{-17}): the extraction consumes *no* mesh-cofinal ladder and *no* transport — H_{cof} **is not needed and is replaced as the target route**. Variant B is the *construction substrate* of A, not an independent route (v749 T1.4b reproduced: every deployed frame-A window *is* a directly rebuilt tower member at rel dev 0.0; the rebuild ladder is PD per stage with honestly falling margins). Variant C is the *quantified density step*, not an independent extraction: on the sealed non-native C^2 class the per-channel quadrature defect is *one-signed* (comb negative, arch/pole positive), falls at the D^2 class rate per channel (ratios 3.6..9.0) and sits inside the rigorous closed-form interpolation envelope $|E_{\mathrm{ch}}| \leq (D^2/8)\,\|F''\|\,K_{\mathrm{ch}}$ — controlled on its own, *no wall margin anywhere*. The anchor-only floor is real (E_{tot} constant nonzero $+1.77 \times 10^{-4}/+1.59 \times 10^{-4}$ — the R319 false-floor mechanism reproduced on this probe's own elements). In v970 S0 the onset formula is re-established as an *exact* rational toy stabilization (the tent-window read equals the exact atom sum 35/16 for every $\alpha \geq \alpha^* = 5/8$, differs below, and is constant under refinement $D_0 \to D_0/4$; the nearest-neighbour mutant is caught).

Round r326 implemented the adjudicated architecture in `rh/lean/RH/Elementwise.lean` (merged and green; consumed by v970 as a report, re-verified by `lake build` and the RH suite). The three **proved kernel pieces** are the load-bearing content:

Theorem (proved, kernel-checked)	Content
`sourceExact_buildPrimeWindow`	the wave-12 reviewer target (Remark every canonical family member satisf `SourceExactSpec` (nodes/comb derived atom-set completeness via `predefined` tivity) — `SourceExact` is *eliminated as* *the extraction route*; built windows prova
`comb_elementwise_stabilization` / `elementwise_finite_stabilization`	the comb channel of the R325 statem onset $a_0(f) = \max(1, \lceil\exp(\text{steps} \cdot D_0)\rceil$ *element's support alone* (corpus gauge quantifier; `GridElement` built for real support; honesty tie: `combRead_eq_wi` read *is* the built window's comb channe poses from the three channels
`weil_nonneg_of_windowlocal`	**the extraction without the ladder**: w of the canonical family implies Weil-fo every grid element by *one finite instan* Euclid prime anchor above the onset + tive mesh) — no refinement tower, no PS plus `weil_nonneg_of_bordered` (the co *named* Prop `BorderedCompressionBri` mitment)

The sorry census moves 5 → 8: the three *new* `sorry`s are *typed Level-C statements that were not formalizable before this round* — `arch_elementwise_stabilization` (classical, S2: the arch kernel transcription `arch_A` is classical integral analysis, R325-measured exact at 1.5×10^{-15}), `pole_elementwise_stabilization` (classical, S2: the v716 pole closed form, 2.0×10^{-17}), and `specFamily_sourceExact_completion` (classical + definitional, opacity-forced; the extraction route does *not* consume it — that is the point of the architecture). The five legacy `sorry`s are byte-identical; the two true *window-local* holes (Level B) are unchanged.

Remark 13.1 (The honest formulation of the new connection — binding, wave 13). The wave-12 honest chain reading (Remark 12.1) is *updated* by the r325/r326 architecture to the following binding formulation: **two window-local lemmata (over the canonical family) + the elementwise architecture + cited classics (density, v912 continuity, the Weil criterion) ⇒ RH; open links: the arch/pole transcription (classical), the compression bridge, the source completion (definitional).** The window-local positivity premise itself (Level B) is proved nowhere; the three open links are the named typed `sorry`s and the named Prop above.

13.2 The measured terminal composition (r324, r324-pre banked) and the coincidence anatomy (r327)

Round r324 ran under a binding mid-round *reviewer course correction*, disclosed in full: the round's original contract (the F_A-boundedness hunt) was executed and sealed *first* (`fa_provenance_probe.py`, 36/36, SPEC `9a6696f8` — verdict FA__BOUNDED__DISTRIBUTIONAL + QMAX__SHARE__OPEN + NOT__COMPOSED: the F_A distribution matches the iid stationary max-statistic null, KS $0.0982 \le 0.125$, $p_{\max} = 0.090$, envelope $C_F = 3.357$; the honest tension disclosed — the per-rank p values at the named spikes are 0.005/0.000/0.000), then *superseded*: the F_A-boundedness statement was demoted as potentially unnecessarily strong, the cleaner decomposition is $Q_{\max} \times M_2$. The pre-work is *banked* as r324-pre, stays sealed and citable, and its record is consumed as disclosed anchors. The binding contract (`qmax_m2_origin_probe.py`, 36/36, SPEC `dc36cacb`) fired **pileup__grows__subcritical** ($e_{\text{tot}} = +0.172 <$ CRIT $= 0.224$, stability halves $+0.141/-0.160$ both subcritical, the second *falling* — decided; by the reviewer contract a sufficient, first-class outcome):

- **S0 exact** (re-proved module-own in v970): $M_3 \le q_{\max} M_2$ and $M_2 \le q_{\max}$ (Fractions; toy slack exactly 1/36, *double* equality at the flat profile; live 69 worlds slack 0.0), and the identity $F_A \cdot B \cdot \log m = m\, q_{\max}$ (2.7×10^{-16} live): F_A **is de-black-boxed** — it is the rank-local normalization of $m\, q_{\max}$, not an independent coordinate; the r321 bracket upper side equals Φ_{H1} and the interpolation improves it by exactly $M_2/q_{\max} \le 1$.

- **S1:** the pointwise M_2 export is *refuted* ($C_2 = 2.2557$ mid-ladder freeze fails 7/39 — exactly the banked spike set, +35..42 %) while the r302 *distributional* export holds GO (stationarity KS 0.023/0.016, tighter than the r302 core 0.043).

- **S2, the mechanism half and half:** the $O(\log m)$ *scale count* certifies pointwise ($C_{\text{NSC}} = 2.0258$, 0/39) but the per-scale mass $O(1/m)$ is refuted ($C_{\text{PIL}} = 9.3583$, 11/39) — the hardness answer: the near-critical windows are *one near-maximal heavy source scale* at the argmax block (kz53: s_0 carries 13.78 of $G = 13.02$ at nsc = 5, the ladder median), not multiscale convergence.

- **S3, the composition (typed MEASURED):** $\sum_j q_j^3 \le 8.941\,(\log m)\, m^{+0.172}/m^2 \Rightarrow N_3 \ge m^{0.914}/\sqrt{8.941 \log m}$, $N_2 \ge N_3$ (the r306 exact chain) $\Rightarrow$ the N_2 density need for all $m \ge m_0^*$ $\Rightarrow$ the r297 target $\Rightarrow$ the v964-S0 vdC theorem $\Rightarrow$ $\delta' > 0.21$ generic $\Rightarrow$ the `terminal_positive_main` edge. All measured $m \le 274$ are closed by the finite certificates; **the gap from 274 to m_0^* is the disclosed extrapolation hypothesis** — m_0^* is astronomical, no cofinal claim. (The m_0^* log-space solver is rebuilt verbatim in v970 S0, and the disclosed pre-freeze linear-space bug — a search capped at 10^{18} that *misses* the root — is rebuilt and caught.)

- **S3′, the chain-honest bar (independent chain audit, r328B):** the record bar CRIT $= 0.224$ was derived from the *atom-level* need 0.888 (R302__ATOM__NEED for $n_{\text{eff,atom}}$); the composition functional $N_2 = n_{\text{eff}}$ is *block-level* — its need is 0.908 in N units, i.e. $908/1002 = 0.9062$ in m units via $\text{sl}_{\text{nact}} = +1.002$. The chain-honest bar is therefore CRIT $= 2\,(1 - 0.9062) = 0.188$: **the verdict survives** ($+0.172 < 0.188$, exact in v970 S0-T5d), but the sealed bar was too loose — any future e_{tot} in the window $(0.188, 0.224)$ would fire falsely, so the going-forward bar is 0.188. The record threshold $N_2 \ge m^{0.888}$ does *not* by itself reach the block-level need ($0.888 \times 1.002 = 0.890 < 0.908$); the inference carries via the full polylog form $1 - e_{\text{tot}}/2 = 0.914 \ge 0.9062$ (margin +0.008, exact). Both onsets are astronomical — $m_0^* = 10^{59.6}$ at the record (atom-level) need, $m_0^* = 10^{238}$ chain-honest, from the same solver — so the qualitative statement (measured rungs plus a typed extrapolation hypothesis) is unchanged. The sealed r324 record is *not* edited: the correction lives as a chain annotation and an exact v970 gate, and the record pressure line attributing the whole e_{tot} (incl. $e_{M_2} = +0.014$) to the $q_{\max}$ factor is correct as a product bound but misleading per-factor.

Round r327 (`group_mass_cap_probe.py`, 34/34, SPEC `11e4fd40` final / freeze `71f8b7b4`) descended one level, from the heavy scale to the fold *group*. Verdict **cap_partial** with anatomy letter **SOURCE** (median named ratio $1.057 \geq 0.5$), and the structural finding of the round: *on all 65 rungs the heaviest group of the argmax block is exactly one β/ω fold pair* (one bulk + one window atom at one position; window-atom histogram $[0, 65, 0]$) with median alignment 1.000 — the two ancestors *reinforce*. At kz53 the single pair carries 88.8 % of the argmax block's atom mass at gap 0.076 (the second group 13× lighter): **the spike is literally one bulk/window coincidence** (kz67 is the exception shape: 6 groups, bshare 0.416); position thirds $[21, 29, 15]$ mid-heavy (the r322 mid/tail echo). The Λ-pair cap is *refuted* as a mid-ladder polylog ($C_{LV} = 1.1838$ fails 19/39, named 0/4 — the exact two-ancestor algebra $g_{\text{abs}} \leq 2\,v_{\max}$ holds at slack 0.0, the *constant* does not close); the direct group cap fails at $A = 1$ and $A = 2$; **the group count certifies** pointwise ($C_{NG} = 2.6351$, 0/39, named 4/4) — the *third* certified $O(\log m)$ count of the lane after the r324 scale count. The route exponent $e_{\text{route}} = +0.106 < 0.224$ but the stability halves straddle $(+0.411/-1.183)$: HEAVY_GROUP_UNBOUNDED does *not* fire — the heavy-group coordinate *decays* with depth (a shallow/mid phenomenon, not growth). Q3: no recomposition — the r324 MEASURED chain stays the honest end state of the route.

Remark 13.2 (The coincidence geometry — the named direction)*.* The typed direction after r327: the spike cap, if it exists, is a statement about *where* the single heavy β/ω coincidence can *sit* — the source geometry of the bulk/window pairing at one position — not about group count (certified) or multiplicity (proven). Alternatively the honest end state of the route is the r324 MEASURED composition with its typed extrapolation gap.

Remark 13.3 (Lane status after wave 13)*.* L^* stays internally **paused** (the r323 fork was cleanly aborted before any write access by the same adjudication). The *terminal* lane ends at the MEASURED composition (Section 13.2) with the coincidence-geometry question (Remark 13.2) as the named direction. The *extraction* lane has its architecture standing (Section 13.1): three proved kernel pieces, three typed Level-C rests (arch/pole transcription, compression bridge, source completion). Lean: 8 `sorry`s, all typed — the two true window-local holes unchanged.

Certification. Ledger: `PRIME.EXTRACTION.ELEMENTWISE.01 [E]`; `PRIME.L2.RENYI3.MEASURED_COMPOSITION.01 [O]`; `PRIME.L2.RENYI3.SLIDING_BOUND.01 [O update]`; `PRIME.L2.RENYI3.PROVENANCE.01 [O update]`. Module: `verification/v970_extraction_and_composition.py`.
Probes (embedded byte-exact, sealed smoke stage; full 16-hex SPEC SHAs pinned in the module and in `rh/INVENTORY.json`): `fa_provenance` (36/36, `9a6696f8`, r324-pre — banked), `qmax_m2_origin` (36/36, `dc36cacb`), `extraction_order` (18/18, `31277f91`), `group_mass_cap` (34/34, `11e4fd40`); rounds r323–r327; the Lean round r326 (`rh/lean/RH/Elementwise.lean`, census 5 → 8) is consumed as a report (artifacts in `rh/lean/`, re-verified by `lake build` and the RH suite); the r323 abort is consumed as a note (clean, before any write access); the window-local positivity premise is proved nowhere; no round in flight at this cut.

14 The wave-14 consolidation (v978–v982): terminal martingale, cover and growth, the closed margin chain, the Borodin duality and the second arithmetic

Wave 14 — reviewer-authorized after the double substance condition (r355 K2 NU-free + r357 second arithmetic) — freezes rounds r334–r357 as *five* modules. Unlike the embedding waves 8–13, each module re-derives its exact layer *from scratch* (sympy symbolic + pure Fractions, no probe imports) and re-runs the sealed round verdicts as exact decision logic on the frozen record aggregates with tipping mutants; the eighteen discovery probes stay sealed in `experiments/tfpt-discovery/`, pinned in `rh/INVENTORY.json` and re-verified by this suite.

Theorem 14.1 (Density martingale and moment dictionary; v978)**.** *On any finite genealogy tree with node mass $A(v)$ and leaf count $n(v)$, the normalized density $X(v) = d(v)/d(\mathrm{root})$ with $d(v) = A(v)/n(v)$ is a nonnegative martingale under the count path measure $P(c \mid v) = n(c)/n(v)$, using* only *mass conservation $\sum_c A_c = A_v$. With m leaves of masses y_i, total Y, $M_k = \sum_i (y_i/Y)^k$ and $q_{\max} = \max_i y_i/Y$:*

$$E[X_\infty^2] = mM_2, \qquad E[X_\infty^3] = m^2 M_3, \qquad \max X_\infty = m\, q_{\max},$$

so the terminal target $\sum q^3 \le C(\log m)^A$ is *a cubic martingale-moment statement. The tilted tower $E[X_\infty^3] = E_Q[\prod_k \Gamma(V_k)]$ holds with tilted steps $\tilde{p}_c = p_c R_c^3/\Gamma(v)$ (the untilted reading is false — exact-Fraction counterexample); the hand-off $E_{3h} \le (mq_{\max})^2\,\mathrm{msh}$ is exact for every heavy leaf subset; every two-child fold obeys the pair ceiling $\Gamma \le 4$ (from $4R - R^3 = R(2-R)(2+R) \ge 0$ on $0 \le R \le 2$), the source of the K1 constant $R_{\mathrm{ALG}} = 4^{1/3}$.*

Theorem 14.2 (Pinning; v980)**.** *On the rank-2 spectral model the pair block of $(I - E)^{-1}$ inverts to $p_2' = m(g_{21}a_2^2 + b_2^2)/\Delta t^2$, $q_2' = m(g_{21}a_1^2 + b_1^2)/\Delta t^2$, $c_2' = m(g_{21}a_1a_2 + b_1b_2)/\Delta t^2$ with $p'q' - c'^2 = m^2 g_{21}/\Delta t^2$ (Lagrange). Under orthonormal top-2 geometry ($\Delta t^2 = 1$) the dressed characteristic polynomial factors as $(t-m)(t-mg_{21})$: the smaller dressed eigenvalue deficit* equals the margin identically, *and $r_2' = 1 - c'^2/(p'q') = g_{21}(t_2 - t_1)^2/((g_{21} + t_1^2)(g_{21} + t_2^2))$ (the r345 two-level formula). Together with the one-line identity $m_2'(p' + q' - m_2') = p'q' - c'^2$ (Vieta), the rate-equality theorem (flat geometry $\Rightarrow$ dressed ratio = margin ratio exactly) and the ρ_K identities, the L^* margin-law chain is complete and typed: weights dictionary, deficits candidate law, pinning theorem, $\rho_r = \varphi$-difference computable pointwise.*

Theorem 14.3 (Borodin particle–hole duality; v981)**.** *On a discrete support of $S = 2N - 1$ atoms with positive weight u (half filling: $\lceil S/2 \rceil = N$, an integer lemma — the r228 law* is *Borodin's complementary rank condition), the rank-N projection kernel and the rank-$(N-1)$ kernel of the reciprocal dual weight $u_j^\vee = 1/(u_j P'(x_j)^2)$ satisfy*

$$\Pi_N^u + G\,\Pi_{N-1}^{u^\vee} G^{-1} = I, \qquad G = \mathrm{diag}\big(1/(u_j P'(x_j))\big)$$

(rational conjugator, no square roots). Restricted to any pair Y, $E = G_Y(R^{-1} - I)G_Y^{-1}$ with $R = (\Pi_{N-1}^{u^\vee})_{Y,Y}$, hence

$$L^* \iff \lambda_{\max}(E) < 1 \iff R > \tfrac{1}{2}I, \qquad \textit{margin} = 2 - 1/\lambda_{\min}(R),$$

and $p/q = 2 - (R^{-1})_{kk}$ exactly. On the cosine lattice the dual weight is exactly $u^\vee \propto c_j(1 - x_j)/|f|$: the original weight carries $+\log|f|$, the dual $-\log|f|$ — the r354 anti-correlation of the two φ blocks is duality algebra *by design.*

Theorem 14.4 (The derived matched Dirichlet frame; v982)**.** *From $\Lambda(s,\chi) = (q/\pi)^{(s+a)/2}\Gamma((s+a)/2)L(s,\chi)$ the archimedean density on the critical line is*

$$F_A^\chi(\xi) = -\log(\pi/q) + \mathrm{Re}\,\psi\Big(\tfrac{1+2a}{4} + \tfrac{i\xi}{2}\Big)$$

for $a \in \{0,1\}$ — parity and conductor enter only through the functional equation; at $a = 0$, $q = 1$ this is *the zeta dictionary (bitwise); the conductor shift is exactly $\log q$; the Lag-space parity kernel is $e^{-w/2}$ (even) $\to e^{-3w/2}$ (odd).*

Remark 14.5 (The binding Frame-A restriction; v979). The first certifying three-arm cover ($0/51$, $m_0^* = 10^{22.6}$), the P02 predictor ($51/51$) and the growth ceiling $C_{\mathrm{FAB}} = 14.93$ (fresh tranche $0/6$; polylog $m_0^* = 10^{18.9}$) are census-certified at frozen constants — and *all of it is*

Frame-A census (binding, r353): the second construction family breaks the FAB ceiling (+21 %) and kills the reserve floor at $m \sim 760$; the sliding coverage is *finite*. The one cross-family survivor is K2 ($F_A B\, g_{\mathrm{rel}} \leq 11.87$), certified NU-free over three window aspects (r355; the gap geometry is the ν-free coordinate, its source chain exactly closed with vacuous census caps) and 0/126 on the second arithmetic (r357) — two families, three aspects, three arithmetics. The data-free K1 formula $R_{\mathrm{ALG}} = 4^{1/3} \to 8/5$ (interior, exact cube comparisons) and the mesh identity $h - \nu u \in (0, 3/2]$ (exact) are the wave's constructional-algebra additions.

Remark 14.6 (Honest verdicts and the lane status after wave 14). The Borodin duality's own sealed verdict is DUALITY-REPARAM-ONLY: no dual block passes the carrier clauses, the equivalence is a *reparametrization* of the open condition — the L* lane goes **final** to the specialist memo (`rh/problem/lstar_problem.tex`), with the one-object question halved: *why is the difference of the by-design-coupled blocks* $\sim 500\times$ *small* — now a question about local dual two-point data (RESERVE-LOCALIZED: the local dual 2×2 block predicts the log reserve without readback), and with the Lubinsky-type AC hard-edge class measurably excluded ($a_{\rho K} = 1.4222$ vs the constant prediction). The matched frame moves the living-world evidence from $n = 1$ to $n = 3$ arithmetics on 126 windows and retypes the r330 splits as frame artifacts, while the zeta-graded signatures (exact half-filling pinning, saturated reserve, φ suppression $439\times$ vs $3\times$, counting constants) type the L* margin legislation as a statement about *how close zeta's world sits to the wall*; the matched scramble breaks — the wall needs the arithmetic. The terminal lane ends unchanged at the MEASURED composition (Section 13.2). Lean: *no change ships with this wave* — the duality is a reparametrization of the existing hole `lstar_subordination`, not a new target form; the two true window-local holes are unchanged. No RH claim, no GRH claim, no L* claim.

Certification. Ledger: `PRIME.TERMINAL.DENSITY_MARTINGALE.01 [E]; PRIME.L2.COVER_GROWTH_K2.01 [E]; PRIME.LSTAR.MARGIN_CHAIN.01 [E]; PRIME.LSTAR.DUAL_HOLE.01 [E]; PRIME.WORLD.DIRICHLET_MATCHED_FRAME.01 [E]; PRIME.LSTAR.SUBORDINATION.01 [O update]; PRIME.L2.RENYI3.SLIDING_BOUND.01 [O update]`. Module: `verification/v978_terminal_density_martingale.py; verification/v979_cover_growth_k2.py; verification/v980_lstar_margin_chain.py; verification/v981_lstar_borodin_duality.py; verification/v982_dirichlet_matched_frame.py`.
Rounds r334–r357 (notes DCLXXXI–DCCIX). The wave-14 modules embed no probes: each re-derives its exact layer from scratch and gates the frozen record aggregates; the sealed probes (fold_capacity r334 `d5698a1d`, edge_packing r335 `950c9b9e`, lstar_parity r336 `9e4bf4f1`, fold_martingale r337 `617a83a6`, fold_density r339 `822d6bff`, cauchybinet_hall r340 `86587b48`, fold_bellman r341 `7a9cf6b3`, pair_extremal r342 `b09f8ccd`, pair_coupling r343 `9ffc2705`, fold_two_scale r344 `06375c3a`, gap_ratio r345 `1f99235a`, fold_cover r346 `95559d5d`, delta_alpha r347 `bd1aa7f3`, delta_source r348 `307814e9`, thirdarm r349 `9b593e63`, alpha_source r350 `c3998c87`, qmax_growth r351 `67102e4c`, rhor_source r352 `dc6bbd2c`, second_family r353 `bd89e331`, phi_wander r354 `f9db84da`, k2_source r355 `1f14bd93`, borodin_dual_hole r356 (freeze `5d277d57`), dirichlet_matched_frame r357 `4bf1a94b`) stay experiments-side, pinned in `rh/INVENTORY.json` and re-verified by this suite; the r338 door-2 revision and the r334–r336/r340/r343/r345/r349/r353 honest negatives are consolidated as prose; no round is in flight at this cut.

15 The measured walls and honest negatives

The program's negative results are load-bearing: they are typed, with break loci, and every future proof attempt must route around them.

Attack class	**Measured failure (break locus)**	**Round**
Compression / spectral compactness	inkompressible on the exact coordinate: no compact eigenspace ($K80 = 20$), participation dimension $\approx 0.195N$; arithmetic worlds are spectrally *spread*, destroyed worlds compact	r253/r254
Global gauge constant	no constant survives the quotient test; the missing layer drifts $-1.1..-2.1$ decades end-to-end and is orientation-relevant	r252
Band locality (node bands)	transitions are dense (0.55–0.63 of degrees), drift anti-correlated with them	r255
Mass majorants (sup / Cauchy–Schwarz / triangle on masses)	0/42 coverage, median $\times 490$ excess, break locus $0.97N$ (terminal h collapse with $+8.2$-decade rebound)	r258
Level crossing	parity branches cross 5–16 degrees too early; MAIN w9 crosses 45 times without a single true flip	r259
One-swap statistics	the critical swap exists (0.2 dec) but MAIN carries the same structure at the same degrees; gap distributions fully overlap	r259
Subtraction-free positivity (LGV)	no-go at depth $k = 1$, single Gram entry, every world including MAIN	r261
Global block involution	4 unpaired negative configurations at $n = 2$ already (65 152 pairings machine-checked)	r261
Transfer cone	best cone certificate is *identical* to terminal budget positivity: wall-equivalent, no new mechanism	r261
Smooth (Szegő) dressing	the D-layer is essentially singular ($p = 0.997$, $\exp(c/(z-s))$ type, up to 380 decades): not σ°-pairable	r251/r253
Bordered spectral bound	$f^\mathsf{T}(I-Q)^{-1}f < 1$ is measured *false*: $g(1) = 6.06..15.41$ on 42/42; both augmented border variants cross the wall	r257/r265
Naive s-coordinate	closed form $D_n(s) = B - s^2 S_{n-1}$: the monotonicity is a trivial corollary and the endpoint *is* the budget inequality (restatement)	r265
IIKS s-family	the cross-ratio ODE $X'(s) = -\\|(I-sQ)^{-1}f\\|^2$ is exact and fully definite (0/17849 indefinite terms) — but its endpoint criterion is exactly the blocked wall bound: DEFINITENESS__WALL__EQUIVALENT	r265
Fixed head (finite-dimensional core)	no fixed $r \leq 8$ exists; the best candidate's reserve 6.5×10^{-6} sits four decades under the sealed bar and decays in the r286 law class; $\lambda_{\min}(S_H(8))$ tracks the wall margin to 4 digits — the hoped-for simplification is *empty*, S_H carries the whole problem	r307
Paired-cone induction (architecture B)	B1–B4 adjudication: the sealed mass-ratio cone is sound but certifies only 56 %; no real induction $K_j \to K_{j+1}$; world-blind at deg 8 — the whole EPST/SCR indefiniteness sits in the negative budget step (= the r243 wall reading)	r309
Block-Green feasibility as a new discriminator	the r308 world separation is *entirely* the budget sign: the Δ-sparsity graph is chordal on every size, the trivial dual $Y_t = e_t e_t^\mathsf{T}$ closes the stalls two-sidedly — a chordal restatement of the known wall reading	r311
Constructive PSD-G rule (Lane A, cone language)	COEFFICIENT__SIGN__WALL: the identity forces negative coefficients exactly on the wall coordinates (Farkas mass on the border/budget row, t-share 0.583/0.725; exact rational certificates on SM1/MINI16); mecha-	r312

Attack class (wave 4)	Measured failure (break locus)	Round
Border-dressed resolvent identities	exhaustive census (four augmented-first candidates + 72 Plücker/Dodgson combinations): no identity in the class; every definite direction is a drain restatement, the square never sits on the target side	r266
Eventual triangle	$\mathrm{sp}(g, N) = +0.31$ under the bar, exceptions in all terciles, the deepest rung kz52 ($N = 878$) itself an exception — no magnitude-separation law	r263
Exact-identity routes for the exception scalar	K1 bordered-Schur = target restatement + support geometry (terminal ratio stays positive on flipped controls); K2 squares on the drain side; K3 paircorr miniature ($d2$ fires 42/42)	r263
Euler mechanism for the cancellation decay	no surgery collapses γ_{true} (all 12 stage medians $\geq +0.20$; free permutation $+1.40$): the $N^{-0.45}$ decay is generic root-scale, not an arithmetic fingerprint	r273
KYP / state-space memory	two-sided structural no-go: every uniform quadratic memory exact-rationally infeasible (o1 fiber forcing $p_{11} = 0 \Rightarrow p_{12} = 0$; o2 descent window $\sum b^2 \leq 0$), and the unique admissible Riccati memory is target-inverse (sp = +1.00, falsely certifies SCRAMBLE)	r275
Raw atom-Sturm winding	MAIN breaks $c_n \equiv n$ at 56/48 at positive h: the positive-measure separation theorem fails genuinely for the signed comb — not the Maslov object (R2 is)	r277
Uniform metric stability constant	the local law exists but θ^* sits 25–170× below the smallest measured dose; L_D window-specific (17.8–309): perturbative only, no uniform lemma candidate	r278

The consistent reading (r259/r261): in both cores the object is a *coherent totality*, not a dominant component — the true sign is carried by coherent cancellation over the configuration ensemble (~ 5.7 decades deep at $n = 6$ already), which no dominance model at any k-swap resolution sees. This is the consistent explanation of why two years of single-component attacks measurably failed. The sharpened residual question after r265: *no admissible s-coordinate carries both the right endpoint and a definite dynamics* — a genuine s-tool must couple the border drive into the resolvent structure without reducing to the wall bound (the r266 question, Section 17).

Round r607 (probe `event_lindblad_twokey_probe.py`, SHA `3c1db81b`, SPEC `3abe6b9b`, 15/15) is a method no-go: KILLED(RECOORDINATIZATION). Lindblad CP $\Leftrightarrow \lambda_{\min}(Q_W)$ to 10^{-15} in TRUE/SCRAMBLE/EPSTEIN/WPERM; the diagonal compensator $\Sigma\lambda$ diverges $\sim 2e^{X/2}$; Euler–Pick Gram needs 10 scalars; labels are decorative. Experiments only. **No RH claim.**

(2026-09-02) Prolate hyperbolicity route. Connes–Consani–Moscovici, `arXiv:2310.18423`, the proposition after Lemma 3.3, gives $F_\mu(\mathcal{E}(\psi))(s) = R(s) \cdot \Xi(s)$, so $\hat{k}_\lambda$ contains Ξ as a factor. Real-rootedness of $\hat{k}_\lambda$ is therefore RH together with real-rootedness of R_λ, and the ratio F_4/F_0 cancels Ξ. Typed KILLED(FACTORIZATION); no probe needed. **No RH claim.**

(2026-09-02) Herglotz closure reduction. The statement "finite Herglotz functions $m_j \to -\Phi'/\Phi$, $\Phi(z) = \xi(\frac{1}{2} + \sqrt{z})$, on a set with an accumulation point in $\mathcal{U} = \{\mathrm{Re}\sqrt{z} > 1/2\} \cap \mathbb{H}$, bounded at one point $\Rightarrow$ RH" is correct classical analysis (normal family of Herglotz functions, identity theorem, pole contradiction), but its hypothesis implies RH. Recorded as a reduction. Round r611 (probe `connes_observable_aubin_nitsche_probe.py`, SHA `93607746`, SPEC `647335a6`, 12/12) returns OBS_NOT_CONVERGING: hybrid Q_λ has $\lambda_{\min} \approx -10^{-5}$ for $L \geq 0.5$ (r606 discretisation crash removed, t_{cut}-stable); $\eta_{\text{obs}}(1.2)$ on TRUE is $0.667/6.3 \cdot 10^{-3}/3.7 \cdot 10^{-4}$ at $L = 0.3/0.4/0.5$ then *rises* $1.6 \cdot 10^{-2}/5.6 \cdot 10^{-2}/9.0 \cdot 10^{-2}/0.150$ at $L = 0.6$–1.0 ($p = -8.1$). OFFLINE $\approx$ TRUE (0.109 at $L = 1.0$), so the observable is blind to an injected off-line zero at $\gamma = 20$; WPERM 0.69, SCRAMBLE 0.20; identity (8) residual $4.6 \cdot 10^{-11}$. The Herglotz/Aubin–Nitsche closure route (Solution 1) is STOP as a lane; `PROLATE.GABOR.ALIGN` is moot. r606b is CLOSED (INCONCLUSIVE(evals-not-converged, disc-disagree> 5pct); superseded by r611 and the KILLED(factorization) typing of the prolate route; diagnostic only). Experiments only. **No RH claim.**

(2026-09-02) Dyadic heat-semigroup criterion. For the Lean pure-Gabor family the cross term $2e^{(\delta^2-\omega^2)/2a}$ carries residual $-\delta^2/(2a^2) \neq 0$, so $U = \sqrt{a}\,Z$ is not a heat solution. The dyadic-in-a heat-semigroup reduction is unavailable and is replaced by the countable rational-packet criterion (r612). Parent analysis 2026-09-02; confirmed numerically at r608 (rel $3 \cdot 10^{-9}$ on traveling lobes). **No RH claim.**

Round r613 (probe `prime_e8_ordered_herglotz_probe.py`, SHA `c89fc1ce`, SPEC `01b4a8be`) returns NO_IDENTIFICATION: the Weyl function is Herglotz (min Im-eig $+3.5 \cdot 10^{-3}$), $(JH)^2 = 0$ exact; at $\varepsilon = 0$ all jumps commute so order is decorative (max 10^{-14}, same lesson as r607); at $\varepsilon = 0.05$ order/labels are non-decorative (16.6/12.0) but residuals at Euler points TRUE 0.66–2.5 agree with SCRAMBLE/WPERM 0.66/0.64 and 2.36/2.36; the twirl chain $E_{\text{full}} < E_7 < E_3$ breaks ($4.5 > 3.1$). The ordered-E8 Herglotz canonical system (Paket C, `PRIME.E8.ORDERED.HERGLOTZ.01`) is dead as a lane. Experiments only. **No RH claim.**

Round r615 (probe `semilocal_firststep_probe.py`, SHA `3a0741de`, SPEC `04f9c9ec`, 26/26) confirms the r496 certificate obstruction on every window that contains a prime: the shift cross-block norm is 0.50 for $N = 40/80/160$ (no decay), and the Schur requirement is $R = 0.25/\lambda^* = 1.4 \cdot 10^3$ at $L = 0.40$ up to $4.45 \cdot 10^6$ at $L = 0.549$. First-prime-step windows are therefore not certifiable by the compact-tail/Schur route. Experiments only. **No RH claim.**

Typed no-go `E8.COXETER.REGULARIZED_SPLIT.NO_GO.01` (round r617, probe `e8_coxeter_euler_completion_` SHA `6ac6bab2`, SPEC `5aa3935b`, 45/45, COMPLETION_EXACT): $\det_2$ removes exactly the trace term; the RH content is the analytic continuation of $P(s) = \sum \mu(k)/k \log \zeta(ks)$; constructions that control only the Hilbert–Schmidt part cannot isolate $1/\zeta(s)$ (this is the conceptual content of r607/r613). Fence: "The trace-free completion is zero-free and pole-free in $\operatorname{Re} s > 1/2$. The splitting into the scalar zeta channel and the Coxeter channel is open and RH-equivalent. No RH claim." Promotion to `verification/` (v1018/v1019) is in progress. Experiments only.

Round r618 (probe `jensen_compiler_rigidity_probe.py`, SHA `29ccf1c7`, SPEC `bc8d8624`, 43/43) returns COMPILERS_STRUCTURALLY_RIGID. Two no-gos: (A) Seifert-compiler rigidity — the Coxeter monodromy spectrum is a congruence invariant (Φ_{30} for 12 random MSM^*); Cayley-transformed Jensen polynomials $\mathcal{C}_{n,8}$ of Ξ vary with n and miss the 30th-root grid by ≥ 0.0168; blocks $d = 2..7$ give Φ_2^2, $\Phi_2\Phi_3$, Φ_5, $\Phi_2\Phi_8$, $\Phi_3\Phi_{12}$, $\Phi_2\Phi_{18}$; (B) rank-one MSS mixed characteristic polynomial equals $\det(xI - \sum A_e)$ exactly; required coefficient ratios c_{n+1}/c_n ($0.0231 \to 0.0169$, $n = 0..6$) are nonlinear in zero power sums. Both compilers are generic real-rootedness certificates; existence for all (n, d) is equivalent to RH; the E8 data are RH-neutral. Experiments only. **No RH claim.**

Round r621 (probe `p2_digamma_duplication_probe.py`, SHA `dff2e1b3`, SPEC `78f47c87`, 37/37) returns DUPLICATION_CONSTANT_ONLY. The duplication identity $\psi(z) + \psi(z + \frac{1}{2}) = 2\psi(2z) - 2\log 2$ holds to $2 \cdot 10^{-25}$ and the operator identity $\mathrm{ARCH}_{1/4} = \mathrm{ARCH}_{\mathbb{C}} - \mathrm{ARCH}_{3/4} - 2\log 2 \cdot G$ to $2 \cdot 10^{-15}$, but the prime-2 operator $P_2 = -w_2 T_{\log 2}$ versus $\operatorname{span}\{\mathrm{ARCH}_{3/4}, G\}$ has residual 0.997–0.998 on the whole odd space (HS cosine ≤ 0.005, fitted $\beta \approx 0.06$ not $-2\log 2$). Translation is not a constant. The $\mathbb{Q}(i)$ form $\mathrm{POLE} + \mathrm{ARCH}_{\mathbb{C}}$ stays positive without its norm-2 prime ($\lambda_{\min} + 0.59$ at 0.549); $\Gamma_{\mathbb{C}}$ without the constant fails at 0.38 like ζ — the conductor constant carries, not the 3/4 channel. No-go: the prime-2 repair is not a duplication factor of the archimedean operator. Experiments only. **No RH claim.**

Round r622 (probe `support_darboux_probe.py`, SHA `845bb937`, SPEC `27ffd017`, PAYLOAD `e1031d17`, 64/64) returns TRANSPORT_FEASIBLE_NORULE. On the scaled family $\widetilde{Q}(L) = U_L^* Q_L U_L$ the transport identity $\partial_L \widetilde{Q} + X^* \widetilde{Q} + \widetilde{Q} X + c\widetilde{Q} = Y^* Y$ is feasible at $X = 0$ for the scalar $c = c_0(L) = -\lambda_{\min}(\widetilde{Q}^{-1/2}(\partial_L \widetilde{Q})\widetilde{Q}^{-1/2})$ (finite: 26 at 0.25, spike 368 at prime entry, 1614 at 0.549), but there is no two-term closed-form law for $c(L)$ (rms 347; fails at the kink and at 0.549); the forbidden $X_0 = -\frac{1}{2}\widetilde{Q}^{-1}\partial_L \widetilde{Q}$ is not in the source algebra (projection residual $0.88 \to 1.0$); the algebra produces rate 0 versus required 31.8/46.0. The prime is not the PSD obstruction — it spikes c_0. No-go: Darboux transport in the source algebra yields no rule and no rate. Experiments only. **No RH claim.**
Round r619 (probe `support_relay_census_probe.py`, SHA `fcfa7771`, SPEC `7299737d`, RESULT `e0379485`, 12/12) returns RELAY_UNIVERSAL, but the RELAY_UNIVERSAL criteria ($\Delta_j \geq 0$, $g < 0$, newest-weight repair) are consequences of $Q_j \succeq 0$ (RH-consistency), not a mechanism. Relay-as-mechanism is therefore a no-go: the informative content is the nearly constant leads, the control breaks at entry+lead, the super-exponential λ^* decay (r619 log-slopes $-77/-79$ at $L = 0.9$–1.4 were measured at fixed $N = 80$ beyond the SVD floor and are N-artifacts, corrected by r630b to -50.6 on 0.3–0.6 and -116 on 0.6–1.0; the RELAY_UNIVERSAL verdict and L_{det} table are unaffected; the constant-lead "greedy law" $\Delta \approx 0.015$ is WITHDRAWN by r635), and L_{det}. Windows with $L \gtrsim 0.5$ are hypersensitive off-line detectors, so relay steps beyond $L \approx 0.5$ are RH-in-a-box statements, not elementary finite problems. Experiments only. **No RH claim.**
Round r623 (probe `semilocal_p2_dilation_probe.py`, SHA `d1bec728`, SPEC `5890676d`, 128/128) returns P2_DILATION_STRUCTURALLY_BLOCKED. The 2-adic channel of the Weil form is the multiplier $m_2(t) = \log 2 \cdot (1 - P_r(t \log 2))$ with $r = 2^{-1/2}$ and P_r the Poisson kernel (boundary value of the de Branges–Rovnyak kernel of the Blaschke factor $b(z) = (z-r)/(1-rz)$); one Euler factor generates the whole rung ladder 2^k (active rungs: [2] for $L \leq 0.65$, $[2,4]+3$ for $L \in \{0.72, 0.80\}$); identity residual $4.9 \cdot 10^{-12}$. Obstructions to the proposed PSD block $\left[\begin{smallmatrix} A_S & B^* \\ B & C \end{smallmatrix}\right]$ with Schur complement Q_L: the ancestor POLE + ARCH has $\lambda_{\min} = -0.078$ ($L = 0.40$), -0.229, -0.367, -0.430 (0.549), -0.536, -0.636, -0.773, -0.926 (0.80) and therefore cannot be a PSD principal block; the 2-channel operator M_2 is indefinite (n_+/n_- 9/9 at 0.40 … 14/10 at 0.80) with numerical rank/N 0.75–1.0 (not rank one) and therefore cannot be $-B^*C^{-1}B$; the generic dilation $\left[\begin{smallmatrix} Q+M_{2-} & \sqrt{M_{2-}} \\ \sqrt{M_{2-}} & I \end{smallmatrix}\right]$ has $\lambda_{\min}$ tracking $\lambda^*(L)$ ($1.82 \cdot 10^{-4}$ vs $1.81 \cdot 10^{-4}$ at 0.40) — tautological. No manifest local SOS in the named factors: $\|N\|/\lambda_{\min}(P) = 2.75$–$5.27$ (negative part never dominated). Literature: Connes–Consani 2021 (Selecta) Thm 1 positivity only for support $[2^{-1/2}, 2^{1/2}]$ (archimedean); CCM arXiv:2310.18423 §4.6 Sonin space without a positivity theorem; arXiv:2403.01247 Jacobi matrices for $S = \{\infty, p\}$ ($q = 1/p$); ζ-cycles numerical — no published unconditional first prime step. The semilocal dilation lane is not opened. Experiments only. **No RH claim.**
Round r625 (probe `iiks_vanishing_metric_probe.py`, SHA `973a1ce2`, SPEC `7a4257e3`, NUM `2d066e2d`, 19/19) returns IIKS_NO_METRIC_IN_CLASS, recorded in prose as NOT_TESTABLE_IN_CORPUS_FORM. For the corpus's discrete IIKS/CD object the jump $J_s = I - 2\pi i s\, f g^{\mathsf{T}}$ on the real CD nodes fails Zhou's requirement $J^* = J^{-1}$ (rel $2.56 \cdot 10^{-2}$ on WPERM kz9), and $\det J = 1$ at s^* while $\tau(s^*) = \det(I - s^* E) = 0$ ($\log|\tau| \approx -38$): this J is not the RHP jump equivalent to $\tau(s) \neq 0$. The $H = I$ "dissipativity" $J + J^* \succeq 0$ (holds on TRUE with min eigenvalue 1.93–1.99 *and* on WPERM despite τ-zeros) is the wrong sufficient condition and is dropped. Best source-explicit contractive metric $H = \operatorname{diag}(\text{arch}^{1.50}, \text{arch}^{1.50})$: TRUE worst margin $-8.17 \cdot 10^{-10}$ (first pass $-3.36 \cdot 10^{-7}$; bar -10^{-12}), WPERM $-7.9 \cdot 10^{-9} \ldots -4.4 \cdot 10^{-4}$, EPSTEIN $-2.7 \cdot 10^{-2} \ldots -6 \cdot 10^{11}$, SCRAMBLE $-3 \cdot 10^{6} \ldots -9 \cdot 10^{57}$; controls fail pointwise before their τ-zero (5/5). The vanishing-lemma route is not testable on the corpus object. Experiments only. **No RH claim.**
Round r626 (probe `xi_finitefree_collision_probe.py`, SHA `c4750063`, SPEC `c7cef974`,

NUM `f533d168`, 51/51) returns COLLISION__DETECTOR__CALIBRATED__NO__BARRIER. The identity $d/dt \log|\mathrm{Disc}(P_t)| = 4 \cdot \Phi_N(P_t)$ along the de Bruijn–Newman heat flow (rel error $8.9 \cdot 10^{-11}$) makes the "collision barrier" $\int \Phi \, dt < \infty$ equivalent to Disc $\neq 0$ along the path (no independent content). Taylor sections of Ξ are not real-rooted at $t = 0$ ($N = 6/8/12$ have 0 real roots; the γ_1 pair becomes real only at $N = 16$–20); $t_0(N)$ is always a spurious far-pair collision; inner collisions occur at $t_c > 0$ for every N. Detector calibrated on the hyperbolic control $S_N = \prod(1 - x^2/\gamma_k^2)$: $\varepsilon_{\mathrm{crit}} = -3.65 \cdot 10^{-4}$ (a_1, $N = 8$), $t_c = 0.597$, $\log_{10}|\Phi|$ up to 7.90. Turán on $c_n = n! a_n$ holds (min ratio 1.02); Newton/Laguerre on the sections fails (0.51); no simple coefficient inequality implies inner Disc > 0. Experiments only. **No RH claim.**

Round r627 (probe `af_twomoment_optimizer_probe.py`, SHA `da1a9118`, SPEC `9b95f564`, 17/17) returns NO__GAIN. Contract `PRIME.INERTIA.TWOMOMENT.OPT.01`. The A–F two-moment functional $R(\psi) = [\int \psi^2 + \iint |u - v| \psi(u)\psi(v)]/(\int \psi)^2$ with $p = 2 - R$ on $\operatorname{supp} \psi \subset [-\frac{1}{2}, \frac{1}{2}]$ has Euler–Lagrange $(I + K)\psi = \lambda \cdot 1$, hence $\psi'' + 2\psi = 0$ and unique critical point $\psi^* = \cos(\sqrt{2}\, t)$ (Montgomery–Taylor), $p^* = 0.6725007037$ ($N = 8$–64 identical, ω-scan optimum $\sqrt{2}$); positivity slack $\min 0.760 > 3/4$. The proved ceiling 0.6818287 bounds the richer two-moment + on/off-block certificate, not the window optimum; the gap $9.3 \cdot 10^{-3}$ is not closable by window optimization. Experiments only. **No RH claim.**

Round r628 (probe `window_box_verifier_probe.py`, SHA `9c99ed6a`, SPEC `36c0e66a`, NUM `84aaf685`, 12/12) returns COMPRESSION__CERTIFIED__BOX__NUMERICAL. Contract `PRIME.WINDOW.BOX.VERIFIER.01`. Interval-certified (40 digits, interval Cholesky/Weyl) positivity of the finite window compression: $\lambda_{\min} \geq 1.05 \cdot 10^{-7}$ ($L = 0.55$, $N = 24$, primes $\{2, 3\}$), $\geq 6.1 \cdot 10^{-11}$ (0.65, $N = 40$), $\geq 9.3 \cdot 10^{-17}$ (0.80, $N = 40$, $\{2, 3, 4\}$) — cancellation depth up to $6 \cdot 10^{14}$. A rigorous zero-free box is *not* closed: the unconditional on-line majorant via Trudgian $|S(T)| \leq 0.112 \log T + 0.278 \log\log T + 2.51$ (JNT 134, 2014) is $\sim 10^7$ too loose; the numerical box is nonempty ($L = 0.80$: $\beta_{\min} = 0.55$ for $\gamma \leq 25$ and 50/60; 0.70 at 30; 0.80–0.90 at 35–45); $\log \Gamma_{\mathrm{num}} = -1.24 + 6.86\, L$ ($\sigma = 0.1$). Missing lemma: a local zero-density bound or enumerated zeros (then Booker/Turing-equivalent). Typed as the explicit-formula verification method (Booker, Exp. Math. 15 (2006)) in window form. Experiments only. **No RH claim.**

Round r629 (probe `certificate_class_atlas_probe.py`, SHA `4dae1fbe`, SPEC `0bf7c64f`, 15/15) returns ATLAS__NO__NEW__UNCONDITIONAL__GAIN. Contract `PRIME.CERTIFICATE.ATLAS.01`. LP over hypothetical zero configurations: A1 two-moment reproduces $1 - p_0 = 0.3181713$ ($p_0 = 0.6818287$); A2 $+N(T)/S(T)$: unchanged; A3 + prime-free window positivity (CC 2021): redundancy 0 (min slack $2.2 \cdot 10^{-4}$); A6 + Huxley zero density: unchanged (off-line mass forced to $\sigma \leq 1/12$, already at $\sigma \to 0$). Conditional rows: GUE $k = 3 \to 5/6$, $k = 4 \to 31/36$; pair correlation $\alpha_{\max} = 1.5/2/\infty \to 0.7046/0.7159/0.7500$. Reading: only correlation input beyond support 1 buys proportion. Experiments only. **No RH claim.**

Round r632 (probe `jet_deflated_ldet_probe.py`, SHA `63ecb68f`, SPEC `31a29760`, RESULT `6299cedc`, 14/14) returns JET__GAIN__SMALL. Contract `PRIME.WINDOW.JETDEFLATION.01` (reviewer-proposed). Baseline reproduces r619 L_{det} (0.55/0.50/0.80). Deflation $\hat{f}^{(r)}(20) = 0$: $m = 0$ no gain ($G = 0$), $m = 1$ $G = 0.40$ (L_{det} $0.55 \to 0.95$ at $(0.6, 20)$); control at $\gamma = 50$ moves 0.10–0.20; deflating the first eight true zeros: $G = 0.65$ and sector Ritz floor at $L = 0.6$ jumps $1.76 \cdot 10^{-9} \to 2.02$. Coverage kill: deflating all zeros up to $\Gamma = 50$ already needs $\dim K/N = 0.50$ (above Nyquist 31.8), $\Gamma = 100 \to 1.45$ (sector collapses), $\Gamma = 200 \to 3.95$ — sectorization is not a cheap RH-free statement (recoordinatization). Deflation at 20 does not touch the odd-sector failure (peak frequency 4.6/3.9). Experiments only. **No RH claim.**

Round r630 (probe `margin_law_symreg_probe.py`, SHA `5b3d1125`, SPEC `87febb97`, RESULT `a0765d7f`, 19/19, 166 s, dual-run identical) returns MARGIN__LAW__UNRESOLVED. Contract `PRIME.MARGIN.LAW.SYMREG.01`. First-run facts: $L_{\mathrm{cross}} = 0.3716$, double-precision SVD floor $L = 0.78$. Addendum r630b (tall QR, 40 digits, $N \in \{40, 60, 80, 120, 160\}$, $L \in [0.30, 1.40]$) decides N__LIMITED__ARTIFACT: $\lambda^*(L)$ converges in N for $L \leq 1$ (35/36 points, relative change $120 \to 160$ median $4.8 \cdot 10^{-2}$) but not for $L > 1$ (0/20); converged log-slopes -50.6 (band 0.3–0.6,

N-stable to 3 digits) and -116 (band 0.6–1.0, stable for $N \geq 80$); $\lambda^*_{160}(0.6) = 1.65 \cdot 10^{-9}$, $\lambda^*_{160}(1.0) = 9.93 \cdot 10^{-30}$; no converged holdout, descriptive quadratic $C = 95.5$ (neither $2\gamma_1 = 28.27$ nor $\gamma_1^2/\pi = 63.6$) — both the Widom-L^2 and the $-2\gamma_1 L^2$ readings unsupported. N2: the converged minimizer nulls 4707–4831 of 5000 zeros (frequency-sparse). N3 synthetic controls (α = fitted decay constant): true 169, constant-density 101, all ordinates scaled to first zero at 10: 105, at 20: 271, only first zero moved to 20: 172 — density-driven (Landau–Widom plunge from the unconditional $N(T)$ profile of the low band), not band-edge; RH-neutral. N4 dBN gap functional B_k on 4999 gaps: max 41.8, $p_{99} = 11.5$, median 2.13, fraction $B_k < 2 = 0.470$ (0.515 after compressing one gap by 0.3) — no universal gap barrier. Experiments only. **No RH claim.**

Round r633 (probe `frontier_followups_probe.py`, SHA `9f5287e7`, SPEC `ee7c7ba7`, RESULT `814abdf5`, 16/16, 18 s, dual-run identical) returns four follow-up verdicts on sealed objects. Contract `PRIME.FRONTIER.FOLLOWUPS.01`. F1 A_COMPACT_OTHER($p \approx 1.02$): $a_{\max}(\sigma, \gamma^*) = c\,\sigma^{1.024}(\log\gamma^*)^{-1.65}$ ($r^2 = 0.945$, $n = 24$), linear in σ, not σ^2 (mechanism: phase threshold $\cos(\sigma(\gamma - \omega)/2a) < 0 \Leftrightarrow a < \sigma|\gamma^* - \omega|/\pi$, not the amplitude gain); $a_0 = 0.720$ at $(\sigma, \gamma^*) = (\frac{1}{2}, 20)$, extrapolated 0.714 at γ_1; true zeros $\min_\omega Z(a, \omega) > 0$ for $a \in \{1, 5, 25, 100, 250\}$; adversarial $\sigma = \frac{1}{2}$ negative only at $a = 1$ ($Z = -3.86 \cdot 10^{-4}$ at γ_1). Parent typing: the compactification removes only the $a \to \infty$ end (RH-blind wide-frequency packets); the RH-relevant end $a \to 0$ stays unbounded; a Lean theorem would need an external explicit $N(T)$/gap bridge — cosmetic, not pursued. F2 FORWARD_INCREASES_MARGIN: Hellmann–Feynman equals Loewner to 10^{-15}; $c(L) = \partial_t \log\lambda^* = +0.315/+0.582/+0.764$ at $L = 0.40/0.55/0.80$ (λ^* $1.89 \cdot 10^{-4}/5.70 \cdot 10^{-8}/2.02 \cdot 10^{-17}$), $t^* = -3.18/-1.72/-1.31$ ($O(1)$, not vacuous); with injection $(\beta, \gamma) = (0.6, 20)$ at $L = 0.55$: $\partial_t\lambda^* < 0$. Typing: the sign discriminates on/off-line but the monotonicity runs in the proof-useless direction — detector, no lever. F3 RANDOM_EULER_REFUTED: $\lambda_{\min}(\mathrm{POLE} + \mathrm{ARCH}) = -0.0783/-0.430/-0.926$ at $L = 0.40/0.549/0.80$ (reproduces r623 to 10^{-3}); MC 200 Haar phase draws at $L = 0.55$: mean $\lambda_{\min} = -0.496$, std 0.317, 99.5% negative, vs true $\lambda^* = 5.70 \cdot 10^{-8}$; Baker: required $\delta = O(1)$ vs toy linear-forms bound $4.6 \cdot 10^{-8}$ — 7.3 decades. The Haar 99.5% negativity stands. The phrase "coherence $C(L) = 1$ / phase coherence" is *not* the mechanism — $C = 1$ is generic for minimizers and holds for the negative Haar draws too (r637). Typing: magnitude-only and Diophantine-approximation arguments are dead as a class (measured no-go; promotion candidate with r615/r619). F4 LEAD_LAW_INVARIANT: VOID (r635/r636) — based on the withdrawn $N = 80$ values $\Delta \approx 0.015$; Δ_q is a corollary of the margin collapse, not a separate invariant. Experiments only. **No RH claim.**

Round r635 (probe `relay_lead_precision_probe.py`, SHA `36c06826`, SPEC `28db9bef`, RESULT `3457e907`, 11/11, 1260 s, dual-run identical) returns LEAD_PARTIAL. Contract `PRIME.RELAY.LEAD.PRECISION.01`. r619's predecessor was the frozen D2 form (events $q' < q$ only); D1 (true minus q) implemented, identical here. mp Cholesky PSD-boundary bisection, $N \in \{80, 120, 160\}$, damped Legendre: converged leads $\Delta_2 = 0.0253$, $\Delta_3 = 0.0083$, $\Delta_4 = 0.0064$ (0.00606 at $N = 160$), $\Delta_5 = 0.0034$ (0.00275 at $N = 160$); $\Delta/s_q = 0.125/0.058/0.058/0.020$; $\lambda^*(L_q) = 1.36 \cdot 10^{-3}/5.93 \cdot 10^{-8}/8.72 \cdot 10^{-13}/8.69 \cdot 10^{-17}$; initial negative rate $r_q = 3.38 \cdot 10^{-2}/3.14 \cdot 10^{-6}/7.35 \cdot 10^{-11}/1.32 \cdot 10^{-15}$. Probe why: "positive N-stable leads only at small L_q (4/18 events; later events already indefinite at entry)". For $q = 7$–32 the probe reports $\Delta = 0$ with $\lambda_{\min} < 0$ already at L_q, conv = 0 ($q = 7$: $L = 0.973$, $\lambda_{\min} = -6.99 \cdot 10^{-11}$); the probe does not mark these as floor-limited. **Parent typing** (verbatim): (i) the r619 "greedy law" $\Delta \approx 0.015$ nearly constant across 18 events was an $N = 80$ double-precision artefact — WITHDRAWN; the converged leads decrease steeply ($0.025 \to 0.003$) and track the collapsing entry margin; (ii) for $q \geq 7$ the value $\lambda_{\min} < 0$ "at entry" is NOT a negativity of the true form (RH-consistency and the r628 certification forbid it: the true form for $L < L_7$ equals the D2 predecessor); it means $\lambda^*(L_7 \approx 0.97) \approx 10^{-25}$ lies below the accuracy floor of the 40-digit prime-side builder ($\sim 10^{-17}$–10^{-20}) $\Rightarrow \Delta_q$ for $q \geq 7$ is UNMEASURABLE at this precision, not zero. Not spacing-limited ($s_{31} = 0.0159$ but Δ

undetermined there). Experiments only. **No RH claim.**

Round r636 (probe `relay_lead_law_probe.py`, SHA `ee2deb46`, SPEC `6bf7a7ee`, RESULT `2a6bd720`, 6/6, 0.5 s, dual-run identical) returns LEAD_LAW_PARTIAL. Contract `PRIME.RELAY.LEAD.LAW.01`. Only four converged leads = training set, holdout empty; peak relative errors: M1 margin/rate linear 18.4 ($q = 5$), M1 quadratic 4.39 ($q = 2$–4 within 12–18%, $q = 5$ fails), M2 edge-sliver competition 1.96, M3 grammar $c_0 + w^3 + \lambda c^2$ 0.0026 (3 parameters on 4 points — interpolant, no evidence), M4 spacing 48.8 / αs 2.70 ($\alpha = 0.074$). No model $< 20\%$ on all four; spacing does not win; with four points there is nothing to derive. Parent typing: the "lead law" as a message is dissolved — Δ_q is a corollary of the margin collapse, not a separate invariant; r633-F4's LEAD_LAW_INVARIANT referred to the withdrawn $N = 80$ values and is void. Experiments only. **No RH claim.**

Round r637 (probe `relay_vote_map_probe.py`, SHA `fc5c1426`, SPEC `44bf67f8`, RESULT `540557ad`, 14/14, 27.7 min dual, mp 40 digits $N = 80$ ($N = 120$ check agrees at $L = 0.55/0.80$)) returns VOTE_MODE_SPECIFIC ($k_{\text{thr}} = 3$) + FLIP_COUNTEREXAMPLE. Contract `PRIME.RELAY.VOTE.MAP.01`. $C_0 = 1$ at all nine $L \in [0.40, 1.00]$; modes $k = 0, 1, 2$ unanimous at all L, $k \geq 3$ split (70/90 (L, k)); h_0 is R_q-EVEN-dominant on every overlap ($\|f_-\|^2/\|f_+\|^2 = 0.33 \ldots 0.99 < 1$): at the critical mode every present prime term ADDS positivity ($g = +(\|f_+\|^2 - \|f_-\|^2)$), consistent with the odd/edge failing mode of the predecessor (r619/r620) where the missing prime would have added. Flip test: 0/19 R_q-odd flips lower Q (first: $L = 0.40$, $q = 2$, $\Delta Q = +4.30 \cdot 10^{-3}$) $\Rightarrow$ FLIP_COUNTEREXAMPLE, the variational lemma in the odd sector is refuted. Controls: SCRAMBLE minimizer $C = 1$ at ALL L (incl. 5 events at $L = 1.0$), POLE + ARCH ancestor $C = 1$ at all L; WPERM splits from $L = 0.90$ (0.715, 0.444), EPSTEIN from 0.70 (0.941). Euler coherence testable only for 2 vs 4 below $L = 1$: agreement $27/36 = 0.75$, h_0 agrees at $L = 0.90/1.00$. Haar MC: $L = 0.45$ frac_neg 0.955 with frac_unanimous 1.000; $L = 0.55$ 1.000/1.000; $L = 0.70$ 1.000/0.865, $\operatorname{corr}(C, \lambda_{\min}) = -0.17$. Parent typing: unanimity is GENERIC — a geometric property of edge-localized low minimizers of translation-type forms (SCRAMBLE reproduces it at every L), not an arithmetic message; the coherence index C is not a positivity predictor (Haar draws unanimous yet negative). Experiments only. **No RH claim.**

The decoding programme (r635–r637) as a whole: the three candidate "messages" of the prime event log beyond the explicit formula (constant lead, unanimity, odd-edge defect type) reduce to margin collapse and edge geometry; no side channel measured. Experiments only. **No RH claim.**

Round r638 (probe `gabor_first_contact_selberg_probe.py`, SHA `2ab532bc`, SPEC `22a92842`, RESULT `58a36aff`, 14/14) returns KILLED(structural: selberg-convolution-is-hankel). The first-contact + Selberg square-completion attack on the pure-Gabor zero side $G = \text{POLE} - \text{PRIME} + \text{ARCH}$ fails its pre-registered Toeplitz gate: the Selberg term $\sum_N (\Lambda * \Lambda)(N)\, \psi(\log N)$ is a Hankel form in $(\log m, \log n)$ and admits no PSD completion over the operator-generated ψ (15/15 cells, best $\lambda_{\min} = -0.0097$ against tolerance -10^{-9}); the only squares compatible with the prime-side frequency support $\{\log n\}$ are Euler-local and cost $\geq 3.28\times$ the POLE+ARCH budget at $a \leq 0.3$, $\omega \geq 30$. At a first-contact point $\partial_a Z = \frac{1}{2}\partial_\omega^2 Z$ (to 10^{-16}), so the contact carries one differential condition, not two, and negativity is a down-set in a. On the true ζ the sign of the Selberg part at near-contact minima is indefinite and world-blind (MAIN 0.25–0.33, SCRAMBLE 0.22–0.31, WPERM 0.20–0.29). Experiments only. **No RH claim.**

16 External position: the rank-trace ceiling (r267)

Round r267 (probe `ranktrace_adjudication_probe.py`, 23/23, SPEC `b0437394`) adjudicated **arXiv:2608.13637** (Alpöge–Furman, Aug. 2026; argument found by an AI system, formalized sorry-free in Lean 4): a rank–trace inequality on a finite Weil-form compression plus Sylvester inertia gives *unconditionally* that at least 2/3 of the nontrivial zeros are simple and on the critical line, together with a *proved ceiling* $p_0 \leq 0.6818$ for every certificate built from only two trace moments against bandwidth-one test functions.

- **Same family, dual side.** Their compression atoms are *zeros* (near-tight frame: aspect 1, atom-norm spread 0, $R = 4/3$); ours is the *prime comb* (aspect ≈ 2 = half-filling, atom-norm spread 7–8 decades, $R \approx 73$–126). No basis change connects the regimes: NO_IMPORT, the frame mismatch is quantified.
- **The ceiling is our wall — in location.** Their binding constraint (off-diagonal prime-sum cancellation beyond diagonal control; pair correlation with support > 1) is exactly the locus of our measured PAIRCORR kill rule: the first *externally proved* form of our measured wall — but only for their two-moment certificate class.
- **Orthogonal in scope.** Our full comb coordinate and the bordered terminal scalar lie outside their certificate class; their ceiling does not close our lane, and their theorem does not bound our objects (CEILING_ORTHOGONAL, TWO_MOMENT_BLIND).
- **Source note.** Their `EnclOK` numerics are interval arithmetic outside the Lean kernel — their unconditional statement is Lean-checked modulo that enclosure layer.

The strategic consequence: the wall the program measured from the inside for two years now has an external, independently proved counterpart in the two-moment class. Passing it *provably requires* either pair correlation beyond support one or a certificate class outside two trace moments — which is precisely where the program's bordered terminal scalar already lives.
Round r616 (probe `inertia_highermoment_probe.py`, SHA `26ee62ae`, SPEC `fef4a72f`, 16/16) returns C_CAPPED as a follow-up to r267. The $k = 2$ reconstruction reproduces Alpöge–Furman 2/3 (ψ_0) and the ceiling 0.6818287. $\mathrm{Tr}(W^3)$ at bandwidth one is not unconditional ($\sum|\xi_j| \leq 3 > 2$ = Rudnick–Sarnak 1996 support for ζ; Hejhal 1994 hexagon under RH; Montgomery 1973 and Goldston–Montgomery 1987 under RH; the Montgomery–Vaughan diagonal $X^k \leq T^{2-\varepsilon}$ fails for $k \geq 3$), hence $p_3^{\mathrm{uncond}} = p_2 = 0.6818$. GUE ideal input n_+: $0.75 \to 5/6 \to 31/36$; simple-on-line $0.6818 \to 0.6818 \to 0.7222$. Finite moments never force negative index 0. Experiments only. **No RH claim.**
Round r639 (probe `lamzouri_hilbert_adjudication_probe.py`, SHA `61fbb4b9`, SPEC `4bc44422`, 23/23) adjudicates **arXiv:2609.02882** (Lamzouri, Sept. 2026), a new proof of the same theorem that replaces the finite Weil-form matrix and the rank–trace inequality by a single Hilbert-space (Bessel) inequality on $L^2(-\frac{1}{2}, \frac{1}{2})$ and applies Montgomery's pair correlation in the unconditional Baluyot–Goldston–Suriajaya–Turnage-Butterbaugh form (test-function support in $[-1, 1]$). Verdict SAME_CLASS + CONSTANTS_IDENTICAL + WALL_LOCATION_CONFIRMED: the key proposition is rebuilt exactly on 921 synthetic conjugation-invariant multisets (bound $n \geq 2|\mathcal{Z}| - \sum K(z-s)^2$, saturating at n); its constant functional $C_\eta = Q(0) + 2\int_0^1 \alpha Q(\alpha)\, d\alpha$ coincides with the r267 functional $R(\eta^2)$ to 1.7×10^{-15}, and the bound $2N - \|\cdot\|_{\mathrm{HS}}^2$ is the Alpöge–Furman chain at $\mathrm{tr}\, G = N$ — the certificate consumes exactly two moments, atlas class A1 is unchanged and the ceiling 0.6818287 applies; $C_0 = 2 - C_{\mathrm{MT}} = 0.672500703679$ is the r627 optimum p^* to 2×10^{-11}, $C_1 = 0.83625$. The sole arithmetic input is the proven pair-correlation range $|\alpha| \leq 1$ ($Q = \eta^2 * \eta^2$ vanishes for $|\alpha| \geq 1$; widening $\mathrm{supp}\, \eta$ to $(-0.6, 0.6)$ puts 2.8% of the Q-mass beyond 1), i.e. the r267 $X \leq T$ boundary and the measured PAIRCORR wall; under the pair-correlation-conjecture form factor $\min(|\alpha|, 1)$ the window optimum rises $p(\lambda) = 0.6725 \to 0.866 \to 0.932 \to 0.976 \to 0.989 \to 0.996$ for $\lambda = \frac{1}{2}, \frac{3}{4}, 1, \frac{3}{2}, 2, 3$. Nothing imports to the prime-comb side (no Bessel target; the r267 frame mismatch stands). Source note: the AxiomProver Lean certificates prove the proposition unconditionally and the theorem modulo BGSTB Lemma 5 and Riemann–von Mangoldt as hypotheses. Experiments only. **No RH claim.**

17 The remaining edges and the campaign

The kernel-Loewner compact-tail lane is not among these edges: Theorem 2.11 machine-checks $\lambda_*(0.3) \geq 2.1\times10^{-3}$ in float64 (not interval, not [E]), and Section 15 records the method no-go at $L = 0.8$.

Open problem 17.1 (Edge A: terminal cross-ratio, cofinal)**.** Prove $q_N < 1$ on the cofinal ladder — equivalently derive the true floor of h_{N-1}/F_{N-1}^2 source-purely (the measured floor is $1.4278 > 7/5$, margin 0.028, not saturated). By Theorem 2.7 the last fiber edge is a *positive tau cross-ratio*. **Wave-4 state:** the 42-rung *surface* of this edge is certified (Section 3.4: 41 mechanism target-blind + kz15 exact-finite) — a census + certificate result, not a cofinal theorem. The cofinal front is the single hypothesis H5 of the r271 universal pair theorem (equivalently lemma L2, pair-sum decay): a non-adjacent/global mechanism at address c3 supplying $N^{-\delta'}$ with $\delta' > 0.21$ of the measured available 0.45; per r273 the mechanism must capture *generic* cancellation, and per r275 the state-space vehicle is excluded. Registered as `PRIME.PORT.COUPLEDTAU.TERMINAL_CROSSRATIO.01` [O].

Open problem 17.2 (Edge B: prefix resummation / oriented midpoint theorem)**.** Find the closed form of the coherent ensemble resummation: the determinant as a signed partition sum, resummed source-purely — no saddle dominance, no k-swap truncation (both measured dead, Section 15). The r261 finite-combinatorial half closed honestly negative (CAMPAIGN_INPUT_FROZEN). **Wave-4 state:** the orientation of the wall follows a *predictable* law — the Maslov census GO of Theorem 4.2 — and the named path is the *oriented midpoint theorem*: the contraposition with an independent index obstruction that forbids the R2 interlacing break before half-filling on MAIN (the raw atom-Sturm count is refuted; per r276/r278 the `MainWindow` arithmetic must enter as *metric* exactness, not Euler bookkeeping). Registered as `PRIME.PORT.FULLSOURCE.PREFIX_RESUMMATION.01` [O] with the successor contract `PRIME.PORT.RHP.MIDPOINT.ORIENTED_THEOREM.01` [O]. **Wave-5 state:** rounds 279–281 are executed and consumed (Section 5): the theorem share is promoted (the parity and budget theorems), the candidate generic two-sided obstruction is refuted exactly, and the center is precisified — the entire open content is the budget localization, i.e. free-window quasi-definiteness (Remark 5.5); the successor lanes `PRIME.PORT.REPRESENTATION.CONTEST.01` and `PRIME.PORT.RHP.FULLSOURCE.QUASIDEFINITENESS.01` are executed and consumed in wave 6 (`v963`, Section 6); the open center continues in canonical form as `PRIME.LSTAR.SUBORDINATION.01` [O].

Open problem 17.3 (Odd-sector edge of the first prime step)**.** For odd h supported on $[-L, L]$,

$$\langle h, (\mathrm{POLE} + \mathrm{ARCH})h\rangle + w_2(\|f_+\|^2 - \|f_-\|^2) \;\geq\; 0,$$

where $f_\pm$ are the even/odd parts of h relative to the reflection $(Rf)(x) = f(d - x)$ at $d = \log 2$ (overlap $I = [d - L, L]$). Round r620 retypes the parent's earlier $|g|$ lemma: the prime term $-w_2 g$ stabilizes the symmetric edge sector and destabilizes the antisymmetric one, so a $|g|$ bound was mis-stated (the prime hurts on P_-). The negative direction of $\mathrm{POLE} + \mathrm{ARCH}$ migrates from the inner region to the R-symmetric edge sector (mass inner/+/−: 0.85/0.15/0.005 at $L = 0.38$ to 0.06/0.94/0.001 at $L = 0.549$); the antisymmetric edge sector is safe without the prime ($\lambda_{\min}(A|_{P_-}) - w_2 = +5.7 \to +0.29$). The repair is a quantitative dominance with margin $\lambda^*(L) = \lambda_{\min}(A^{\mathrm{odd}} + w_2(P_+ - P_-))$, not an exact sector identity. Registered as `PRIME.SEMILOCAL.ODDSECTOR.EDGE.01` [O](r615; retyped r620 to the oriented form). Window forms with $L \lesssim 16$ cannot detect off-line zeros (r605 scaling $a = \sigma^2/64$ needs u-support $\sim 8/\sigma$), so the first prime step is far weaker than RH. **No RH claim.**

Relay architecture (r619–r622). The support-relay organizes the window by a canonical interval per prime power q: entry $L_j = \frac{1}{2}\log q$, with continuity $Q_j(L_j) = Q_{j-1}(L_j)$ (the newest event is added at the endpoint of the previous interval). The conjunction $\forall j\, R_j$ of the finite relay steps is equivalent to Weil positivity of the window form. After r619–r622 the honest status is: the RELAY_UNIVERSAL criteria ($\Delta_j \geq 0$, $g < 0$, newest-weight repair) are *consequences* of $Q_j \succeq 0$ (RH-consistency), not a mechanism; there is no source-algebra Darboux rule for the rate (r622); the prime-2 repair is not a duplication factor of the archimedean operator (r621); and relay steps beyond $L \approx 0.5$ are RH-in-a-box statements (hypersensitive off-line detectors, r619 L_{det}), not elementary finite problems. The r619 log-slopes $-77/-79$ at $L = 0.9$–1.4 (fixed $N = 80$ beyond the SVD floor) are N-artifacts, corrected by r630b to -50.6 (0.3–0.6) and -116 (0.6–1.0); the RELAY_UNIVERSAL verdict, the entry census, the odd/edge failing mode, and L_{det} table are unaffected. The r619 "greedy law" $\Delta \approx 0.015$ nearly constant across 18 events is WITHDRAWN (r635): an $N = 80$ double-precision artefact; the four converged leads are $\Delta_2 = 0.0253$, $\Delta_3 = 0.0083$, $\Delta_4 = 0.0064$ (0.00606 at $N = 160$), $\Delta_5 = 0.0034$ (0.00275 at $N = 160$), and Δ_q for $q \geq 7$ is UNMEASURABLE at the 40-digit prime-side floor. **No RH claim.**

Prime-place dilation and IIKS vanishing (r623, r625, r626). The proposed principal-block / Schur-complement dilation of the first 2-adic Weil channel is structurally blocked (r623 P2_DILATION_STRUCTURALLY_BLOCKED): POLE + ARCH is not a PSD principal block, M_2 is indefinite and not rank one, the generic dilation tracks $\lambda^*(L)$ tautologically, and the literature stops at the archimedean window $\sqrt{2}$ (Connes–Consani 2021, Thm 1). The IIKS vanishing-lemma metric is not an open lane: the corpus discrete jump is the wrong RHP object (r625, enum IIKS_NO_METRIC_IN_CLASS, prose NOT_TESTABLE_IN_CORPUS_FORM), and no source-explicit metric in the named class is contractive. The finite-free collision barrier is tautological (r626). Together with the first cut (r619–r622) every tested positivity route reduces to Weil positivity in a new coordinate or to an RH-strength hypothesis. The standing assets remain the unconditional rational Gabor criterion (r612C), the three RH-neutral exact theorems (Eisenstein bridge, commensurability, Coxeter completion), the first-prime-step mechanism (r615/r620, oriented lemma [O]) and the relay census with L_{det} (r619). Recommendation: pause the active positivity campaign pending a genuinely new mechanism. Subsequent sealed scouts r627 (A–F window optimum = Montgomery–Taylor, NO_GAIN), r628 (box verifier: compression certified, rigor blocked by $S(T)$ looseness), r629 (certificate-class atlas: no unconditional gain), and r632 (jet deflation: coverage collapse) do not reopen a positivity lane. **No RH claim.**

17.1 The master target form (r273)

Since r273 the two edges are carried by *one* augmented positivity statement (the formal home is `rh/lean/RH/Window.lean`). For a window w with Hankel moment matrices H_n, border vector u and budget B, define the augmented matrix

$$A_{w,n} = \begin{pmatrix} H_n & u_n \\ u_n^{\mathsf{T}} & B \end{pmatrix}, \qquad n \leq N_w = \lceil S/2 \rceil.$$

Master target (`augmented_prefix_positive`): *for a MAIN window, $A_{w,n} \succ 0$ for every $n \leq N_w$.* The derivation chain is finite matrix algebra and is *proved* in Lean: the upper principal block gives $H_n \succ 0$, hence all leading principal minors are positive and the prefix chain $h_k = \det H_{k+1}/\det H_k > 0$ follows (former edge B: the wall derived, not assumed); the Schur complement gives $D_n = \det A_n/\det H_n = B - u^{\mathsf{T}} H_n^{-1} u > 0$, hence $q_N < 1$ at the terminal (former edge A). The budget enters only through B, so the 5/7 floor falls out as a *consequence* of the master positivity instead of entering as an axiom. The open content is concentrated in the honest source predicate `MainWindow` (the von-Mangoldt comb arithmetic); the kill lists of Section 15 enumerate what provably does *not* suffice. The r273 retype was forced by three machine-checked counterexamples (`rh/lean/RH/Counterexamples.lean`): the previous universal formulations of both edges and of the r271 margin law H5 over bare bookkeeping structures are *refutable* by trivial models ($q_N = 4$ ladders, $\tau_1 = -1$ chains, $2 < 1$ margins) — the guards make any future "proof from bookkeeping" formally contradictory.

17.2 The campaign state (charter r264)

Frozen charter (SHA `7b9e751d`): goal S6 verbatim on the seven exception rungs; input = the r261 package (frozen G_n, t_n, B; scaled Chebyshev basis); kill list armed (PAIRCORR detector, target-inverse, selection-by-answer, support-geometry); blocked paths: smooth dressing (r253 essentiality) and the bordered spectral bound (r257/r265). The campaign's deformation lever is the *border-drive functional near terminal* (data signature S1_TERMDRIVE, separation 0.959, class PREFIX_DRIVE).

17.3 The named questions in flight (wave-13 state)

- **The oriented midpoint theorem (rounds 279–281, consumed in wave 5).** The step-5 obstruction question is answered in decomposed form (Section 5): the generic two-sided obstruction is refuted exactly, and the demanded independent index obstruction is precisified to the localization of the fixed budget — free-window quasi-definiteness (contract `PRIME.PORT.RHP.MIDPOINT.ORIENTED_THEOREM.01` [O], rounds consumed by `v962`).

- **The representation contest (r282, consumed in wave 6).** Contract `PRIME.PORT.REPRESENTATION.CONTEST.01`: adjudicated CONTEST_ALL_DEAD and consumed by `v963` (Remark 6.4) — the four classical languages are closed with exact reasons; a non-classical "fifth language" would be genuinely new mathematics and is not round-ready without a candidate.

- **The full-source quasi-definiteness (r283, consumed in wave 6).** Contract `PRIME.PORT.RHP.FULLSOURCE.QUASIDEFINITENESS.01`: adjudicated MECHANISM_PARTIAL(A2) and consumed by `v963` (Theorem 6.1) — L* is established as the canonical reduction.

- **The L* subordination (the canonical open center).** Contract `PRIME.LSTAR.SUBORDINATION.01` [O]: $\int p^2\,d\nu_z < \int p^2\,d\mu_z$ for every real $p \neq 0$ with $\deg p < N_w$ on the admissible anchor family (Theorem 6.2); state after wave 7: 57/57 measured (the 42 plus the 15 extension anchors, N_w up to 1218), the margin decay resolved *harmless* (Remark 7.2), the coherence carrier class named (Remark 7.3), the diophantine route excluded (Remark 7.4); state after wave 8 (rounds r290–r295 consumed, Section 8): the working-set geometry is *measured* (tube, ridge, rank-1 DENS curvature valley, simple flip zeros) and the profile-functional question is *functionally negative-sealed* over five class families (Remark 8.4) — the successor question is the closed-functional contract below; standalone problem statement in `rh/problem/lstar_problem.tex` (updated "what is known"); state after wave 11 (Section 11): the surrounding architecture is fully adjudicated (Lane A closed as cone language, B not overtaking, C parked) and the *formal* frame is restructured — the master theorem is now *proved from* L* + terminal (`lstar_terminal_implies_master`, r305), the window source is a real Lean construction (r310/r310b), and the contract stands as one of the two TRUE formal window-local holes (sorries $9 \to 5$); state after wave 12 (Section 12): the indefinite fork is adjudicated — the P1 index language is banked (a total restatement, no lever), the P2 antiphase fingerprint is closed as ALGORITHM_ARTIFACT by the r322 dig (the selection-geometry question is the named residual), and the contract's Lean statement class is red-team hardened (the r319

U-findings repaired by the r320 retypes + guards; the honest chain reading of Remark 12.1 is binding).

- **The closed functional (r290–r295, registered in wave 8).** Contract `PRIME.LSTAR.CLOSED_FUNCTIONAL.01` [O]: exhibit a closed, source-pure functional of the tent-split fraction profile that predicts the survival depth of the working set at HARDENED grade (the sealed r295 three-clause bar) with a stable partial channel — or prove non-existence in a named class; the sealed kill catalog (Remark 8.4) binds the candidate space. State after wave 9 (r296 consumed, Section 9): the DENS-identity fork is *closed* (DENS_WORLD_BLIND — the rank-2 DENS core is not an arithmetic identity and not the L*-gradient direction at the sealed bar; the lane routed to L2); the remaining open fronts are the loss-corpus forensics (K05/K07/K19), the substantive "why L2" question and the w7/w11 knife edge. State after wave 11: **Lane A (the constructive PSD-G rule) is closed as a cone language** — r311 STRICT_SOURCE_CONE + r312 COEFFICIENT_SIGN_WALL with the mechanism named (block-psd membership without rank-one SDD, obstructed at the budget/border sign); the PSD-G rule stays open as a question, the resources moved to the fiber.

- **The tensor mechanism (r309/r313/r316, registered in wave 11).** Contract `PRIME.SOURCE.TENSOR.MECHANISM.01` [O]: does a source-pure cofinal control arise from the banked rank-one update identities (`PRIME.SOURCE.RANKONE.UPDATE.IDENTITY.01` [E])? Registered *after* the documented negatives (Remark 11.1): no transfer-operator contraction at chain level (FLOQUET_EXPANDING), no type-majorant composition, no two-regime pointwise majorant under a mid-ladder freeze.

- **The Rényi-3 provenance (r306/r313–r316, registered in wave 11).** Contract `PRIME.L2.RENYI3.PROVENANCE.01` [O]: the pointwise GO ($C = 1.069$, $A = 2$, 57/57, growing reserve) stands as a certified finite statement; *where does the law come from?* Trilogy state: the identity is exact (r314), the functional is blind (r315), the two-regime form is dead (r316). State after wave 12 (Section 12.2): the R317 fork is executed both ways and resolved — fork (b) sealed WHAC_A_MOLE(kz53, kz83) (the exception-family form is the wrong statement shape for a spike continuum), fork (a) fired SLIDING_BOUND_GO(G_{SQ}) (Theorem 12.4, registered as `PRIME.L2.RENYI3.SLIDING_BOUND.01` [O]); the provenance question is now the $F_A/q_{\max}$ two-part split. State after wave 13 (Section 13.2): the split is executed both ways — r324-pre banked (FA_BOUNDED_DISTRIBUTIONAL + QMAX_SHARE_OPEN + NOT_COMPOSED), r324 composed MEASURED (PILEUP_GROWS_SUBCRITICAL +0.172, subcritical also at the r328B chain-honest bar 0.188; m_0^* record $10^{59.6}$, chain-honest 10^{238}), r327 the SOURCE anatomy; the provenance question is now the *coincidence geometry* (Remark 13.2; registered as `PRIME.L2.RENYI3.MEASURED_COMPOSITION.01` [O]).

- **The sliding cubic bound (r317/r321, registered in wave 12).** Contract `PRIME.L2.RENYI3.SLIDING_BOUND.01` [O]: the theorem candidate $\sum q^3 \le 1.3056\, F_A(w)^2(\log m)^2/m^2$ for $m \ge 73$ (Theorem 12.4) is certified on the measured rungs with all four named violators inside; the demand is the two-part provenance question — is F_A source-pure bounded (measured max 2.47), and what bounds the $q_{\max}$-share $\rho_2/(F_A^2 B^2)$ (the median-of-max shape route; the r302 M_2 stationarity pins the second moment of the same shape)? Honest: $C_{\mathrm{impl}} = 7.97$ is 7.5× looser than r306 — form, not sharpness; the bound is not world-separating by itself. State after wave 13 (Section 13.2): the two-part demand is executed — F_A is *de-black-boxed* by the exact identity $F_A B \log m = m\, q_{\max}$ (the rank-local normalization of $m\, q_{\max}$, no independent coordinate) and distributionally bounded (r324-pre banked, envelope $C_F = 3.357$); the $q_{\max}$-share route composed MEASURED (r324); the sliding bound itself stays the certified theorem candidate on the measured rungs.

- **The vdC L2 lemma (r287, registered in wave 7).** Contract `PRIME.PORT.L2.`

VDC_LEMMA.01 **[O]**: prove the van der Corput bound of Theorem 7.1 at $H = \lceil\sqrt{m}\rceil$ as a *chain* statement (source-pure, window-independent) — the remaining input is the chain origin of the P-variance scaling; the inequality itself is exact and world-blind valid (admissible: L2 is the generic half), and the kz15 razor needs no new mechanism (exact-finite per r270). State after wave 9 (rounds r297–r300 consumed, Section 9): the remaining input stands *reduced to one inequality* — the target level σ^* is frozen (Theorem 9.1), the transfer to the difference measure is exact (Theorem 9.2), the diagonal decay is an exact participation statement (Theorem 9.3), the ratio half is structurally settled (Theorem 9.4). State after wave 10 (rounds r301–r304 consumed, Section 10): the reduction route is *documented stopped* — the cascade is retyped as an exact dictionary around one measured core (Theorem 10.2) and the global-profile mixing route is closed (Remark 10.6); the contract stays **[O]** at the stop state.

- **The NEFF target (r300, registered in wave 9; retyped in wave 10).** Contract PRIME.L2.NEFF_TARGET.01 **[O]**: registered as $\mathrm{slope}(n_{\mathrm{eff}}) \geq +0.908$ for the participation number of the aggregated difference profile (measured $+0.963$, margin 0.055; exactly equivalent to $\mathrm{sl}_D \leq \sigma^*$). State after wave 10 (rounds r301–r304 consumed, Section 10): the demand is *retyped to the documented stop state* — the r303 regress audit shows NEFF/UNIF/ATOM are one algebraic slack $S = \sigma^* - \mathrm{sl}_D = +0.0547$ (Theorem 10.2); the honest state is L2 generic ⇔ anti-concentration of the explicit dc block field; the global-profile mixing route is closed (period-4 comb, LAW_LONGRANGE, Open problem 10.5); the mechanism is two-scale (Remark 10.4); return later with new tools — per the hard reviewer rule a round counts only with new information, no further coordinate change counts.

- **The global metric firewall (r276/r278 follow-up).** Contract PRIME.PORT.WALL.METRIC_FIREWALL.01 **[O]**: the global form of the wall's metric stability — a wall lemma quantitatively stable under family/assignment permutations at small dose while consuming the metric data exactly; the certified state is graded-continuous with exact gradients and a bottom-loaded u-profile, but perturbative-only (θ^* tiny, L_D window-specific).

- **The border-dressed resolvent question (r266, answered negative).** The exhaustive census found *no identity in the class* (Section 15); the campaign needs a genuine local RHP asymptotic at the terminal degree — with the important positive by-product that no candidate reduces to the proven-false wall bound (WALL_DETECTOR_ARMED: target and wall stay measurably different statements).

- **The frame renormalization question (r267 follow-up).** The external ceiling lives in a near-tight zero-atom frame ($R = 4/3$); our comb frame is spread over 7–8 decades ($R \approx 73$–126). Named question: does a source-pure renormalization of the comb frame exist that trades atom-norm spread against certificate class — i.e. can any part of the bordered terminal scalar be moved into a frame where an external-style rank–trace argument applies without crossing the two-moment ceiling? (Adjudicated NO_IMPORT for the verbatim import; the renormalized variant is open.)

17.4 The Gabor reduction chain (r541–r634L)

The window-L* / compact Weil extraction of the preceding subsections remains the program's live mincut (base 4 / refined 5; Level C of Remark 12.2). In parallel, rounds r541–r555 assembled a *conditional* Gabor-wavepacket reduction, formalized in `RH/GaborSeparation.lean`. The historical endpoint `gabor_inputs_to_mathlib_rh` is a sorry-free logical wrapper: the two named analytic inputs `GaborExplicitFormula` and `GaborSeparationInequality` imply Mathlib's `RiemannHypothesis`. As of the r599/r600 vacuity audit that wrapper is **VACUOUS**: it consumed positivity of `gaborSpectralFormula`, which is not the Weil functional. The live object is `gaborZeroSide`. The implication is conditional; it is not a proof of RH. Round r612 deleted the asserting theorem `gabor_separationForAllZeros`; the Prop `GaborSeparationForAllZeros` is retained unasserted (OVERSPECIFIED/believed-false). Separation stays open ([O]). **No unconditional claim.**

Proved bricks (sorry-free). Three analytic pieces are closed as Lean theorems and are not the remaining mincut: (i) the Gaussian integral representation of the pure Gabor test (`RH/GaborIntegral.lean`, r543); (ii) the unit-increment bound `GaborIncrementBound`, discharged by `gaborIncrementBound_holds` with explicit constant $C = 2 \cdot$`zetaZerosInDiskCardBoundInner` (`RH/ZeroIncrement.lean`, r546/r547); (iii) the Jacobi-theta majorant of unit-bin Gaussian maxima, together with bin partial summation, giving a γ-uniform on-line Gaussian mass bound $\leq C \cdot \Theta_{\text{lobe}}$ (`RH/GaborThetaBound.lean`, theorems `gauss_online_mass_uniform` and companions, r552). None of these three statements implies the uniform Gabor inequality.

Explicit formula for the Gabor class. The arithmetic landing site of `GaborExplicitFormula` is the classical prime comb $2\Lambda(n)/\sqrt{n}$ (r550), not the compact honest weight $\Lambda(n)(1+n^{-1})$ refuted for $\hat{h}_W$ at r548. The identity is numerically confirmed on a sealed 36-cell battery (r548, verdict EF_CONFIRMED, 36/36, residuals $\sim 10^{-15}$). Round r581 discharged the Lean remainder: `gabor_vertical_arithmetic_remainder` is a theorem (no `sorryAx`).

Quantifier distinction (binding). Round r541 (SEPARATION_GO_GABOR) is a *certified numeric precheck* against one synthetic functional-equation quadruple: 42/42 cells through $\gamma = 10^4$. It is *not* a discharge of `gabor_separationInequality`, and it is not a configuration-first statement. The configuration-first game — $\forall$ increment-conforming off-line configurations, $\exists$ a Gabor witness — is a strictly stronger quantifier. Round r553 (WITNESS_LOSES) shows that both the one-packet and the two-packet Gabor classes lose that game under pure counting control (clusters of equally extreme quadruples). Round r551 shows that extremal selection alone does not close the gap: the equal-extremal tie factor is $e^{\pi^2/128} \approx 1.080$ against a counting budget 141× the available margin. A spectral-gap hypothesis $\sigma'/\sigma \leq 0.9194$ would compress that counting budget; it is not assumed here. The live attack is mixed-sign / adaptive-nulling witnesses (r554). Reading r541 as a completed separation `sorry` is a category error.

All Gabor probes remain sealed in `rh/INVENTORY.json`; the RH suite is green. Nothing in this subsection is evidence for or against RH. **No RH claim.**

Vacuity audit (r599/r600, 2026-09-01/02). `gaborSpectralFormula` is *not* the Weil functional: it subtracts the pole term $\hat{h}(1)$. For $F_0 =$ `pureGaborTest` $1\,0$ the zero-side sum is $\approx 3 \cdot 10^{-43}$ while $\operatorname{Re}\hat{h}(F_0, 1) = \pi e^{1/8} \approx 3.5599$, so `gaborSpectralFormula` $F_0 \approx -3.56$ unconditionally. The old hypothesis hpos $:= \forall F$ admissible, $0 \leq$ `gaborSpectralFormula` F is therefore **false** (not merely unproven), and every Mathlib endpoint that consumed it was vacuous. The live object is `gaborZeroSide` (spectral sum without pole subtraction); hpos* $:= \forall F$ admissible, $0 \leq$ `gaborZeroSide` F is the RH-core. Through r611 this was equivalently the asserting `sorry gabor_separationForAllZeros`; r612 deleted that theorem and retained the Prop unasserted. `GaborSpectralToArithmetic` is a proved theorem (`GaborSpectralToArithmetic_holds`): the pole-sign lemma $\hat{h}(1) \geq 0$, *not* the RH-core. Round r605A records that `gabor_zeroSide_pure_criterion_iff_rh` takes a separator premise `hsep` (`GaborZeroSideForAllZeros`, prescribed packet `scalingGaborTest`$(a = \sigma^2/64, \omega = \gamma - \pi a/\sigma)$) and is FORALL_ZERO_OVERSPECIFIED. The live unconditional statement is `gabor_zeroSide_pure_criterion_iff_rh_unconditional` (Section 17.5). For the full quartic family the iff still carries `hline`. Old endpoints kept as a VACUOUS (r600) audit trail: `GaborSeparatesOffCriticalZeros`, `gabor_separation_of_inputs`, `gabor_criterion_to_mathlib_rh`, `gabor_inputs_to_mathlib_rh`, `gabor_separation_of_arithmetic_sep` `gabor_arithmetic_inputs_to_mathlib_rh`, `gabor_window_adaptive_inputs_to_mathlib_rh`. Live replacements: `GaborZeroSideSeparatesOffCriticalZeros`, `gabor_zeroSide_inputs_to_mathlib_rh`, `gabor_arithmetic_zeroSide_inputs_to_mathlib_rh`, `gabor_window_adaptive_zeroSide_inputs_to_mat` Named open (no claim): a rigorous tsum bound $Z(F_0) < \pi e^{1/8}$ for an unconditional spectral refutation; on-line positivity for general even quartics (beyond $\langle 1, 0, 0\rangle$) so the iff holds without `hline`. Tactic-sorry census stayed 8 through r605; $8 \to 7$ at r612; no `sorryAx`. **No RH claim.**

Numerical evidence (experiments only, no RH claim). Round r601 (probe `gabor_zeroside_dominance`, SHA `b7089c71`) returns DOMINANCE_FAILS: 132 cells, explicit-formula sanity $\leq 1.4 \cdot 10^{-15}$ with 2000 zeros, no negative cell. Positivity is certified only for $\omega \geq \omega_0(a)$ ($\omega_0 \approx 500$ at $a = 0.5$, down to 10 at $a \geq 3$); no $a \leq 5$ is ω-uniform. The cancellation exponent $e^{(1/8+d^2/2)/a}$ near zeros is confirmed. Conclusion: the Gabor family with fixed a is a Bombieri / Connes–Consani-type sub-family, not a proof of hpos*. Round r603 (probe `prime_inequality_evosearch`, SHA `06d2ac9a`, 13/13, byte-identical) is a two-key genetic search over $\Lambda(n)$, $n \leq 10^6$: mechanically green, discriminatively empty. The 1378 grid-survivors are comparisons of convergent constants on $\operatorname{Re} s = 1$ ($|\mathrm{Sc}(1,2)| \leq |\mathrm{Sc}(1,14.13)| \leftrightarrow 0.42 \leq 1.62$ via $-\operatorname{Re}\zeta'/\zeta(1+it)$) or grid artefacts. Controls C1–C4 are all density-only: local jitter $p^{0.6}$ moves ψ by only $\sim 10^2$, and a $\beta = 0.75$ fake zero has amplitude $2100 < 6900$ (visible only from $X \approx 10^{10}$). Lesson recorded: at accessible X, RH-level and 0.75-level are indistinguishable by proven-constant inequalities. Round r604 (probe `tfpt_spectrum_zero_crosscorr_probe.py`, SHA `748ae509`, 7/7) returns ARTIFACT: TFPT-native spectra (Moran-kernel complex dimensions, comb tones $\omega_1, \omega_2, \omega_3$, E8-clock integers) versus the first 4000 zeta ordinates. Positive controls $\log 2, 3, 5, 7$ fire at $z \geq 5$; every candidate with $z \geq 5$ is a $\log p^k$ alias (the only raw hit $\omega_1 + 2\omega_2 = \log 89$ dies under unfolding). Nearest-neighbour and clock-moduli tests are null. Experiments only. **No RH claim.** Round r606 (probe `connes_prolate_residual_gap_probe.py`, SHA `aeef52a6`, SPEC `115c0409`; r606 first-run 6–9 mode, $t_{\mathrm{cut}} = 150$) is INCONCLUSIVE: the CCM Poisson candidate k_λ has Davis–Kahan ratio $\eta = 0.13/0.093/0.029$ at $L = 0.3/0.4/0.5$, with breakdown for $L \geq 0.6$ attributed to the uniform position grid (small blocks fail at $L = 0.8$; cf. r478); μ jumped to 5.9 and $g < 0$ at $L \geq 0.6$ (discretisation artefact). Addendum r606b (two runs byte-identical, ~ 0.4 s) returns INCONCLUSIVE(evals-not-converged, disc-disagree> 5pct): no L meets successive-$N < 1\%$ and Slepian/Fourier agreement $< 5\%$; the wall is discretisation (Slepian λ_2 pollution beyond $N = 12$–24, Fourier odd-mode $\sim 2\%$, $t_{\mathrm{cut}} = 1000$ blocked by spatial Nyquist), not a GO/STOP on η. The r606 $\mu \approx 5.9$ jump is gone (k tracks λ_1); λ_1^e slightly negative ($-3 \cdot 10^{-6}$–$-8 \cdot 10^{-6}$) at $L \geq 0.5$ but unconverged — not a Chuk contradiction; $\widehat{\theta}[0, 30]$ NREAL $2, 3, 3, 4, 4, 4$ vs Riemann count 3 (plateaus at 4). Lane CLOSED — superseded by r611 (OBS_NOT_CONVERGING) and the KILLED(factorization) typing of the prolate route; diagnostic only. Experiments only. **No RH claim.** Round r608 (probe `gabor_tropical_heat_probe.py`, SHA `574c1cc2`, 9/9) returns TROPICAL_CONFIRMED: $2a \cdot \log|Z| \to F(\omega)$ with error $\leq 6.77 \cdot a \cdot \log(1/a)$. On-line, $F = -\mathrm{dist}(\omega, \{\gamma\})^2 \leq 0$; a synthetic $\beta = 0.9$ orbit gives $F = +\sigma^2 = 0.16$. The hat obeys $\partial_a \hat{h} = \frac{1}{2}\partial_\omega^2 \hat{h} - \hat{h}/(2a)$ on the two Gaussian lobes only — the cross lobe $2e^{(\delta^2-\omega^2)/2a}$ carries residual $-\delta^2/(2a^2)$, so $\sqrt{a}\, Z$ is *not* a heat solution and a dyadic-in-a reduction via the heat semigroup is unavailable for this family. Source-side cancellation budget $B(a) = 2.05/4.18/13.6$ at $a = 0.2/0.1/0.05$, i.e. $B \sim e^{1/(8a)}$: deriving $F \leq 0$ from the prime side needs cancellation to relative precision $e^{-(1/8+|F|/2)/a}$ with primes up to $n = e^{1/(2a)}$ — the quantitative form of the NOGO-T1 wall. Consequence: the HEAT.SOURCE lane is a NOGO T1 (RH-strength cancellation) and is STOP. Experiments only. **No RH claim.** Round r609 (probe `e8_directed_readout_probe.py`, SHA `9ab0a638`, SPEC `3edfb824`, 49/49) returns E8_READOUT_SEALED(7/8): Seifert $S + S^T = A$, $\mathrm{charpoly}(-S^{-1}S^T) = \Phi_{30}$, $\mathrm{rank}(S - S^T) = 8$; H_8 weight enumerator $1 + 14y^4 + y^8$; $240 = 16 + 14 \cdot 16$; Steiner $S(3, 4, 8)$; $\mathrm{srg}(120, 56, 28, 24)$ eigenvalues $56/8/-4$ with multiplicities $1/35/84$; $N(n) = 240\sigma_3(n)$ for

$n \le 10$; $\tau(n)$ for $n \le 8$ ($A_P(1) = 1658880$); $E_4^3 - E_6^2 = 1728\Delta$; $(\mathbb{Z}/30)^\times \cong C_2 \times C_4 \not\cong C_2^3$, character table $= H_2 \otimes F_4$; $E_6 \oplus A_2$ index 3, Smith $(1^6, 3, 3)$, roots $72 + 6 + 162$. C7 was NOT_TESTED at probe time and was later supplied in `note_e8_gaussian_code.tex` (explicit $8 \times 8 = H_2 \otimes F_4$, permutation $[0, 1, 4, 3, 5, 2, 7, 6]$); the Gauss-code transform remains undefined (open remark `rem:c7-audit`). TFPT-side exact structure; no prime bridge is shown; promotion to `verification/` is a pending user decision. Experiments only. **No RH claim.** Round r611 is recorded in Section 15 (OBS_NOT_CONVERGING; Herglotz/Aubin–Nitsche STOP). Round r613 is recorded in Section 15 (NO_IDENTIFICATION; ordered-E8 canonical system dead). These are sealed experiments (r606/r606b CLOSED as a diagnostic on the same file; superseded by r611), not verification modules.

Round r615 (probe `semilocal_firststep_probe.py`, SHA `3a0741de`, SPEC `04f9c9ec`, 26/26) returns FIRSTSTEP_LEAKAGE_CONTROLLED. The Weil window form with exactly one prime ($L \in (0.3466, 0.5493)$, $w_2 = \sqrt{2}\log 2 = 0.98026$) has Ritz $\lambda^*(L)$ equal to zero-side leakage to 10^{-4}–10^{-5} at all tested L; the extremal vector is Slepian-like (overlap with $\psi_0 \ge 0.9989$, $\ge 99.8\%$ spectral mass below $\gamma_1 = 14.1347$) with no character change at the prime entry. $\lambda^*(0.30) = 7.57 \cdot 10^{-3}$ (v1017 padded floor $2.1 \cdot 10^{-3}$, factor 3.6), $\lambda^*(0.40) = 1.85 \cdot 10^{-4}$, $\lambda^*(0.549) = 5.80 \cdot 10^{-8}$ with POLE $+1.30$ / ARCH -1.29 / PRIME $1.78 \cdot 10^{-2}$ (cancellation depth $2.25 \cdot 10^7$). Pinning at $L = 0.52$: $\delta_{\text{crit}} = 2.5 \cdot 10^{-5}$ relative on w_2, $\varepsilon_{\text{crit}} = +4.1 \cdot 10^{-5} / -1.7 \cdot 10^{-6}$ on the atom position, WPERM -0.135, SCRAMBLE $-0.33/-0.18/-0.047$. Mechanism (T9): the minimizer of the fictitious prime-free form POLE+ARCH is purely odd, with POLE $= -2\langle h, \sinh(u/2)\rangle^2 < 0$ and ARCH < 0, and $g(\log 2) < 0$, so $-w_2 g(\log 2)$ carries the positivity ($L = 0.40$: $-0.018 - 0.060 + 0.130 = +0.053$). The odd sector turns negative at $L \approx 0.372$; prime 2 enters at 0.3466 (margin $\Delta L \approx 0.025$). Named open lemma `PRIME.SEMILOCAL.ODDSECTOR.EDGE.01` [O], retyped at r620 to the oriented form: for odd h on $[-L, L]$, $\langle h, (\text{POLE+ARCH})h\rangle + w_2(\|f_+\|^2 - \|f_-\|^2) \ge 0$; the negative direction of POLE+ARCH migrates from the inner region to the R-symmetric edge sector; the antisymmetric edge sector is safe without the prime. The repair is a quantitative dominance with margin $\lambda^*(L)$, not an exact sector identity. Fence: window forms with $L \lesssim 16$ cannot detect off-line zeros (r605 scaling $a = \sigma^2/64$ needs u-support $\sim 8/\sigma$), so the first prime step is far weaker than RH. Experiments only. **No RH claim.**

Round r617 (probe `e8_coxeter_euler_completion_probe.py`, SHA `6ac6bab2`, SPEC `5aa3935b`, 45/45) returns COMPLETION_EXACT. $\operatorname{Tr} C = -1$; the Φ_{30} Möbius identity holds; $(1-x)\Phi_{30} = 1 - x^2 - x^3 + x^6 + x^7 - x^9$; $D_{E8}(s) = \zeta(6s)\zeta(10s)\zeta(15s)/[\zeta(2s)\zeta(3s)\zeta(5s)\zeta(30s)]$ has residuals at $X = 10^6$ of $1.28 \cdot 10^{-4}$ ($s = 0.75$) and $3.5 \cdot 10^{-14}$ (1.5); $\det_2$ scalar control $e^{P(s)}/\zeta(s)$ residual $1.75 \cdot 10^{-14}$. Class statement: scalar -1 gives $A_1 \to 1/\zeta(2s)$, $A_2 \to 1/\zeta(3s)$, $E_6 \to \zeta(4s)\zeta(6s)/[\zeta(2s)\zeta(3s)\zeta(12s)]$; negative control B_2 (trace 0) gives $\zeta(2s)/[\zeta(s)\zeta(4s)]$, divergent at 0.75. The identity is Beurling-generic. Typed no-go `E8.COXETER.REGULARIZED_SPLIT.NO_GO.01` is recorded in Section 15. Fence: "The trace-free completion is zero-free and pole-free in $\operatorname{Re} s > 1/2$. The splitting into the scalar zeta channel and the Coxeter channel is open and RH-equivalent. No RH claim." Promotion to `verification/` (v1018/v1019) is in progress. Experiments only.

Round r618 (probe `jensen_compiler_rigidity_probe.py`, SHA `29ccf1c7`, SPEC `bc8d8624`, 43/43) returns COMPILERS_STRUCTURALLY_RIGID. Data: $a_0 = \xi(1/2) = 0.497120778$, Turán inequalities for $n \le 12$, $J_{n,d}$ hyperbolic for $d \le 8$, $n \le 4$ (consistent with GORZ 2019; nothing new). The two structural no-gos are recorded in Section 15. Fence: both compilers are generic real-rootedness certificates; existence for all (n, d) is equivalent to RH; the E8 data are RH-neutral. Experiments only. **No RH claim.**

Round r619 (probe `support_relay_census_probe.py`, SHA `fcfa7771`, SPEC `7299737d`, RESULT `e0379485`, 12/12) returns RELAY_UNIVERSAL. Prime powers $q \le 32$ (18 events), entry $L_j = \frac{1}{2}\log q$. Every ancestor form Q_{j-1} (without the newest event) fails a finite lead after entry:

q	2	3	4	5	7	8	9	11	13
Δ_j	0.0279	0.0123	0.0146	0.0111	0.0105	0.0140	0.0125	0.0116	0.0131
q	16	17	19	23	25	27	29	31	32
Δ_j	0.0166	0.0126	0.0144	0.0141	0.0170	0.0181	0.0151	0.0147	0.0221

$\Delta_j = 0.011$–0.028 (nearly constant at $N = 80$). The constant-lead "greedy law" $\Delta \approx 0.015$ is WITHDRAWN by r635 (an $N = 80$ double-precision artefact); converged leads $\Delta_2 = 0.0253$, $\Delta_3 = 0.0083$, $\Delta_4 = 0.0064$ (0.00606 at $N = 160$), $\Delta_5 = 0.0034$ (0.00275 at $N = 160$); Δ_q for $q \geq 7$ is UNMEASURABLE at the 40-digit prime-side floor (not zero). The RELAY_UNIVERSAL verdict, the entry census, the odd/edge failing mode, and the $L_{\det}$ table are unaffected. The failing mode is always odd, reflection-symmetric edge sector w.r.t. the newest event ($g < 0$), edge mass 0.07–0.65; the newest weight restores positivity (all 18). λ^* decay slopes per interval $-50/-77/-79$ at fixed $N = 80$ (global -68.6; Slepian -28.3; r615 -44.8) — the $-77/-79$ values at $L = 0.9$–1.4 are N-artifacts (corrected by r630b: trustworthy slopes -50.6 on 0.3–0.6 and -116 on 0.6–1.0; RELAY_UNIVERSAL and $L_{\det}$ unaffected; greedy $\Delta \approx 0.015$ WITHDRAWN by r635) — super-exponential (Landau–Widom plunge); λ^* underflows the SVD range beyond $L \approx 1.4$. Controls: SCRAMBLE first failure 0.376, WPERM 0.402, EPSTEIN (2^k only) $0.564 \approx L_3 + 0.015$ — each control breaks at entry+lead.

(β, γ)	$L_{\det}$
$(0.6, 20)$	0.55
$(0.75, 20)$	0.55
$(0.6, 50)$	0.80
$(0.9, 20)$	0.50

Parent typing. The RELAY_UNIVERSAL criteria ($\Delta_j \geq 0$, $g < 0$, repair) are consequences of $Q_j \succeq 0$ (RH-consistency), not a mechanism. The informative results are the control breaks at entry+lead (the constant-lead "greedy law" $\Delta \approx 0.015$ is WITHDRAWN by r635), the super-exponential decay, and $L_{\det}$ — which shows that windows with $L \gtrsim 0.5$ are hypersensitive off-line detectors (a nulled zero displaced by σ contributes $-\sigma^2|\partial \hat{f}|^2$, beating the tiny λ^*), hence relay steps beyond $L \approx 0.5$ are RH-in-a-box statements, not elementary finite problems. Fence: window positivity for the scanned L is RH-consistent. Experiments only. **No RH claim.**
Round r620 (probe `p2_reflection_factor_probe.py`, SHA `194fe9e5`, SPEC `7c8e73b7`, 73/73) returns MIXED_SECTOR. Exact identity for odd h, $d = \log 2$, overlap $I = [d-L, L]$, reflection $(Rf)(x) = f(d-x)$: $g_h(d) = -(\|f_+\|^2 - \|f_-\|^2)$; the prime term $-w_2 g = w_2(\|f_+\|^2 - \|f_-\|^2)$ stabilizes the symmetric edge sector and destabilizes the antisymmetric one. $\dim E_-(A^{\text{odd}}) = 1$ for $L \geq 0.38$ ($\lambda_{\min}$ $-2.4 \cdot 10^{-2}$ at 0.38 to -0.43 at 0.549). Mass of the negative vector inner/+/−: 0.85/0.15/0.005 (0.38), 0.71/0.28/0.01 (0.40), 0.50/0.48/0.02 (0.43), 0.06/0.94/0.001 (0.549) — migrates from inner to R-symmetric edge. The antisymmetric sector is safe without the prime ($\lambda_{\min}(A|_{P_-}) - w_2 = +5.7 \to +0.29$); cross term 0.12–0.28. Oriented inequality $\lambda_{\min}(A^{\text{odd}} + w_2(P_+ - P_-)) = \lambda^*(L) \geq 0$ everywhere. The contract `PRIME.SEMILOCAL.ODDSECTOR.EDGE.01` [O]is retyped to this oriented statement; the repair is a quantitative dominance with margin $\lambda^*(L)$, not an exact sector identity. Experiments only. **No RH claim.**
Round r621 (probe `p2_digamma_duplication_probe.py`, SHA `dff2e1b3`, SPEC `78f47c87`, 37/37) returns DUPLICATION_CONSTANT_ONLY (recorded as a no-go in Section 15). Experiments only. **No RH claim.**
Round r622 (probe `support_darboux_probe.py`, SHA `845bb937`, SPEC `27ffd017`, PAYLOAD `e1031d17`, 64/64) returns TRANSPORT_FEASIBLE_NORULE (recorded as a no-go in Section 15). The transport formulation relocates the tiny margins into the scalar $c_0(L)$ (relative form of the wall); no source-side rule. Experiments only. **No RH claim.**

Round r623 (probe `semilocal_p2_dilation_probe.py`, SHA `d1bec728`, SPEC `5890676d`, 128/128) returns P2__DILATION__STRUCTURALLY__BLOCKED (recorded as a no-go in Section 15). The 2-adic identity $m_2(t) = \log 2 \cdot (1 - P_r(t \log 2))$ holds to $4.9 \cdot 10^{-12}$; the proposed PSD dilation is obstructed at the principal block, at M_2, and as a tautology of $\lambda^*(L)$. The lane is not opened. Experiments only. **No RH claim.**

Round r625 (probe `iiks_vanishing_metric_probe.py`, SHA `973a1ce2`, SPEC `7a4257e3`, NUM `2d066e2d`, 19/19) returns IIKS__NO__METRIC__IN__CLASS (recorded as a no-go in Section 15; prose NOT__TESTABLE__IN__CORPUS__FORM). Zhou's vanishing lemma is not testable on the corpus discrete IIKS/CD object: the jump is not the RHP jump equivalent to $\tau(s) \neq 0$, and no metric in the source-explicit class is contractive. Experiments only. **No RH claim.**

Round r626 (probe `xi_finitefree_collision_probe.py`, SHA `c4750063`, SPEC `c7cef974`, NUM `f533d168`, 51/51) returns COLLISION__DETECTOR__CALIBRATED__NO__BARRIER (recorded as a no-go in Section 15). The collision barrier $\int \Phi\, dt < \infty$ is equivalent to Disc $\neq 0$ along the path; Taylor sections of Ξ are not real-rooted at $t = 0$; the detector is calibrated on a hyperbolic control. Experiments only. **No RH claim.**

Round r627 (probe `af_twomoment_optimizer_probe.py`, SHA `da1a9118`, SPEC `9b95f564`, 17/17) returns NO__GAIN (recorded as a no-go in Section 15). Window optimum in the A–F two-moment class is Montgomery–Taylor $p^* = 0.6725007037$; the proved ceiling 0.6818287 is a richer certificate, not a window bound. Gap $9.3 \cdot 10^{-3}$ not closable by window optimization. Experiments only. **No RH claim.**

Round r628 (probe `window_box_verifier_probe.py`, SHA `9c99ed6a`, SPEC `36c0e66a`, NUM `84aaf685`, 12/12) returns COMPRESSION__CERTIFIED__BOX__NUMERICAL (recorded as a no-go in Section 15). Finite-window positivity is interval-certified; a rigorous zero-free box is blocked by Trudgian $|S(T)|$ looseness ($\sim 10^7$). Experiments only. **No RH claim.**

Round r629 (probe `certificate_class_atlas_probe.py`, SHA `4dae1fbe`, SPEC `0bf7c64f`, 15/15) returns ATLAS__NO__NEW__UNCONDITIONAL__GAIN (recorded as a no-go in Section 15). Unconditional rows do not improve on $p_0 = 0.6818287$; only pair-correlation input beyond support 1 buys proportion. Experiments only. **No RH claim.**

Round r632 (probe `jet_deflated_ldet_probe.py`, SHA `63ecb68f`, SPEC `31a29760`, RESULT `6299cedc`, 14/14) returns JET__GAIN__SMALL (recorded as a no-go in Section 15). Local L_{det} gain collapses under coverage ($\dim K/N$ above Nyquist). Experiments only. **No RH claim.**

Round r630 (probe `margin_law_symreg_probe.py`, SHA `5b3d1125`, SPEC `87febb97`, RESULT `a0765d7f`, 19/19) returns MARGIN__LAW__UNRESOLVED with r630b addendum N__LIMITED__ARTIFACT (recorded as a no-go in Section 15). Converged log-slopes -50.6 (0.3–0.6) and -116 (0.6–1.0); the margin law is density-driven from the unconditional $N(T)$ profile (RH-neutral); the dBN gap functional B_k is not a universal barrier (fraction $B_k < 2 = 0.470$). Experiments only. **No RH claim.**

Round r633 (probe `frontier_followups_probe.py`, SHA `9f5287e7`, SPEC `ee7c7ba7`, RESULT `814abdf5`, 16/16) returns four follow-up verdicts (recorded as a no-go in Section 15). RANDOM__EULER__REFUTED is a measured class no-go (magnitude-only and Diophantine-approximation arguments dead; Haar 99.5% negativity stands); $C(L) = 1$ is *not* the mechanism (generic for minimizers, including negative Haar draws; r637); a-compactification is cosmetic (the RH-relevant end $a \to 0$ stays unbounded); forward Loewner is a detector, not a lever; LEAD__LAW__INVARIANT is VOID (withdrawn $N = 80$ values). Experiments only. **No RH claim.**

Round r635 (probe `relay_lead_precision_probe.py`, SHA `36c06826`, SPEC `28db9bef`, RESULT `3457e907`, 11/11) returns LEAD__PARTIAL (recorded as a no-go in Section 15). The r619 greedy law $\Delta \approx 0.015$ is WITHDRAWN; four converged leads 0.0253/0.0083/0.0064/0.0034; $q \geq 7$ UNMEASURABLE at the 40-digit floor. Experiments only. **No RH claim.**

Round r636 (probe `relay_lead_law_probe.py`, SHA `ee2deb46`, SPEC `6bf7a7ee`, RESULT `2a6bd720`, 6/6) returns LEAD__LAW__PARTIAL (recorded as a no-go in Section 15). No model

$< 20\%$ on all four points; the lead law as a message is dissolved. Experiments only. **No RH claim.**
Round r637 (probe `relay_vote_map_probe.py`, SHA `fc5c1426`, SPEC `44bf67f8`, RESULT `540557ad`, 14/14) returns VOTE__MODE__SPECIFIC+FLIP__COUNTEREXAMPLE (recorded as a no-go in Section 15). Unanimity is GENERIC (edge geometry, not arithmetic); C is not a positivity predictor. Experiments only. **No RH claim.**
The decoding programme (r635–r637) as a whole: the three candidate "messages" of the prime event log beyond the explicit formula (constant lead, unanimity, odd-edge defect type) reduce to margin collapse and edge geometry; no side channel measured. Experiments only. **No RH claim.**
Round r638 (probe `gabor_first_contact_selberg_probe.py`, SHA `2ab532bc`, SPEC `22a92842`, RESULT `58a36aff`, 14/14) returns KILLED(structural: selberg-convolution-is-hankel). The first-contact + Selberg square-completion attack on the pure-Gabor zero side $G = \mathrm{POLE} - \mathrm{PRIME} + \mathrm{ARCH}$ fails its pre-registered Toeplitz gate: the Selberg term $\sum_N (\Lambda * \Lambda)(N)\,\psi(\log N)$ is a Hankel form in $(\log m, \log n)$ and admits no PSD completion over the operator-generated ψ (15/15 cells, best $\lambda_{\min} = -0.0097$ against tolerance -10^{-9}); the only squares compatible with the prime-side frequency support $\{\log n\}$ are Euler-local and cost $\geq 3.28\times$ the POLE+ARCH budget at $a \leq 0.3$, $\omega \geq 30$. At a first-contact point $\partial_a Z = \frac{1}{2}\partial_\omega^2 Z$ (to 10^{-16}), so the contact carries one differential condition, not two, and negativity is a down-set in a. On the true ζ the sign of the Selberg part at near-contact minima is indefinite and world-blind (MAIN 0.25–0.33, SCRAMBLE 0.22–0.31, WPERM 0.20–0.29). Experiments only. **No RH claim.**
Round r631L (Lean-only, no probe). Classification of the seven tracked `sorry`s: R-class Canonical.lean:264 `lstar_canonical`, :273 `terminal_q_canonical`; T-class but blocked Canonical.lean:396 `pair_terminal_dictionary`, Source.lean:1008 `mainWindow_iff_builtFromPrimeSource`; A-class Elementwise.lean:1589 `arch_gauss_mellin_digamma_identity`, ExternalBridges.lean:15641 `standard_explicit_formula_identification`, :15708 `fullWeil_separates_offCritical_zeros`. No closure. Docstring fix: `GaborContourLimitRemainder` is proved (`gaborContourLimitRemainder_holds`, r576). Census stays 7. **No RH claim.**
Round r634L (Lean-only, no probe). `gabor_explicitFormula_pure` is unconditional for admissible pure F; `gabor_primeSide_rational_criterion_iff_rh` is the prime-side rational Π_1 form of RH (axioms `propext/Classical.choice/Quot.sound`). Remaining EF brick: `GaborHatQuarticExplicitRemainder` (non-pure quartics only). Census stays 7. **No RH claim.**
Round r638L (Lean-only claim-boundary repair, no new probe). Two A-class assertions from r631L are retired rather than "proved": the exact eventual `arch_gauss_mellin_digamma_identity` quantifier is overstrong (fixed mesh becomes coarser as the anchor grows), and `standard_explicit_formula_ide` mixes the $2\Lambda/\sqrt{n}$ corpus gauge with the $\Lambda(n)(1 + n^{-1})$ contour gauge. Their Props remain unasserted; historical arch extraction theorems now take the former explicitly, while `standard_explicit_formula_identification_honest` remains proved. `lake build RH` completes 3578/3578 jobs; the tactic-sorry census falls $7 \to 5$. Seven affected probes passed twice byte-identically. This is a claim-boundary repair, not progress on RH. **No RH claim.**

Two-key filter (externalization package). A proposed route counts as RH-relevant only if it (key 1) fails under prime-position scrambling that preserves densities and (key 2) fails when the functional equation / an off-line zero quadruple is injected. Round r565 showed the Gabor dominance margin is scramble-invariant (key 1 fails $\Rightarrow$ it is zero-geometry, not arithmetic).

Outlook (2026). arXiv:2608.13637 (Aug. 2026, Lean-verified, `github.com/anthropics/zeta-23-lean`, Apache-2.0) obtains that at least 2/3 of the zeros are simple and on the line (0.6725 with a Montgomery–Taylor window) via a rank–trace inequality plus Sylvester inertia on a finite compression of Weil's Hermitian form; the prime side is the Montgomery–Vaughan diagonal only; §7.2 proves a "bandwidth-one ceiling" ≈ 0.6819, beyond which prime-pair information (Hardy–Littlewood strength, Fourier support > 1) or higher trace moments in the Rudnick–Sarnak range are required. arXiv:2602.04022 (Connes, Feb. 2026) together with Connes–Consani–Moscovici "Zeta spectral triples" (EMS 2026) show that for each cutoff c the even ground state of the truncated Weil form has Mellin zeros on the critical line (Connes–van Suijlekom); convergence to ζ-zeros is the open RH-core. arXiv:2608.24827 (Aug. 2026) gives unconditional window positivity $Q(f) \geq 8.9 \cdot 10^{-18}\|f\|^2$ for support 1.6, with $-\ln \lambda^*(L) \simeq 2\pi^2 N(T^*)/\ln N(T^*)$, $T^* = 2\pi e^{2L}$; pointwise certificates need frequencies up to $2\pi \exp(4e^L)$ (a doubly exponential barrier). Our reading: all three meet the same wall as our r565/r603 findings — local (Euler-factor) information is RH-blind; progress beyond the ceiling needs prime-pair (support > 1) control. **No RH claim.**

17.5 The exposed-orbit separator (r605)

Round r605A: `GaborZeroSideForAllZeros` is the prescribed-packet pointwise form $\forall \rho$ with `scalingGaborTest`$(a = \sigma^2/64,\ \omega = \gamma - \pi a/\sigma)$ and is FORALL__ZERO__OVERSPECIFIED; `gabor_zeroSide_pure_criterion_iff_rh` takes `hsep` as a premise.

Round r605N (probe `gabor_exposed_orbit_probe.py`, SHA `c99998c4`, SPEC `1a4bf949`, 40/40, byte-identical) returns EXPOSED__ORBIT__HOLDS: 9/9 configs negative at $n = 0$, normalised limit $-4m(\rho^*)$ exact, tail bound holds.

Rounds r605B+C add `RH/GaborExposedOrbit.lean` (1734 lines, sorry-free). The module supplies `gaborScore`, the `ExposedOrbit` structure, `exists_exposedOrbit_of_not_rh`, phase lock $\cos = -1$, the three-lobe majorant $\operatorname{Re}\hat{h}(\rho^*) \leq -(\pi/4a)\, e^{M/2a} + (3\pi/4a)\, e^{(M-\Delta)/2a}$, `gaborExposedOrbitAssembly_holds`, and the endpoints `exists_negative_pureGabor_of_not_rh` (`hasm`), `gabor_zeroSide_pure_criterion_iff_rh_of_assembly` (`hasm`), and

$$(\forall a, \omega\ (a > 0):\ 0 \leq \texttt{gaborZeroSide}(\texttt{pureGaborTest}\, a\, \omega)) \iff \texttt{RiemannHypothesis}$$

(`gabor_zeroSide_pure_criterion_iff_rh_unconditional`; #print axioms = {`propext`, `Classical.choice`, Tactic-sorry census at this landing 8 (Canonical 3, Source 1, Elementwise 1, GaborSeparation 1, ExternalBridges 2); live census 7 after r612 (GaborSeparation 0). Independent audit r605C-AUDIT: STATEMENT__SOUND (`gaborZeroSide` is $\operatorname{Re}\sum'$ over `IsCriticalStripZetaZero` weighted by `riemannZetaMultiplicity`; summable for every pure packet by `gaborMultiplicityWeightedHatSummable`; $\leftarrow$ is on-line positivity $\hat{h}(\frac{1}{2} + it) =$ a real square ≥ 0; every `ExposedOrbit` field is discharged from $\neg$RH).

Mandatory wording. RH $\iff$ nonnegativity of the pure Gabor zero-side on the two-parameter family (a, ω). The sharpest live formal statement (r634L) is the prime-side rational criterion `gabor_primeSide_rational_criterion_iff_rh`: RH $\iff \forall$ rational (a, ω) with $a > 0$, POLE $-$ PRIME $+$ ARCH ≥ 0 on `pureGaborTest` $a\, \omega$. Positivity of `gaborZeroSide` itself remains open. This is not an RH proof. The Weil test space is reduced to a two-parameter pure Gabor family. **No RH claim.**

Low-height zero-free and spectral refutation (r610). Lean-only, in `RH/GaborSeparation.lean`. `GaborLowHeightZeroFree` is an unasserted Prop at $T_0 = 4$ (classically true: the first zero is 14.1347, not in Mathlib). The pole term is exactly $\pi e^{1/8}$ (`gaborHat_pure_one_zero`); the zero side is $\leq 19/20$ of it, and `gaborSpectralFormula_refuted_of_lowHeight` is a theorem. #print axioms = {`propext`, `Classical.choice`, `Quot.sound`}. Conditional/finite only. **No RH claim.**

Countable rational-packet criterion (r612/r612C). The asserting theorem `gabor_separationForAllZer` is deleted. The Prop `GaborSeparationForAllZeros` is retained unasserted with an OVER-

SPECIFIED / believed-false docstring. Module `RH/GaborCountableCriterion.lean` (551 lines, imported in `RH.lean`). Round r612C proves `gaborZeroSideContinuous_holds` (joint continuity of $Z(a,\omega) := \texttt{gaborZeroSide}(\texttt{pureGaborTest}\, a\, \omega)$ on $\{a > 0\} \times \mathbb{R}$ via a local uniform Gaussian majorant $\|\hat{h}\| \le (\pi/a_0)\exp((1/8 + \omega_1^2/2)/a_0)\exp(-t^2/(4a_1))$ and `continuousOn_tsum`; summability via `summable_gauss_over_zeros`). The rational-packet criterion is now *unconditional*:

$$\texttt{gabor\_zeroSide\_rational\_criterion\_iff\_rh\_unconditional} : (\forall a, \omega \in \mathbb{Q},\ a > 0 :\ 0 \le \texttt{gaborZeroSide}(\texttt{p}$$

#print axioms = {`propext`, `Classical.choice`, `Quot.sound`}. Named-open list shrinks by one (`GaborZeroSideContinuous` is proved). Tracked sorry census stays 7. Module count $53 \to 54$ at r612. Conditional/finite only in the sense that positivity of `gaborZeroSide` itself remains open. **No RH claim.**

Prime-side rational criterion (r634L). The pure-class explicit formula `gabor_explicitFormula_pure` is now a theorem (unconditional: both remainders are theorems). Transporting the r612C rational-packet criterion across that identity yields the sharpest live Π_1 form `gabor_primeSide_rational_criterion_iff_rh` (568 lines in `RH/GaborCountableCriterion.lean`):

$$(\forall a, \omega \in \mathbb{Q},\ a > 0 :\ 0 \le \texttt{gaborPoleSide}(\texttt{pureGaborTest}\, a\, \omega) - \texttt{gaborPrimeComb}(\texttt{pureGaborTest}\, a\, \omega) + \texttt{gaborA}$$

#print axioms = {`propext`, `Classical.choice`, `Quot.sound`}. The remaining EF brick `GaborHatQuarticExplicitRemainder` applies only to non-pure even quartics and is not used by the criterion. Positivity of `gaborZeroSide` itself remains open. This is not an RH proof. **No RH claim.**

18 Lean 4 formalization status

Following the r267 recommendation (layer by provability), the workspace `rh/lean` contains a Lean 4 project (toolchain v4.29.1, mathlib pinned; builds green on the reference machine). Status at the time of writing: fifty-six modules under `rh/lean/RH/` (r617L: $54 \to 56$). Tactic-sorry census 5 after r638L (Canonical 3, Source 1, ExternalBridges 1; Elementwise and GaborSeparation 0); all other modules 0. Round r631L classified the previous seven as **R-class** (RH-core, Canonical.lean:264 `lstar_canonical`, :273 `terminal_q_canonical` — L^*-positivity / terminal gate $q_N < 1$; opaque `canonicalCompletion`); **T-class but blocked** (Canonical.lean:396 `pair_terminal_dictionary`, OP dictionary $Z = r_{N-1}$ missing; Source.lean:1008 `mainWindow_iff_builtFromPrimeSource`, both sides opaque); **A-class** (Elementwise.lean:1589 `arch_gauss_mellin_digamma_identity`; ExternalBridges.lean:15641 `standard_explicit_formula_identification`; :15708 `fullWeil_separates_offCritical_zeros`). r638L retires the first two as wrong-type asserting theorems; the sole remaining A-class `sorry` is the classical compact Weil separator. No closure is claimed. The inventory below is the live census. Historical status notes that follow it retain their original wording and are not a second count.

Gabor explicit-formula brick chain (r555–r598). The Gabor route's analytic spine is a sorry-free brick chain, recorded module-by-module in the inventory. In order: hat analyticity (`GaborHatAnalytic`, r555); horizontal edges (`GaborHorizontalEdges`, r557); right and left vertical identification (`GaborVertical`, r564; `GaborLeftVertical`, r570); fixed-T contour residue (`GaborContourResidue`, r557); arch contour shift and Digamma fold (`GaborArchContour`/`GaborArchDigamma`, r577–r581); autocorrelation closed form (`GaborAutocorrelation`, r572); zero summability (`GaborZeroSummable`, r557); functional-equation multiplicity (`GaborFEMultiplicity`, r582); spectral bridge with on/off split and fundamental-domain exhaustion (`GaborSpectralBridge`, r579–r589); dominance Log/Log2 (`GaborDominanceLog`, r575; `GaborDominanceLog2`, r583/r589); outer tail (`GaborOuterTail`, r593/r594); window glue and anchored witness (`GaborWindowGlue`, r595; `GaborAnchoredWitness`, r598). Round r617L adds two sorry-free modules outside this Gabor spine: `CoxeterCompletion` (Φ_{30} identities and $\operatorname{Tr} C = -1$) and `PrimeLogIndependence` ($\mathbb{Q}$-linear independence of $\{\log p\}$; Mathlib had no such lemma). Round r634L discharges the pure-class explicit formula as a theorem: `gabor_explicitFormula_pure` states that for every admissible pure F (coeffs $\langle 1, 0, 0\rangle$)

$$\texttt{gaborZeroSide}\, F = \texttt{gaborPoleSide}\, F - \texttt{gaborPrimeComb}\, F + \texttt{gaborArchSide}\, F,$$

unconditionally — both named remainders `gaborContourLimitRemainder_holds` (r576) and `gabor_vertical_arithmetic_remainder` (r581, via `gaborArchDigammaIdentification_holds`) are theorems. The implication wrappers `gabor_explicitFormula_pure_of_remainders` and `gabor_explicitFormula_of_remainders` remain sorry-free. The only remaining EF brick is `GaborHatQuarticExplicitRemainder` (non-pure even quartics; the prime-side criterion does not need it). The sharpest live Π_1 statement is `gabor_primeSide_rational_criterion_iff_rh`: $(\forall a, \omega \in \mathbb{Q},\ a > 0 :\ 0 \le \texttt{gaborPoleSide} - \texttt{gaborPrimeComb} + \texttt{gaborArchSide}$ of $\texttt{pureGaborTest}\, a\, \omega) \iff \texttt{RiemannHypothesis}$, axioms $\{\texttt{propext}, \texttt{Classical.choice}, \texttt{Quot.sound}\}$. **No RH claim.**

Module	**Role**	**Round(s)**	`sorry`
`Audit`	axiom audit of the main theorem chain	C1 / r638L	0
`Augmented`	$L^\dagger$ unification of L^* and the terminal statement	r332	0
`Basic`	abstract coupled-tau chain data	r256–r259	0
`Canonical`	canonical final domain (three typed holes)	C1 / r326	3
`Closure`	reconstruction of master positivity from L^* and terminal	r305	0
`Counterexamples`	permanent r273 reviewer refutation guards	r273 / r320	0
`CoxeterCompletion`	Φ_{30} identities and $\operatorname{Tr} C = -1$	r617L	0
`DualResolvent`	$L^\dagger \Leftrightarrow R^\dagger \succ \frac{1}{2} I$ as finite matrix identity	r362 / r430	0
`EdgeBalance`	edge-balance chain as finite real algebra	r417–r426	0
`Elementwise`	conditional elementwise extraction without a mesh ladder	r326 / r638L	0
`ExternalBridges`	honest-contour bridge plus compact separation target	r463 / r638L	1
`FaithfulFold`	production-window transcription (source to grid fold)	r459 / r463	0
`FlankEntry`	flank-entry lemma against the pivot layer	r384	0
`FrequentlySelected`	semidefinite + infinitely-often conditional extraction	r430 / r638L	0
`GaborAnchoredWitness`	anchored singleton fragment and tail modulus	r598	0
`GaborArchContour`	χ'/χ contour shift onto the critical line	r578 / r581	0
`GaborArchDigamma`	left-edge Digamma identification of χ'/χ	r577 / r581	0
`GaborArithmeticSeparator`	existential arithmetic Gabor separator	r590 / r600	0
`GaborAutocorrelation`	closed form of the pure Gabor ACF	r572	0
`GaborContourResidue`	fixed-T residue identity for $(\zeta'/\zeta)\hat{h}$	r557	0
`GaborCountableCriterion`	rational-packet zero-side iff RH (unconditional, r612C); prime-side iff r634L; 568 lines	r612/r612C/r634L	0
`GaborDominance`	canonical isolation-shrink dominance	r569	0

Complete Lean module inventory (continued)

Module	Role	Round(s)	`sorry`
`GaborDominanceAssembly`	packing and remainder-budget assembly	r571	0
`GaborDominanceLog`	counting-theorem (log occupancy) dominance	r575	0
`GaborDominanceLog2`	log-compatible remainder BoundLog2	r583 / r589	0
`GaborDominanceProof`	remaining isolation-shrink dominance bricks	r569	0
`GaborExplicitFormula`	EF implication from named remainders; pure EF unconditional (r634L); 220 lines	r558 / r581 / r634L	0
`GaborExposedOrbit`	existential pure-Gabor zero-side separator	r605	0
`GaborFEMultiplicity`	FE-invariant analytic order of zeros	r582	0
`GaborHatAnalytic`	entirety and FE-symmetry of `gaborHat`	r555	0
`GaborHorizontalEdges`	horizontal contour edges vanish	r557	0
`GaborInequality`	symbolic Gabor-inequality boundary	r544 / r552	0
`GaborIntegral`	bilateral Gaussian integral for the pure test	r543	0
`GaborLeftVertical`	left-edge prime reflection	r570	0
`GaborOuterTail`	window-adaptive outer quadrupole tail	r593 / r594	0
`GaborSeparation`	Gabor-wavepacket separation architecture; 2181 lines	r542–r634L	0
`GaborSpectralBridge`	spectral on/off split and FD exhaustion	r579–r589	0
`GaborThetaBound`	Gauss-bin majorant and partial summation	r552	0
`GaborVertical`	right/left vertical $T \to \infty$ identification	r564	0
`GaborVerticalLimit`	Landau-gap $T \to \infty$ contour glue	r576	0
`GaborWindowGlue`	three-term window-plus-tail comparison	r595	0
`GaborZeroSummable`	summability of $\hat{h}_W$ over nontrivial zeros	r557 / r587	0
`GraphResolvent`	graph-resolvent dictionary (finite algebra)	r412	0
`Haynsworth`	Haynsworth inertia additivity	r373	0
`InnerBridges`	finite inner-bridge algebra of the A-cap	r464	0
`Inertia`	matrix-theorem / Sylvester layer	r305	0
`OneDefect`	one-defect absorption theorems	r406	0
`Open`	ladder bookkeeping of the former open edges	r273	0
`PairBound`	finite pair-bound theorem of the coupled-tau lane	r269 / r271 / r320	0
`PivotCoordinate`	finite-algebra lemmas in source pivot coordinates	r380	0
`PrimeLogIndependence`	$\mathbb{Q}$-linear independence of $\{\log p\}$	r617L	0
`Recursion`	proved coupled-tau core identities	r256–r263	0
`Selected`	exact real selected-window domain and conditional extraction	r397 / r638L	0
`Source`	explicit prime-window source interface	r310 / r320	1
`Window`	`VonMangoldtWindow` structure and master theorem	r273 / r305	0
`ZeroIncrement`	local zero-count increment bound	r546 / r547	0

Detail on the historically named spine (live sorry counts are those of the inventory, not the older notes below):

File	Status
`RH/Basic.lean`	proved

File	**Status**
`RH/Recursion.lean`	**proved, no `sorry`**
`RH/Inertia.lean`	**proved** since r305 (1 `sorry`: the mathlib gap)
`RH/Closure.lean`	**reconstruction** (r305); 1 `sorry`

File	Status
`RH/Source.lean`	**source interface** (r310/r310b); **red-team retyped** (r320); 1 `sorry`

File	Status
`RH/Window.lean`	**master theorem** (r273) + **the fog-free hole** (wave 5) + **the canonical forr**
`RH/Open.lean`	retyped (r273, no `sorry`)
`RH/PairBound.lean`	proved + 1 `sorry`; **red-team retyped** (r320)

File	**Status**
`RH/Counterexamples.lean`	**proved, no `sorry`** (guards)
`RH/Elementwise.lean`	**Level-C architecture** (r326); 3 typed `sorry`s

File	Status
`RH/Selected.lean`	**exact selected domain** (r397/r638L); 0 `sorry`
`RH/GaborIntegral.lean`	**proved, no `sorry`**
`RH/ZeroIncrement.lean`	**proved, no `sorry`**
`RH/GaborThetaBound.lean`	**proved, no `sorry`**
`RH/GaborSeparation.lean`	0 named `sorry`; census 7 (r634L, 2026-09-02); 2181 lines
`RH/GaborExposedOrbit.lean`	**proved, no `sorry`** (r605B+C)

File	Status
`RH/GaborCountableCriterion.lean`	**proved, no sorry** (r612/r612C/r634L)
`RH/CoxeterCompletion.lean`	**proved, no sorry** (r617L)
`RH/PrimeLogIndependence.lean`	**proved, no sorry** (r617L)

r397: `lstar_canonical/terminal_q_canonical` are degraded to the alternative (rational-certificate) route; the named open kernel is `selected_augDualResolvent_gt_half`. Sorry census stays five. NO RH CLAIM.
r556/r600: a parallel Gabor route (`RH/GaborSeparation.lean`) is documented in Section 17.4. The historical wrapper `gabor_inputs_to_mathlib_rh` is **VACUOUS** (r600); the live wrapper is `gabor_zeroSide_inputs_to_mathlib_rh`. The former asserting Gabor sorry `gabor_separationForAllZeros` was deleted at r612 (Prop retained unasserted) and is *not* part of the window-local census below. r541 is a numeric precheck, not a configuration statement. NO RH CLAIM.
The `sorry` declarations (**eight in total** since the r326 Level-C round: five window-local since the r305/r310/r310b restructuring — census 5 → 5 across the r320 red-team repair, with two of the five *retyped* — plus the three *named typed Level-C statements* of `RH/Elementwise.lean`, census 5 → 8) are honest markers, not deferred work of a known proof. The r305 reconstruction proved the master theorem *from* its two true inputs (`lstar_terminal_implies_master`) and cleaned the Inertia layer (four of five statements proved, incl. a from-scratch Sylvester criterion), so the census now separates cleanly: (i) `lstar_subordination` — the base/wall *window-local* TRUE hole, the program's open center in canonical two-measure form; (ii) `terminal_positive_main` — the border/fiber *window-local* TRUE hole ($0 < B \wedge q < 1$, faithful to the r258 TERMINAL_Q_LAW + r243 budget form); (iii) `pair_margin_main` — the same fiber hole in r271 pair coordinates (the r263 dictionary not yet formalized); (iv) `crossing_budget` — a mathlib gap (Jacobi inertia / minor dictionary), exact-rationally certified in v962; (v) `mainWindow_iff_builtFromPrimeSource` — the definitional source bridge, opacity-forced by the honest `MainWindow` predicate (the r310 form `mainWindow_explicit_bridge` is proved from it); and, since r326, the three named typed Level-C statements (vi) `arch_elementwise_stabilization` and (vii) `pole_elementwise_stabilization` — classical kernel transcription (the `arch_A` GL-48 integrals and the v716 pole closed form; R325-measured exact) — and (viii) `specFamily_sourceExact_completion` — classical + definitional, opacity-

forced, *not* consumed by the extraction route. The two true *window-local* holes (Level B of Remark 12.2) *are* the program's open window-family problem, stated in truth-capable form: conditioned on the deliberately opaque `MainWindow` predicate they can neither be refuted by toy models (the r273 and wave-5 counterexample guards verify that the old universal forms could) nor proved without supplying the von-Mangoldt arithmetic. The two wave-12 caveats are updated in wave 13: the named open Lean target `CanonicalPrimeWindow` plus exactly one construction theorem `sourceExact_buildPrimeWindow` (Remark 12.3) is *implemented and proved* (r326, Section 13.1), and the census now *names* the Level-C distance (the three typed statements above) without closing it. No axioms, no `native_decide`, no statement weakening; builds green.

Reproducibility

All objects referenced here are pinned in `rh/INVENTORY.json` (SHA-256 per file) and exercised by the RH suite `rh/verification/run_rh.py` (integrity check, sealed probes r250–r327 in smoke mode, the fifteen verification modules v955/ v956/v958/v959/v960/v961/v962/v963/v964/v965/v966/v967/v968 v970 by path, and the Lean build gate); since wave 4 the suite's fast mode is a gate of `bash build.sh audit`. The canonical claim state is the repository ledger; the canonical prose is `tfpt_prime_front.tex`.

Closing status. Research documentation. The certified content is finite and window-level ([**E**]); the two edges of Section 17 are open ([**O**]); the mincut is base 4 / refined 5. The Gabor chain of Section 17.4 is a parallel conditional reduction: the live criterion is positivity of `gaborZeroSide` (hpos*); the historical `gabor_inputs_to_mathlib_rh` is VACUOUS (r600); r541 is a precheck, not a configuration statement; separation stays open. **Nothing in this document is evidence for or against the Riemann Hypothesis in either direction. NO RH CLAIM.**